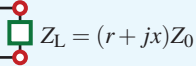
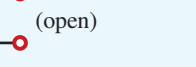
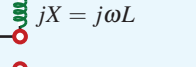
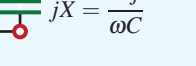


Table 2-3 Magnitude and phase of the reflection coefficient for various types of loads. The normalized load impedance $z_L = Z_L/Z_0 = (R + jX)/Z_0 = r + jx$, where $r = R/Z_0$ and $x = X/Z_0$ are the real and imaginary parts of z_L , respectively.

Load	$ \Gamma $	θ_r
 $Z_L = (r + jx)Z_0$	$\left[\frac{(r-1)^2 + x^2}{(r+1)^2 + x^2} \right]^{1/2}$	$\tan^{-1} \left(\frac{x}{r-1} \right) - \tan^{-1} \left(\frac{x}{r+1} \right)$
 Z_0	0 (no reflection)	irrelevant
 (short)	1	$\pm 180^\circ$ (phase opposition)
 (open)	1	0 (in-phase)
 $jX = j\omega L$	1	$\pm 180^\circ - 2 \tan^{-1} x$
 $jX = \frac{-j}{\omega C}$	1	$\pm 180^\circ - 2 \tan^{-1} x$

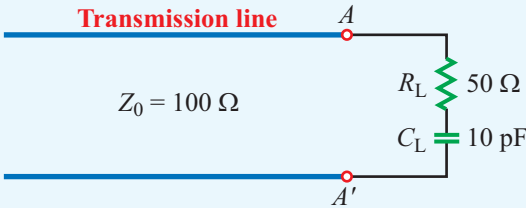


Figure 2-13 RC load (Example 2-3).

From Eq. (2.59), the voltage reflection coefficient is

$$\begin{aligned}
 \Gamma &= \frac{z_L - 1}{z_L + 1} \\
 &= \frac{0.5 - j1.59 - 1}{0.5 - j1.59 + 1} \\
 &= \frac{-0.5 - j1.59}{1.5 - j1.59} \\
 &= \frac{-1.67e^{j72.6^\circ}}{2.19e^{-j46.7^\circ}} \\
 &= -0.76e^{j119.3^\circ}.
 \end{aligned}$$

This result may be converted into the form of Eq. (2.62) by replacing the minus sign with e^{-j180° . Thus,

$$\begin{aligned}
 \Gamma &= 0.76e^{j119.3^\circ}e^{-j180^\circ} \\
 &= 0.76e^{-j60.7^\circ} \\
 &= 0.76/\underline{-60.7^\circ},
 \end{aligned}$$

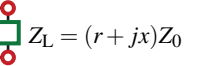
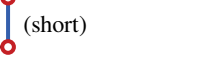
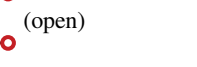
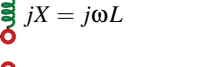
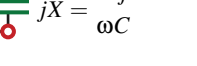
or

$$|\Gamma| = 0.76, \quad \theta_r = -60.7^\circ.$$

$$\begin{aligned}
 z_L &= \frac{Z_L}{Z_0} \\
 &= \frac{R_L - j/(\omega C_L)}{Z_0} \\
 &= \frac{1}{100} \left(50 - j \frac{1}{2\pi \times 10^8 \times 10^{-11}} \right) \\
 &= (0.5 - j1.59) \Omega.
 \end{aligned}$$

表 2-3 各种负载的反射系数的大小和相位。归一化负载阻抗 $z_L = Z_L/Z_0 = (R + jX)/Z_0 = r + jx$ ，其中 $r = R/Z_0$ 和 $x = X/Z_0$ 分别是 z_L 的实部和虚部。

CE

Load	$ \Gamma $	θ_r
 $Z_L = (r + jx)Z_0$	$\left[\frac{(r-1)^2 + x^2}{(r+1)^2 + x^2} \right]^{1/2}$	$\tan^{-1} \left(\frac{x}{r-1} \right) - \tan^{-1} \left(\frac{x}{r+1} \right)$
 Z_0	0 (no reflection)	irrelevant
 (short)	1	$\pm 180^\circ$ (phase opposition)
 (open)	1	0 (in-phase)
 $jX = j\omega L$	1	$\pm 180^\circ - 2 \tan^{-1} x$
 $jX = \frac{-j}{\omega C}$	1	$\pm 180^\circ - 2 \tan^{-1} x$

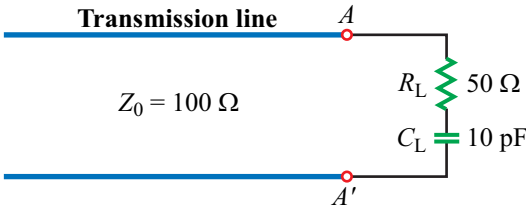


图 2-13 RC 负载（示例 2-3）。

从方程。 (2.59) ， 电压反射系数为

$$\begin{aligned}
 \Gamma &= \frac{z_L - 1}{z_L + 1} \\
 &= \frac{0.5 - j1.59 - 1}{0.5 - j1.59 + 1} \\
 &= \frac{-0.5 - j1.59}{1.5 - j1.59} \\
 &= \frac{-1.67e^{j72.6^\circ}}{2.19e^{-j46.7^\circ}} \\
 &= -0.76e^{j119.3^\circ}.
 \end{aligned}$$

该结果可以转化为式 (1) 的形式。 (2.62) 用 e^{-j180° 替换减号。因此,

$$\begin{aligned}
 \Gamma &= 0.76e^{j119.3^\circ}e^{-j180^\circ} \\
 &= 0.76e^{-j60.7^\circ} \\
 &= 0.76/\underline{-60.7^\circ},
 \end{aligned}$$

或者

$$|\Gamma| = 0.76, \quad \theta_r = -60.7^\circ.$$

$$\begin{aligned}
 z_L &= \frac{Z_L}{Z_0} \\
 &= \frac{R_L - j/(\omega C_L)}{Z_0} \\
 &= \frac{1}{100} \left(50 - j \frac{1}{2\pi \times 10^8 \times 10^{-11}} \right) \\
 &= (0.5 - j1.59) \Omega.
 \end{aligned}$$

Example 2-4: $|\Gamma|$ for Purely Reactive Load

Show that $|\Gamma| = 1$ for a lossless line connected to a purely reactive load.

Solution: The load impedance of a purely reactive load is $Z_L = jX_L$. From Eq. (2.59), the reflection coefficient is

$$\begin{aligned}\Gamma &= \frac{Z_L - Z_0}{Z_L + Z_0} \\ &= \frac{jX_L - Z_0}{jX_L + Z_0} \\ &= \frac{-(Z_0 - jX_L)}{(Z_0 + jX_L)} = \frac{-\sqrt{Z_0^2 + X_L^2} e^{-j\theta}}{\sqrt{Z_0^2 + X_L^2} e^{j\theta}} = -e^{-j2\theta} = 1e^{j(\pi-2\theta)},\end{aligned}$$

where $\theta = \tan^{-1} X_L/Z_0$. Hence, for $\Gamma = |\Gamma|e^{j\theta_r}$,

$$|\Gamma| = 1$$

and

$$\theta_r = \pi - 2\theta = \pi - 2\tan^{-1}\left(\frac{X_L}{Z_0}\right).$$

Exercise 2-8: A $50\ \Omega$ lossless transmission line is terminated in a load with impedance $Z_L = (30 - j200)\ \Omega$. Calculate the voltage reflection coefficient at the load.

Answer: $\Gamma = 0.93/-27.5^\circ$. (See [EM](#).)

Exercise 2-9: A $150\ \Omega$ lossless line is terminated in a capacitor with impedance $Z_L = -j30\ \Omega$. Calculate Γ .

Answer: $\Gamma = 1/-157.4^\circ$. (See [EM](#).)

Exercise 2-10: Given that the reflection coefficient at the load is $\Gamma = 0.6 - j0.3$, determine the normalized load impedance z_L .

Answer: $z_L = 2.2 - j2.4$. (See [EM](#).)

2-6.2 Standing Waves

Using the relation $V_0^- = \Gamma V_0^+$ in Eqs. (2.54a) and (2.54b) yields

$$\tilde{V}(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z}), \quad (2.63a)$$

$$\tilde{I}(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z}). \quad (2.63b)$$

These expressions now contain only one (yet to be determined) unknown, V_0^+ . Before we proceed to solve for V_0^+ , however, let us examine the physical meaning underlying these expressions. We begin by deriving an expression for $|\tilde{V}(z)|$, which is the magnitude of $\tilde{V}(z)$. Upon using Eq. (2.62) in Eq. (2.63a) and applying the relation $|\tilde{V}(z)| = [\tilde{V}(z)\tilde{V}^*(z)]^{1/2}$, where $\tilde{V}^*(z)$ is the complex conjugate of $\tilde{V}(z)$, we have

$$\begin{aligned}|\tilde{V}(z)| &= \left\{ \left[V_0^+(e^{-j\beta z} + |\Gamma|e^{j\theta_r}e^{j\beta z}) \right] \right. \\ &\quad \cdot \left. \left[(V_0^+)^*(e^{j\beta z} + |\Gamma|e^{-j\theta_r}e^{-j\beta z}) \right] \right\}^{1/2} \\ &= |V_0^+| \left[1 + |\Gamma|^2 + |\Gamma|(e^{j(2\beta z + \theta_r)} + e^{-j(2\beta z + \theta_r)}) \right]^{1/2} \\ &= |V_0^+| \left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta z + \theta_r) \right]^{1/2},\end{aligned} \quad (2.64)$$

where we have used the identity

$$e^{jx} + e^{-jx} = 2\cos x \quad (2.65)$$

for any real quantity x . To express the magnitude of $\tilde{V}$ as a function of d instead of z , we replace z with $-d$ on the right-hand side of Eq. (2.64):

$$|\tilde{V}(d)| = |V_0^+| \left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta d - \theta_r) \right]^{1/2}. \quad (2.66)$$

By applying the same steps to Eq. (2.63b), a similar expression can be derived for $|\tilde{I}(d)|$, which is the magnitude of the current $\tilde{I}(d)$:

$$|\tilde{I}(d)| = \frac{|V_0^+|}{Z_0} \left[1 + |\Gamma|^2 - 2|\Gamma|\cos(2\beta d - \theta_r) \right]^{1/2}. \quad (2.67)$$

The variations of $|\tilde{V}(d)|$ and $|\tilde{I}(d)|$ as a function of d , which is the position on the line relative to the load (at $d = 0$), are illustrated in **Fig. 2-14** for a line with $|V_0^+| = 1\text{ V}$, $|\Gamma| = 0.3$, $\theta_r = 30^\circ$, and $Z_0 = 50\ \Omega$. The sinusoidal patterns are called *standing waves* and are caused by the *interference* of the two traveling waves. The *maximum value* of the *standing-wave pattern* of $|\tilde{V}(d)|$ corresponds to the position on the line at which the incident and reflected waves are *in-phase* [$2\beta d - \theta_r = 2n\pi$ in Eq. (2.66)]. Therefore, they add constructively to give a value equal to $(1 + |\Gamma|)|V_0^+| = 1.3\text{ V}$. The *minimum value* of $|\tilde{V}(d)|$ occurs when the two waves interfere destructively, which occurs when the incident and reflected waves are in *phase-opposition* [$2\beta d - \theta_r = (2n + 1)\pi$]. In this case, $|\tilde{V}(d)| = (1 - |\Gamma|)|V_0^+| = 0.7\text{ V}$.

► Whereas the repetition period is λ for the incident and reflected waves considered individually, the repetition period of the standing-wave pattern is $\lambda/2$. ◀

示例 2-4: 纯无功负载的 $|\Gamma|$

证明连接到纯无功负载的无损线路的 $|\Gamma| = 1$ 。

解: 纯无功负载的负载阻抗为 $Z_L = jX_L$ 。从方程。 (2.59) , 反射系数为

$$\begin{aligned}\Gamma &= \frac{Z_L - Z_0}{Z_L + Z_0} \\ &= \frac{jX_L - Z_0}{jX_L + Z_0} \\ &= \frac{-(Z_0 - jX_L)}{(Z_0 + jX_L)} = \frac{-\sqrt{Z_0^2 + X_L^2} e^{-j\theta}}{\sqrt{Z_0^2 + X_L^2} e^{j\theta}} = -e^{-j2\theta} = 1e^{j(\pi-2\theta)},\end{aligned}$$

其中 $\theta = \tan^{-1} X_L/Z_0$ 。因此, 对于 $\Gamma = |\Gamma|e^{j\theta_r}$,

$$|\Gamma| = 1$$

和

$$\theta_r = \pi - 2\theta = \pi - 2\tan^{-1}\left(\frac{X_L}{Z_0}\right).$$

练习 2-8: $50\ \Omega$ 无损传输线的端接阻抗为 $Z_L = (30 - j200)\ \Omega$ 的负载。计算负载处的电压反射系数。

Answer: $\Gamma = 0.93/-27.5^\circ$. (See [EM](#).)

练习 2-9: $150\ \Omega$ 无损线路的端接阻抗为 $Z_L = -j30\ \Omega$ 的电容器。计算 Γ 。

Answer: $\Gamma = 1/-157.4^\circ$. (See [EM](#).)

练习 2-10: 假设负载处的反射系数为 $\Gamma = 0.6 - j0.3$, 请确定归一化负载阻抗 z_L 。

Answer: $z_L = 2.2 - j2.4$. (See [EM](#).)

2-6.2 驻波

使用等式中的关系 $V_0^- = \Gamma V_0^+$ (2.54a) 和 (2.54b) 产量

$$\tilde{V}(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z}), \quad (2.63a)$$

$$\tilde{I}(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z}). \quad (2.63b)$$

这些表达式现在仅包含一个 (尚未确定) 未知数 V_0^+ 。然而, 在我们继续求解 V_0^+ 之前, 让我们检查一下这些表达式背后的物理意义。我们首先推导 $|\tilde{V}(z)|$ 的表达式 , 它是 $\tilde{V}(z)$ 的大小。使用方程式后。 (2.62) 式(2.63a) 并应用关系式 $|\tilde{V}(z)| = [\tilde{V}(z)\tilde{V}^*(z)]^{1/2}$, 其中 $\tilde{V}^*(z)$ 是 $\tilde{V}(z)$ 的复共轭, 我们有

$$\begin{aligned}|\tilde{V}(z)| &= \left\{ \left[V_0^+(e^{-j\beta z} + |\Gamma|e^{j\theta_r}e^{j\beta z}) \right] \right. \\ &\quad \cdot \left. \left[(V_0^+)^*(e^{j\beta z} + |\Gamma|e^{-j\theta_r}e^{-j\beta z}) \right] \right\}^{1/2} \\ &= |V_0^+| \left[1 + |\Gamma|^2 + |\Gamma|(e^{j(2\beta z + \theta_r)} + e^{-j(2\beta z + \theta_r)}) \right]^{1/2} \\ &= |V_0^+| \left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta z + \theta_r) \right]^{1/2},\end{aligned} \quad (2.64)$$

我们在哪里使用了身份

$$e^{jx} + e^{-jx} = 2\cos x \quad (2.65)$$

对于任何实际数量 x 。为了将 $\tilde{V}$ 的大小表示为 d 而不是 z 的函数, 我们将等式右侧的 z 替换为 $-d$ 。 (2.64) :

$$|\tilde{V}(d)| = |V_0^+| \left[1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta d - \theta_r) \right]^{1/2}. \quad (2.66)$$

通过对等式应用相同的步骤。 (2.63b), 对于 $|\tilde{I}(d)|$ 可以导出类似的表达式, 它是电流 $\tilde{I}(d)$ 的幅度:

$$|\tilde{I}(d)| = \frac{|V_0^+|}{Z_0} \left[1 + |\Gamma|^2 - 2|\Gamma|\cos(2\beta d - \theta_r) \right]^{1/2}. \quad (2.67)$$

$|\tilde{V}(d)|$ 和 $|\tilde{I}(d)|$ 的变化作为 d 的函数, d 是线上相对于负载的位置 (在 $d = 0$ 处) , 如图 2-14 所示, 其中一条线为 $|V_0^+| = 1\text{ V}$ 、 $|\Gamma| = 0.3$ 、 $\theta_r = 30^\circ$ 和 $Z_0 = 50\ \Omega$ 。正弦曲线称为驻波, 是由两个行波干涉引起的。 $|\tilde{V}(d)|$ 驻波图的最大值对应于等式中入射波和反射波同相的线上位置 [$2\beta d - \theta_r = 2n\pi$ 。 (2.66)]。因此, 它们相加后得到的值等于 $(1 + |\Gamma|)|V_0^+| = 1.3\text{ V}$ 。当两个波相消干涉时 , 即当入射波和反射波相位相反时, 会出现 $|\tilde{V}(d)|$ 的最小值 [$2\beta d - \theta_r = (2n + 1)\pi$]。在这种情况下, $|\tilde{V}(d)| = (1 - |\Gamma|)|V_0^+| = 0.7\text{ V}$ 。

► 单独考虑入射波和反射波的重复周期为 λ , 而驻波图案的重复周期为 $\lambda/2$ 。 ◀

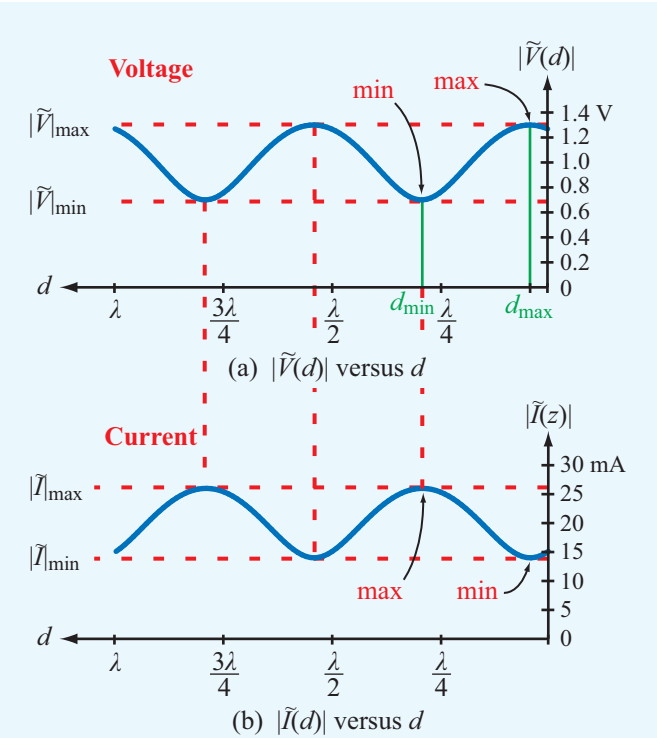


Figure 2-14 Standing-wave pattern for (a) $|\tilde{V}(d)|$ and (b) $|\tilde{I}(d)|$ for a lossless transmission line of characteristic impedance $Z_0 = 50 \, \Omega$, terminated in a load with a reflection coefficient $\Gamma = 0.3e^{j30^\circ}$. The magnitude of the incident wave $|V_0^+| = 1 \, \text{V}$. The standing-wave ratio is $S = |\tilde{V}|_{\max}/|\tilde{V}|_{\min} = 1.3/0.7 = 1.86$.

The standing-wave pattern describes the spatial variation of the magnitude of $\tilde{V}(d)$ as a function of d . If one were to observe the variation of the instantaneous voltage as a function of time at location $d = d_{\max}$ in **Fig. 2-14**, that variation would be as $\cos \omega t$ and would have an amplitude equal to 1.3 V [i.e., $v(t)$ would oscillate between $-1.3 \, \text{V}$ and $+1.3 \, \text{V}$]. Similarly, the instantaneous voltage $v(d, t)$ at any location d will be sinusoidal with amplitude equal to $|\tilde{V}(d)|$ at that d .

► Interactive **Module 2.4^a** provides a highly recommended simulation tool for gaining better understanding of the standing-wave patterns for $\tilde{V}(d)$ and $\tilde{I}(d)$ and the dynamic behavior of $v(d, t)$ and $i(d, t)$. ◀

^aAt em8e.eecs.umich.edu

Close inspection of the voltage and current standing-wave patterns shown in **Fig. 2-14** reveals that the two patterns are

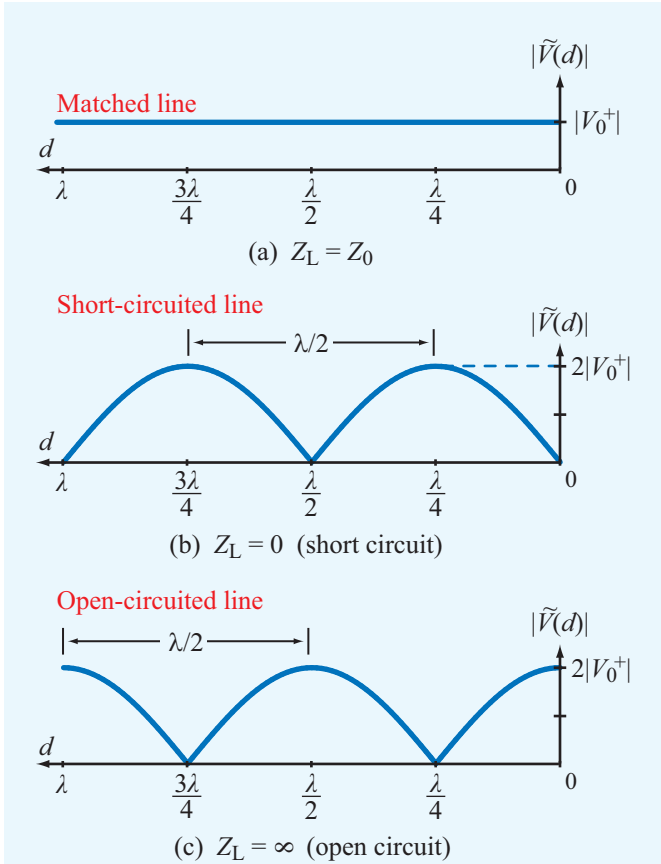


Figure 2-15 Voltage standing-wave patterns for (a) a matched load, (b) a short-circuited line, and (c) an open-circuited line.

in phase opposition (when one is at a maximum, the other is at a minimum, and vice versa). This is a consequence of the fact that the third term in Eq. (2.66) is preceded by a plus sign, whereas the third term in Eq. (2.67) is preceded by a minus sign.

The standing-wave patterns shown in **Fig. 2-14** are for a line with $\Gamma = 0.3e^{j30^\circ}$. The peak-to-peak variation of the pattern ($|\tilde{V}|_{\min} = (1 - |\Gamma|)|V_0^+|$ to $|\tilde{V}|_{\max} = (1 + |\Gamma|)|V_0^+|$) depends on $|\Gamma|$. For the special case of a matched line with $Z_L = Z_0$, we have $|\Gamma| = 0$ and $|\tilde{V}(d)| = |V_0^+|$ for all values of d , as shown in **Fig. 2-15(a)**.

► With no reflected wave present, there is no interference and no standing waves. ◀

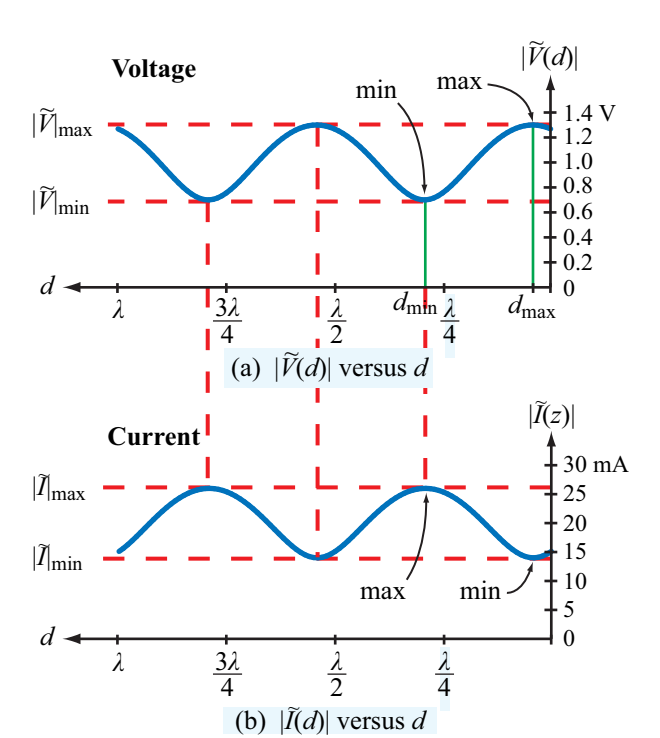


图 2-14 特性阻抗 $Z_0 = 50 \, \Omega$ 的无损传输线的 (a) $|\tilde{V}(d)|$ 和 (b) $|\tilde{I}(d)|$ 的驻波图，端接于反射系数 $\Gamma = 0.3e^{j30^\circ}$ 的负载。入射波的幅度为 $|V_0^+| = 1 \, \text{V}$ 。驻波比为 $S = |\tilde{V}|_{\max}/|\tilde{V}|_{\min} = 1.3/0.7 = 1.86$ 。

驻波图将 $|\tilde{V}(d)|$ 幅度的空间变化描述为 d 的函数。如果观察图 2-14 中位置 $d = d_{\max}$ 处的瞬时电压随时间的变化，则该变化将是 $\cos \omega t$ ，并且幅度等于 1.3 V [即， $v(t)$ 将在 $-1.3 \, \text{V}$ 和 $+1.3 \, \text{V}$ 之间振荡]。类似地，任何位置 d 处的瞬时电压 $v(d, t)$ 将是正弦波，其幅度等于该 d 处的 $|\tilde{V}(d)|$ 。

► 交互模块 2.4^a 提供了强烈推荐的模拟工具，可以更好地理解 $\tilde{V}(d)$ 和 $\tilde{I}(d)$ 的驻波模式以及 $v(d, t)$ 和 $i(d, t)$ 的动态行为。 ◀

aat em8e.eecs.umich.edu

仔细检查图 2-14 所示的电压和电流驻波模式可以发现，这两种模式是

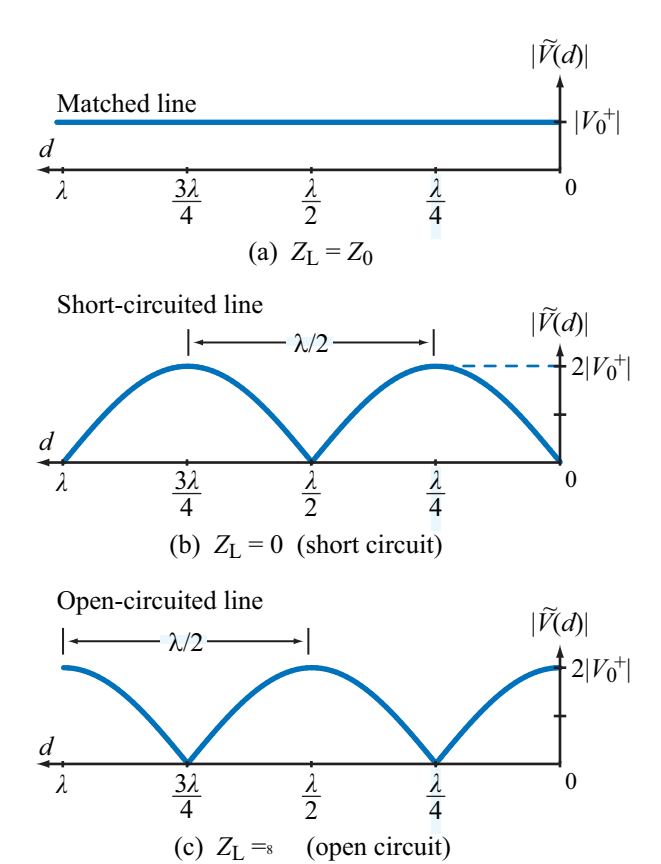


图 2-15 (a) 匹配负载、(b) 短路线路和 (c) 开路线路的电压驻波模式。

反相（当一个处于最大值时，另一个处于最小值，反之亦然）。这是方程中第三项的结果。(2.66) 前面有一个加号，而等式中的第三项。(2.67) 前面有一个减号。

图 2-14 所示的驻波图是针对 $\Gamma = 0.3e^{j30^\circ}$ 的线。模式的峰到峰变化 ($|\tilde{V}|_{\min} = (1 - |\Gamma|)|V_0^+|$ 到 $|\tilde{V}|_{\max} = (1 + |\Gamma|)|V_0^+|$) 取决于 $|\Gamma|$ 。对于具有 $Z_L = Z_0$ 的匹配线的特殊情况，对于所有 d 值，我们有 $|\Gamma| = 0$ 和 $|\tilde{V}(d)| = |V_0^+|$ ，如图 2-15(a) 所示。

► 由于不存在反射波，因此不存在干扰，也不存在驻波。 ◀

The other end of the $|\Gamma|$ scale, at $|\Gamma| = 1$, corresponds to when the load is a short circuit ($\Gamma = -1$) or an open circuit ($\Gamma = 1$). The standing-wave patterns for those two cases are shown in **Figs. 2-15(b)** and **(c)**; both exhibit maxima of $2|V_0^+|$ and minima equal to zero, but the two patterns are spatially shifted relative to each other by a distance of $\lambda/4$. A purely reactive load (capacitor or inductor) also satisfies the condition $|\Gamma| = 1$, but θ_r is generally neither zero nor 180° (**Table 2-3**). Exercise 2.9 examines the standing-wave pattern for a lossless line terminated in an inductor.

Now let us examine the maximum and minimum values of the voltage magnitude. From Eq. (2.66), $|\tilde{V}(d)|$ is a maximum when the argument of the cosine function is equal to zero or a multiple of 2π . Let us *denote $d_{\max}$ as the distance from the load at which $|\tilde{V}(d)|$ is a maximum*. It then follows that

$$|\tilde{V}(d)| = |\tilde{V}|_{\max} = |V_0^+|[1 + |\Gamma|], \quad (2.68)$$

when

$$2\beta d_{\max} - \theta_r = 2n\pi, \quad (2.69)$$

with $n = 0$ or a positive integer. Solving Eq. (2.69) for $d_{\max}$, we have

$$d_{\max} = \frac{\theta_r + 2n\pi}{2\beta} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}, \quad (2.70)$$

$$\begin{cases} n = 1, 2, \dots & \text{if } \theta_r < 0, \\ n = 0, 1, 2, \dots & \text{if } \theta_r \geq 0, \end{cases}$$

where we have used $\beta = 2\pi/\lambda$. The phase angle of the voltage reflection coefficient, θ_r , is bounded between $-\pi$ and π radians. If $\theta_r \geq 0$, the *first voltage maximum* occurs at $d_{\max} = \theta_r \lambda / 4\pi$, corresponding to $n = 0$, but if $\theta_r < 0$, the first physically meaningful maximum occurs at $d_{\max} = (\theta_r \lambda / 4\pi) + \lambda / 2$, corresponding to $n = 1$. Negative values of $d_{\max}$ correspond to locations past the end of the line; therefore, they have no physical significance.

Similarly, the minima of $|\tilde{V}(d)|$ occur at distances $d_{\min}$ where the argument of the cosine function in Eq. (2.66) is equal to $(2n + 1)\pi$, which gives the result

$$|\tilde{V}|_{\min} = |V_0^+|[1 - |\Gamma|], \quad (2.71)$$

$$\text{when } (2\beta d_{\min} - \theta_r) = (2n + 1)\pi,$$

with $-\pi \leq \theta_r \leq \pi$. The first minimum corresponds to $n = 0$. The spacing between a maximum $d_{\max}$ and the adjacent minimum $d_{\min}$ is $\lambda/4$. Hence, the *first minimum* occurs at

$$d_{\min} = \begin{cases} d_{\max} + \lambda/4, & \text{if } d_{\max} < \lambda/4, \\ d_{\max} - \lambda/4, & \text{if } d_{\max} \geq \lambda/4. \end{cases} \quad (2.72)$$

► The locations on the line corresponding to voltage maxima correspond to current minima, and vice versa. ◀

The ratio of $|\tilde{V}|_{\max}$ to $|\tilde{V}|_{\min}$ is called the *voltage standing-wave ratio* S , which from Eqs. (2.68) and (2.71) is given by

$$S = \frac{|\tilde{V}|_{\max}}{|\tilde{V}|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad (\text{dimensionless}). \quad (2.73)$$

This quantity is often referred to by its acronym, VSWR or the shorter acronym *SWR*, and it provides a measure of the mismatch between the load and the transmission line:

► For a matched load with $\Gamma = 0$, we get $S = 1$, and for a line with $|\Gamma| = 1$, $S = \infty$. ◀

Concept Question 2-5: The attenuation constant α represents ohmic losses. In view of the model given in **Fig. 2-6(c)**, what should R' and G' be in order to have no losses? Verify your expectation through the expression for α given by Eq. (2.25a).

Concept Question 2-6: How is the wavelength λ of the wave traveling on the transmission line related to the free-space wavelength λ_0 ?

Concept Question 2-7: When is a load matched to a transmission line? Why is it important?

Concept Question 2-8: What is a standing-wave pattern? Why is its period $\lambda/2$ and not λ ?

Concept Question 2-9: What is the separation between the location of a voltage maximum and the adjacent current maximum on the line?

$|\Gamma|$ 标度的另一端 $|\Gamma| = 1$ 处对应于负载短路 ($\Gamma = -1$) 或开路 ($\Gamma = 1$) 的情况。这两种情况的驻波模式如图 2 和 3 所示。2-15(b)和(c); 两者都表现出最大值 $2|V_0^+|$ 和最小值等于 0, 但是这两种模式在空间上相对于彼此偏移了距离 $\lambda/4$ 。纯无功负载 (电容器或电感器) 也满足条件 $|\Gamma| = 1$, 但 θ_r 通常既不为零也不 180° (表 2-3)。练习 2.9 检查终止于电感器的无损线路的驻波模式。

现在让我们检查电压幅度的最大值和最小值。从方程 (2.66), 当余弦函数的自变量等于0或 2π 的倍数时, $|\tilde{V}(d)|$ 是最大值。让我们将 $d_{\max}$ 表示为与 $|\tilde{V}(d)|$ 达到最大值时的负载的距离。那么接下来

$$|\tilde{V}(d)| = |\tilde{V}|_{\max} = |V_0^+|[1 + |\Gamma|], \quad (2.68)$$

什么时候

$$2\beta d_{\max} - \theta_r = 2n\pi, \quad (2.69)$$

其中 n 为 0 或正整数。求解方程。 (2.69) 对于 $d_{\max}$, 我们有

$$d_{\max} = \frac{\theta_r + 2n\pi}{2\beta} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}, \quad (2.70)$$

$$\begin{cases} n = 1, 2, \dots & \text{if } \theta_r < 0, \\ n = 0, 1, 2, \dots & \text{if } \theta_r \geq 0, \end{cases}$$

我们使用了 $\beta = 2\pi/\lambda$ 。电压反射系数的相角 θ_r 介于 $-\pi$ 和 π 弧度之间。如果 $\theta_r \geq 0$, 则第一个电压最大值出现在 $d_{\max} = \theta_r \lambda / 4\pi$ 处, 对应于 $n = 0$, 但如果 $\theta_r < 0$, 则第一个有物理意义的最大值出现在 $d_{\max} = (\theta_r \lambda / 4\pi) + \lambda / 2$ 处, 对应于 $n = 1$ 。 $d_{\max}$ 的负值对应于超过行尾的位置; 因此, 它们没有物理意义。

类似地, $|\tilde{V}(d)|$ 的最小值出现在距离 $d_{\min}$ 处, 其中等式中余弦函数的参数如下: (2.66) 等于 $(2n + 1)\pi$, 得出结果

$$|\tilde{V}|_{\min} = |V_0^+|[1 - |\Gamma|], \quad (2.71)$$

$$\text{when } (2\beta d_{\min} - \theta_r) = (2n + 1)\pi,$$

与 $-\pi \leq \theta_r \leq \pi$ 。第一个最小值对应于 $n = 0$ 。最大值 $d_{\max}$ 与相邻最小值 $d_{\min}$ 之间的间距为 $\lambda/4$ 。因此, 第一个最小值出现在

$$d_{\min} = \begin{cases} d_{\max} + \lambda/4, & \text{if } d_{\max} < \lambda/4, \\ d_{\max} - \lambda/4, & \text{if } d_{\max} \geq \lambda/4. \end{cases} \quad (2.72)$$

► 线上对应于电压最大值的位置对应于电流最小值, 反之亦然。 ◀

$|\tilde{V}|_{\max}$ 与 $|\tilde{V}|_{\min}$ 之比称为电压驻波比 S , 由等式 1 得出。 (2.68) 和 (2.71) 由下式给出

$$S = \frac{|\tilde{V}|_{\max}}{|\tilde{V}|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad (\text{dimensionless}). \quad (2.73)$$

该量通常用其缩写词 VSWR 或更短的缩写词 SWR 来表示, 它提供了负载与传输线之间不匹配的度量:

► 对于 $\Gamma = 0$ 的匹配负载, 我们得到 $S = 1$, 对于 $|\Gamma| = 1$ 的线, 我们得到 $S = \infty$ 。 ◀

概念问题 2-5: 衰减常数 α 代表欧姆损耗。鉴于图2-6(c)给出的模型, 为了没有损失, R' 和 G' 应该是多少? 通过式 (1) 给出的 α 表达式验证您的期望。 (2.25a) 。

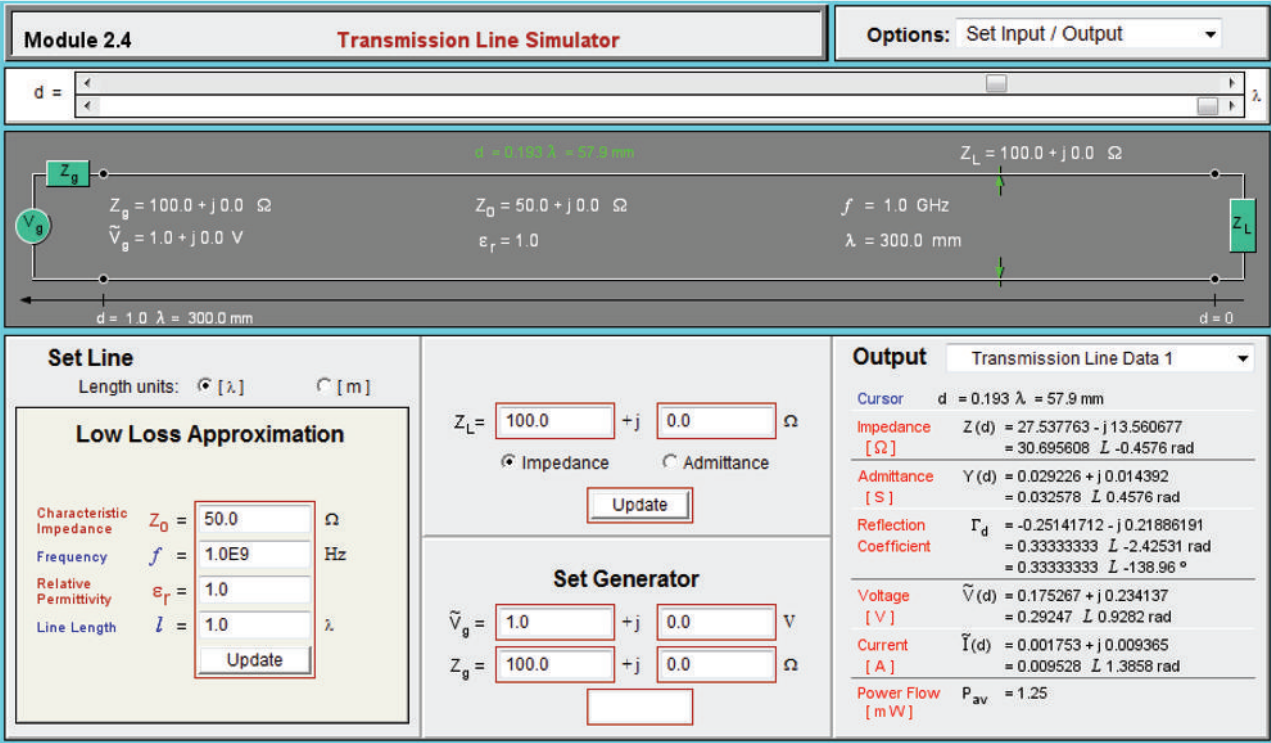
概念问题 2-6: 在传输线上传播的波的波长 λ 与自由空间波长 λ_0 有何关系?

概念问题 2-7: 负载何时与传输线匹配? 为什么它很重要?

概念问题 2-8: 什么是驻波模式? 为什么它的周期是 $\lambda/2$ 而不是 λ ?

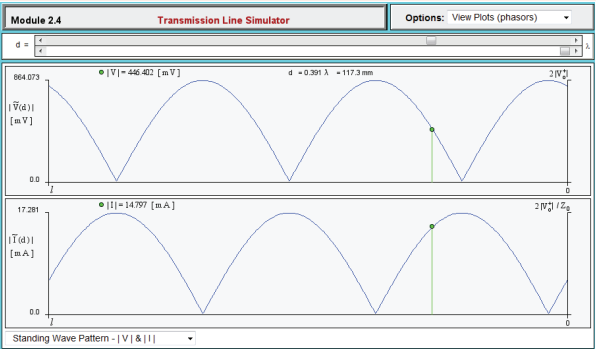
概念问题 2-9: 线路上电压最大值的位置与相邻电流最大值的位置之间的间隔是多少?

Module 2.4 Transmission Line Simulator Upon specifying the requisite input data—including the load impedance at $d = 0$ and the generator voltage and impedance at $d = l$, this module provides a wealth of output information about the voltage and current waveforms along the transmission line. You can view plots of the standing-wave patterns for voltage and current, the time and spatial variations of the instantaneous voltage $v(d, t)$ and current $i(d, t)$, and other related quantities.



Exercise 2-11: Use Module 2.4 to generate the voltage and current standing-wave patterns for a $50\ \Omega$ line of length 1.5λ terminated in an inductance with $Z_L = j140\ \Omega$.

Answer: See Module 2.4 display.



Example 2-5: Standing-Wave Ratio

A $50\ \Omega$ transmission line is terminated in a load with $Z_L = (100 + j50)\ \Omega$. Find the voltage reflection coefficient and the voltage standing-wave ratio.

Solution: From Eq. (2.59), Γ is given by

$$\Gamma = \frac{z_L - 1}{z_L + 1} = \frac{(2 + j1) - 1}{(2 + j1) + 1} = \frac{1 + j1}{3 + j1}.$$

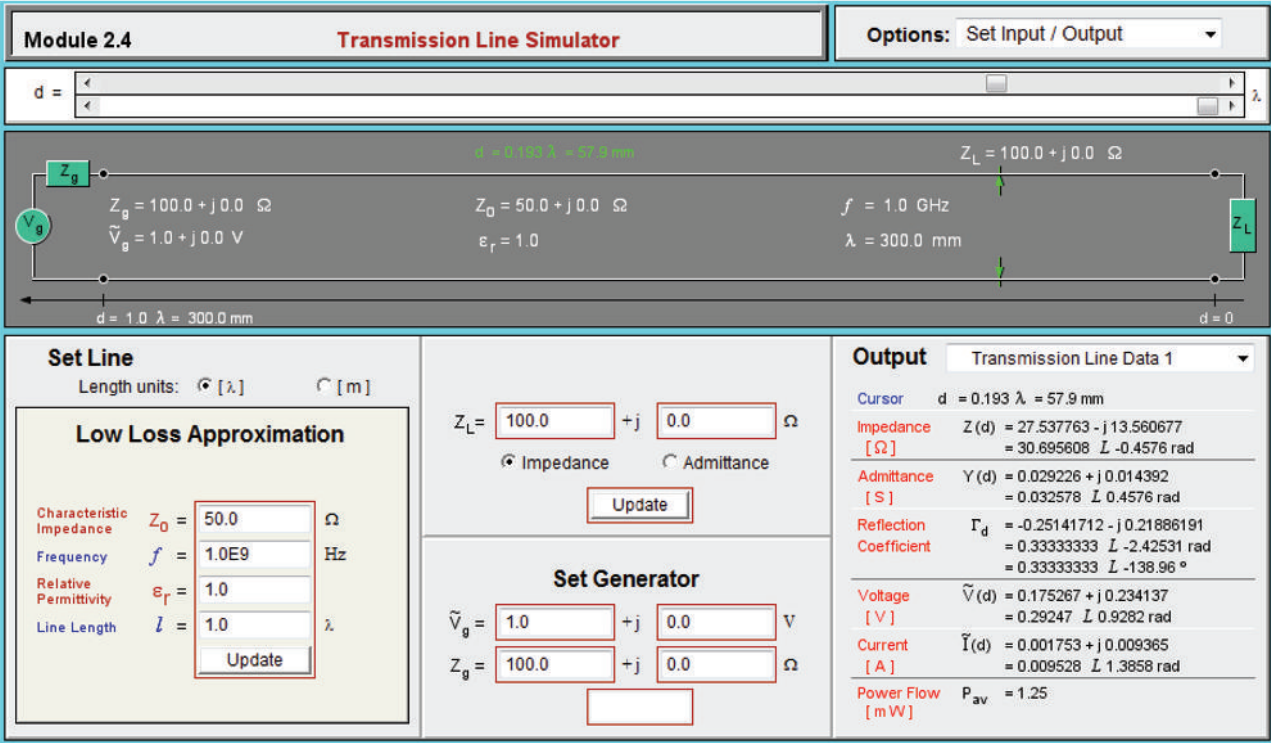
Converting the numerator and denominator to polar form yields

$$\Gamma = \frac{1.414e^{j45^\circ}}{3.162e^{j18.4^\circ}} = 0.45e^{j26.6^\circ}.$$

Using the definition for S given by Eq. (2.73), we have

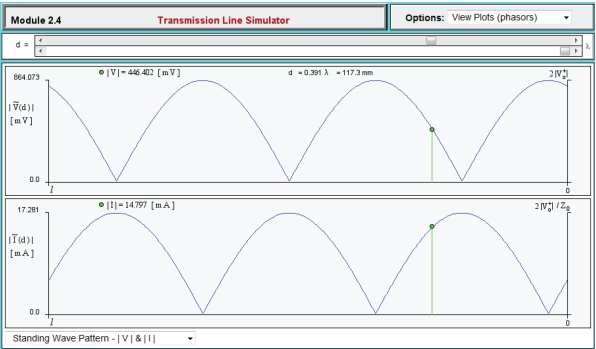
$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.45}{1 - 0.45} = 2.6.$$

模块 2.4 传输线模拟器 在指定必要的输入数据（包括 $d = 0$ 处的负载阻抗以及 $d = l$ 处的发电机电压和阻抗）后，该模块提供了有关沿传输线的电压和电流波形的丰富输出信息。您可以查看电压和电流的驻波模式图、瞬时电压 $v(d, t)$ 和电流 $i(d, t)$ 的时间和空间变化以及其他相关量。



练习 2-11：使用模块 2.4 生成长度为 1.5λ 的 $50\ \Omega$ 线的电压和电流驻波模式，该线终止于 $Z_L = j140\ \Omega$ 的电感。

答：参见模块2.4显示。



例 2-5：驻波比

$50\ \Omega$ 传输线的端接负载为 $Z_L = (100 + j50)\ \Omega$ 。求电压反射系数和电压驻波比。

解决方案：从方程。 (2.59), Γ 由下式给出

$$\Gamma = \frac{z_L - 1}{z_L + 1} = \frac{(2 + j1) - 1}{(2 + j1) + 1} = \frac{1 + j1}{3 + j1}.$$

将分子和分母转换为极坐标形式得到

$$\Gamma = \frac{1.414e^{j45^\circ}}{3.162e^{j18.4^\circ}} = 0.45e^{j26.6^\circ}.$$

使用方程式给出的 S 定义： (2.73) ， 我们有

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.45}{1 - 0.45} = 2.6.$$

Example 2-6: Measuring Z_L

A **slotted-line** probe is an instrument used to measure the unknown impedance of a load, Z_L . A coaxial slotted line contains a narrow longitudinal slit in the outer conductor. A small probe inserted in the slit can be used to sample the magnitude of the electric field and, hence, the magnitude $|\tilde{V}(d)|$ of the voltage on the line (**Fig. 2-16**). By moving the probe along the length of the slotted line, it is possible to measure $|\tilde{V}|_{\max}$ and $|\tilde{V}|_{\min}$ and the distances from the load at which they occur. Use of Eq. (2.73), namely $S = |\tilde{V}|_{\max}/|\tilde{V}|_{\min}$, provides the voltage standing-wave ratio S . Measurements with a $Z = 50\ \Omega$ slotted line terminated in an unknown load impedance determined that $S = 3$. The distance between successive voltage minima was found to be 30 cm, and the first voltage minimum was located at 12 cm from the load. Determine the load impedance Z_L .

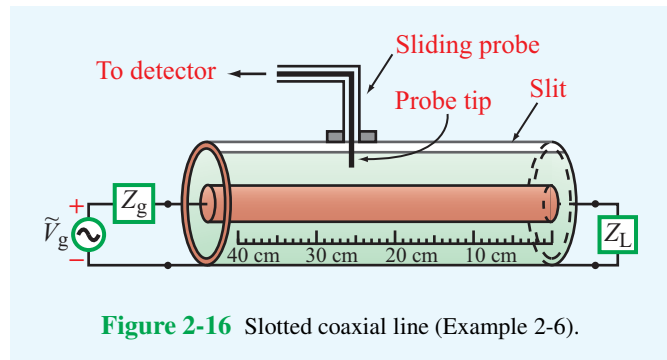


Figure 2-16 Slotted coaxial line (Example 2-6).

Solution: The following quantities are given:

$$Z_0 = 50\ \Omega, \quad S = 3, \quad d_{\min} = 12\ \text{cm}.$$

Since the distance between successive voltage minima is $\lambda/2$,

$$\lambda = 2 \times 0.3 = 0.6\ \text{m},$$

and

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{0.6} = \frac{10\pi}{3} \quad (\text{rad/m}).$$

From Eq. (2.73), solving for $|\Gamma|$ in terms of S gives

$$|\Gamma| = \frac{S-1}{S+1} = \frac{3-1}{3+1} = 0.5.$$

Next, we use the condition given by Eq. (2.71) to find θ_r :

$$2\beta d_{\min} - \theta_r = \pi, \quad \text{for } n = 0 \text{ (first minimum),}$$

which gives

$$\theta_r = 2\beta d_{\min} - \pi = 2 \times \frac{10\pi}{3} \times 0.12 - \pi = -0.2\pi \text{ (rad)} = -36^\circ.$$

Hence,

$$\Gamma = |\Gamma|e^{j\theta_r} = 0.5e^{-j36^\circ} = 0.405 - j0.294.$$

Solving Eq. (2.59) for Z_L , we have

$$Z_L = Z_0 \left[\frac{1+\Gamma}{1-\Gamma} \right] = 50 \left[\frac{1+0.405-j0.294}{1-0.405+j0.294} \right] = (85-j67)\ \Omega.$$

Exercise 2-12: If $\Gamma = 0.5\angle-60^\circ$ and $\lambda = 24\ \text{cm}$, find the locations of the voltage maximum and minimum nearest to the load.

Answer: $d_{\max} = 10\ \text{cm}$, $d_{\min} = 4\ \text{cm}$. (See [EM](#).)

Exercise 2-13: A $140\ \Omega$ lossless line is terminated in a load impedance $Z_L = (280 + j182)\ \Omega$. If $\lambda = 72\ \text{cm}$, find (a) the reflection coefficient Γ , (b) the voltage standing-wave ratio S , (c) the locations of voltage maxima, and (d) the locations of voltage minima.

Answer: (a) $\Gamma = 0.5\angle 29^\circ$, (b) $S = 3.0$, (c) $d_{\max} = 2.9\ \text{cm} + n\lambda/2$, (d) $d_{\min} = 20.9\ \text{cm} + n\lambda/2$, where $n = 0, 1, 2, \dots$ (See [EM](#).)

2-7 Wave Impedance of the Lossless Line

The standing-wave patterns indicate that on a mismatched line the voltage and current magnitudes are oscillatory with position along the line and in phase opposition with each other. Hence, the voltage to current ratio, called the **wave impedance** $Z(d)$, must vary with position also. Using Eqs. (2.63a) and (2.63b) with $z = -d$,

$$\begin{aligned} Z(d) &= \frac{\tilde{V}(d)}{\tilde{I}(d)} = \frac{V_0^+[e^{j\beta d} + \Gamma e^{-j\beta d}]}{V_0^+[e^{j\beta d} - \Gamma e^{-j\beta d}]} Z_0 \\ &= Z_0 \left[\frac{1 + \Gamma e^{-j2\beta d}}{1 - \Gamma e^{-j2\beta d}} \right] = Z_0 \left[\frac{1 + \Gamma_d}{1 - \Gamma_d} \right] \quad (\Omega), \end{aligned} \quad (2.74)$$

where we define

$$\Gamma_d = \Gamma e^{-j2\beta d} = |\Gamma|e^{j\theta_r}e^{-j2\beta d} = |\Gamma|e^{j(\theta_r-2\beta d)} \quad (2.75)$$

as the **phase-shifted voltage reflection coefficient**, meaning that Γ_d has the same magnitude as Γ , but its phase is shifted by $2\beta d$ relative to that of Γ .

示例 2-6: 测量 Z_L

槽线探头是一种用于测量未知负载阻抗 Z_L 的仪器。同轴开槽线在外导体中包含一个狭窄的纵向狭缝。插入狭缝中的小探针可用于对电场强度进行采样，从而对线路上电压的强度 $|\tilde{V}(d)|$ 进行采样 (图 2-16)。通过沿着槽线的长度移动探头，可以测量 $|\tilde{V}|_{\max}$ 和 $|\tilde{V}|_{\min}$ 以及距它们出现的负载的距离。使用等式 (2.73)，即 $S = |\tilde{V}|_{\max}/|\tilde{V}|_{\min}$ ，提供电压驻波比 S 。使用终止于未知负载阻抗的 $Z = 50\ \Omega$ 槽线进行测量，确定 $S = 3$ 。发现连续电压最小值之间的距离为 30 厘米，第一个电压最小值位于距负载 12 厘米处。确定负载阻抗 Z_L 。

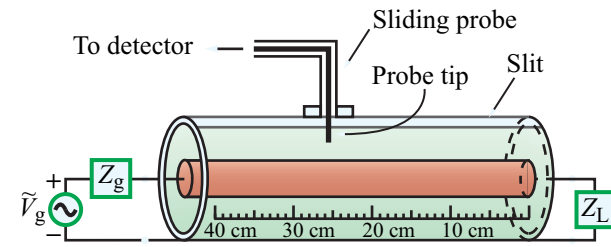


图 2-16 开槽同轴线 (示例 2-6)。

解: 给出下列数量:

$$Z_0 = 50\ \Omega, \quad S = 3, \quad d_{\min} = 12\ \text{cm}.$$

由于连续电压最小值之间的距离为 $\lambda/2$,

$$\lambda = 2 \times 0.3 = 0.6\ \text{m},$$

和

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{0.6} = \frac{10\pi}{3} \quad (\text{rad/m}).$$

从方程 (2.73)，用 S 求解 $|\Gamma|$ 给出

$$|\Gamma| = \frac{S-1}{S+1} = \frac{3-1}{3+1} = 0.5.$$

接下来，我们使用方程给出的条件 (2.71) 求 θ_r :

$$2\beta d_{\min} - \theta_r = \pi, \quad \text{for } n = 0 \text{ (first minimum),}$$

这给出了

$$\theta_r = 2\beta d_{\min} - \pi = 2 \times \frac{10\pi}{3} \times 0.12 - \pi = -0.2\pi \text{ (rad)} = -36^\circ.$$

因此,

$$\Gamma = |\Gamma|e^{j\theta_r} = 0.5e^{-j36^\circ} = 0.405 - j0.294.$$

求解方程 (2.59) 对于 Z_L ，我们有

$$Z_L = Z_0 \left[\frac{1+\Gamma}{1-\Gamma} \right] = 50 \left[\frac{1+0.405-j0.294}{1-0.405+j0.294} \right] = (85-j67)\ \Omega.$$

练习 2-12: 如果 $\Gamma = 0.5\angle-60^\circ$ 和 $\lambda = 24\ \text{cm}$ ，找到最接近负载的电压最大值和最小值的位置。

Answer: $d_{\max} = 10\ \text{cm}$, $d_{\min} = 4\ \text{cm}$. (See [EM](#).)

练习 2-13: 一条 $140\ \Omega$ 无损线路的端接负载阻抗为 $Z_L = (280 + j182)\ \Omega$ 。如果 $\lambda = 72\ \text{cm}$ ，求 (a) 反射系数 Γ ，(b) 电压驻波比 S ，(c) 电压最大值的位置，以及 (d) 电压最小值的位置。

答案: (a) $\Gamma = 0.5\angle 29^\circ$ (b) $S = 3.0$, (c) $d_{\max} = 2.9\ \text{cm} + n\lambda/2$, (d) $d_{\min} = 20.9\ \text{cm} + n\lambda/2$, 其中 $n = 0, 1, 2, \dots$ 。(参见 [EM](#).)

2-7 无损线路的波阻抗

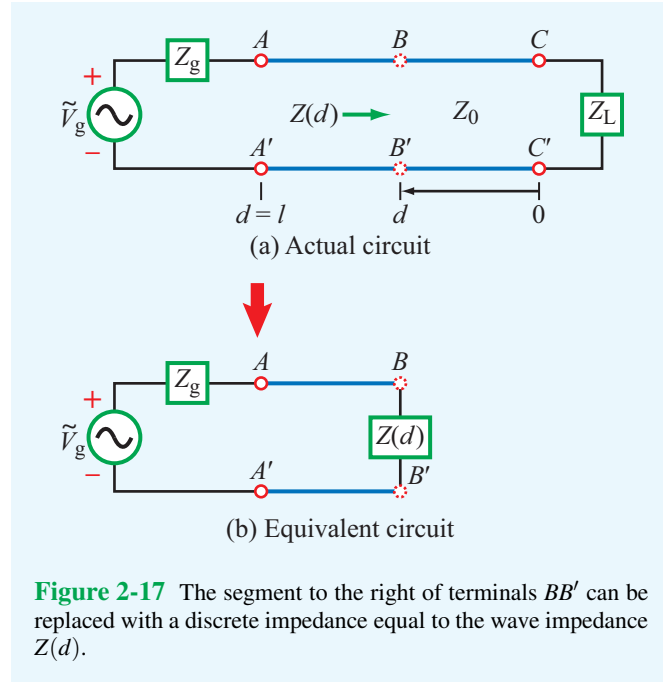
驻波模式表明，在不匹配的线路上，电压和电流幅度随着沿线路的位置而振荡，并且彼此反相。因此，电压与电流之比 (称为波阻抗 $Z(d)$) 也必须随位置而变化。使用方程式 (2.63a) 和 (2.63b) 与 $z = -d$,

$$\begin{aligned} Z(d) &= \frac{\tilde{V}(d)}{\tilde{I}(d)} = \frac{V_0^+[e^{j\beta d} + \Gamma e^{-j\beta d}]}{V_0^+[e^{j\beta d} - \Gamma e^{-j\beta d}]} Z_0 \\ &= Z_0 \left[\frac{1 + \Gamma e^{-j2\beta d}}{1 - \Gamma e^{-j2\beta d}} \right] = Z_0 \left[\frac{1 + \Gamma_d}{1 - \Gamma_d} \right] \quad (\Omega), \end{aligned} \quad (2.74)$$

我们在哪里定义

$$\Gamma_d = \Gamma e^{-j2\beta d} = |\Gamma|e^{j\theta_r}e^{-j2\beta d} = |\Gamma|e^{j(\theta_r-2\beta d)} \quad (2.75)$$

作为相移电压反射系数，这意味着 Γ_d 与 Γ 具有相同的幅度，但其相位相对于 Γ 移动了 $2\beta d$ 。



► $Z(d)$ is the ratio of the total voltage (incident- and reflected-wave voltages) to the total current at any point d on the line, in contrast with the characteristic impedance of the line Z_0 , which relates the voltage and current of each of the two waves individually ($Z_0 = V_0^+ / I_0^+ = -V_0^- / I_0^-$). ◀

In the circuit of **Fig. 2-17(a)** at terminals BB' at an arbitrary location d on the line, $Z(d)$ is the wave impedance of the line when “looking” to the right, i.e., towards the load. Application of the equivalence principle allows us to replace the segment to the right of terminals BB' with a lumped impedance of value $Z(d)$, as depicted in **Fig. 2-17(b)**. From the standpoint of the input circuit to the left of terminals BB' , the two circuit configurations are electrically identical.

Of particular interest in many transmission-line problems is the **input impedance** at the source end of the line where $d = l$, which is given by

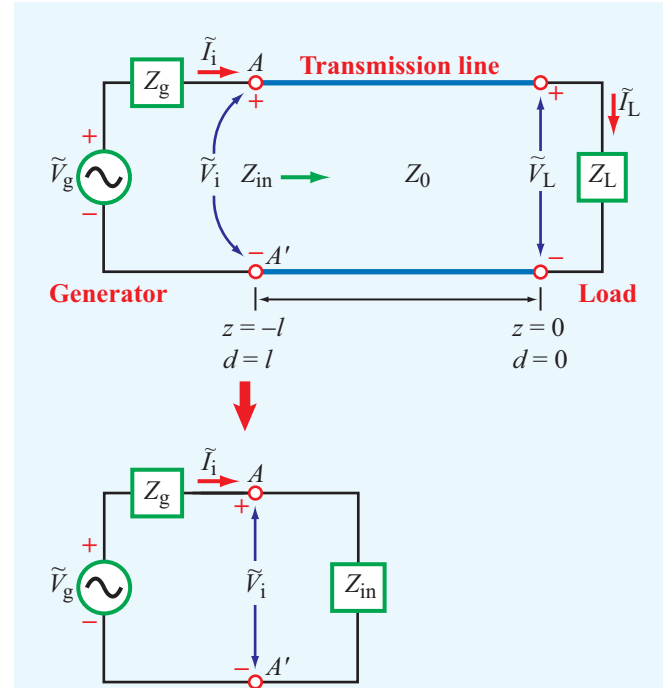
$$Z_{in} = Z(d = l) = Z_0 \left[\frac{1 + \Gamma_l}{1 - \Gamma_l} \right] \quad (2.76)$$

with

$$\Gamma_l = \Gamma e^{-j2\beta l} = |\Gamma| e^{j(\theta_r - 2\beta l)}. \quad (2.77)$$

By replacing Γ with Eq. (2.59) and using the relations

$$e^{j\beta l} = \cos \beta l + j \sin \beta l, \quad (2.78a)$$



Eq. (2.76) can be written in terms of the normalized load impedance z_L as

$$e^{-j\beta l} = \cos \beta l - j \sin \beta l, \quad (2.78b)$$

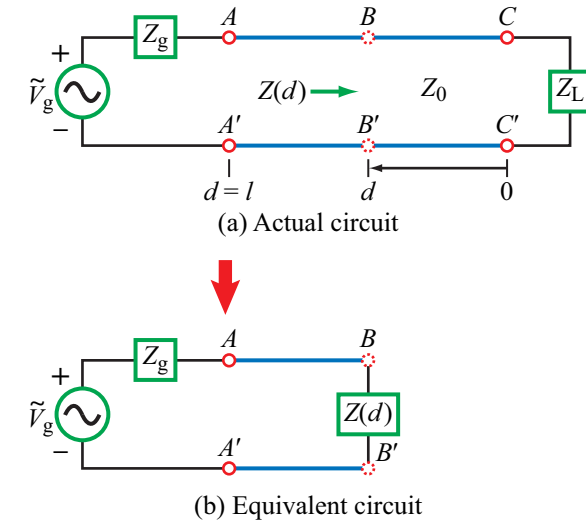
$$\begin{aligned} Z_{in} &= Z_0 \left(\frac{z_L \cos \beta l + j \sin \beta l}{\cos \beta l + j z_L \sin \beta l} \right) \\ &= Z_0 \left(\frac{z_L + j \tan \beta l}{1 + j z_L \tan \beta l} \right). \end{aligned} \quad (2.79)$$

From the standpoint of the generator circuit, the transmission line can be replaced with an impedance Z_{in} , as shown in **Fig. 2-18**. The phasor voltage across Z_{in} is given by

$$\tilde{V}_i = \tilde{I}_i Z_{in} = \frac{\tilde{V}_g Z_{in}}{Z_g + Z_{in}}. \quad (2.80)$$

Simultaneously, from the standpoint of the transmission line, the voltage across it at the input of the line is given by Eq. (2.63a) with $z = -l$:

$$\tilde{V}_i = \tilde{V}(-l) = V_0^+ [e^{j\beta l} + \Gamma e^{-j\beta l}]. \quad (2.81)$$



► $Z(d)$ 是线路上任意点 d 处的总电压（入射波电压和反射波电压）与总电流之比，与线路 Z 的特性阻抗相反，它将两个波的电压和电流分别关联起来（ $Z_0 = V_0^+ / I_0^+ = -V_0^- / I_0^-$ ）。◀

在图 2-17(a) 的电路中，线路上任意位置 d 处的端子 $B B'$ 处， $Z(d)$ 是向右“看”（即朝负载）时线路的波阻抗。应用等效原理，我们可以将端子 BB' 右侧的段替换为 $Z(d)$ 值的集总阻抗，如图 2-17(b) 所示。从端子 BB' 左侧的输入电路的角度来看，两个电路配置在电气上是相同的。

在许多传输线问题中特别令人感兴趣的是线路源端的输入阻抗，其中 $d = l$ ，由下式给出

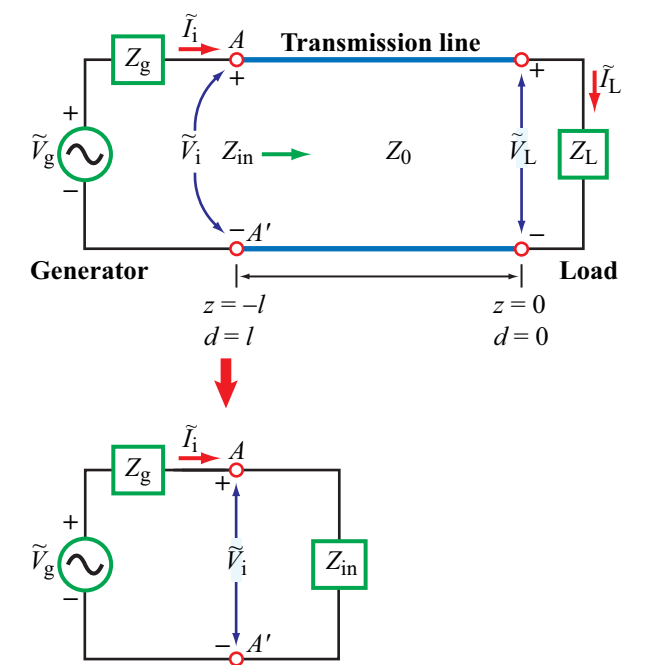
$$Z_{in} = Z(d = l) = Z_0 \left[\frac{1 + \Gamma_l}{1 - \Gamma_l} \right] \quad (2.76)$$

和

$$\Gamma_l = \Gamma e^{-j2\beta l} = |\Gamma| e^{j(\theta_r - 2\beta l)}. \quad (2.77)$$

将 Γ 替换为等式 (2.59) 并使用关系式

$$e^{j\beta l} = \cos \beta l + j \sin \beta l, \quad (2.78a)$$



等式 (2.76) 可以用归一化负载阻抗 z_L 写成

$$e^{-j\beta l} = \cos \beta l - j \sin \beta l, \quad (2.78b)$$

$$\begin{aligned} Z_{in} &= Z_0 \left(\frac{z_L \cos \beta l + j \sin \beta l}{\cos \beta l + j z_L \sin \beta l} \right) \\ &= Z_0 \left(\frac{z_L + j \tan \beta l}{1 + j z_L \tan \beta l} \right). \end{aligned} \quad (2.79)$$

从发生器电路的角度来看，传输线可以用阻抗 Z_{in} 代替，如图 2-18 所示。 Z_{in} 上的相量电压由下式给出

$$\tilde{V}_i = \tilde{I}_i Z_{in} = \frac{\tilde{V}_g Z_{in}}{Z_g + Z_{in}}. \quad (2.80)$$

同时，从传输线的角度来看，传输线输入端的电压由式 (1) 给出。(2.63a) 与 $z = -l$:

$$\tilde{V}_i = \tilde{V}(-l) = V_0^+ [e^{j\beta l} + \Gamma e^{-j\beta l}]. \quad (2.81)$$

Equating Eq. (2.80) to Eq. (2.81) and then solving for V_0^+ leads to

$$V_0^+ = \left(\frac{\tilde{V}_g Z_{in}}{Z_g + Z_{in}} \right) \left(\frac{1}{e^{j\beta l} + \Gamma e^{-j\beta l}} \right). \quad (2.82)$$

This completes the solution of the transmission-line wave equations given by Eqs. (2.21) and (2.23) for the special case of a lossless transmission line. We started out with the general solutions given by Eq. (2.26), which included four unknown amplitudes, V_0^+, V_0^-, I_0^+ , and I_0^- . We then determined that $Z_0 = V_0^+ / I_0^+ = -V_0^- / I_0^-$, thereby reducing the unknowns to the two voltage amplitudes only. Upon applying the boundary condition at the load, we were able to relate V_0^- to V_0^+ through Γ , and finally, by applying the boundary condition at the source, we obtained an expression for V_0^+ .

Example 2-7: Complete Solution for $v(d, t)$ and $i(d, t)$

A 1.05 GHz generator circuit with series impedance $Z_g = 10 \, \Omega$ and voltage source given by

$$v_g(t) = 10 \sin(\omega t + 30^\circ) \quad (\text{V})$$

is connected to a load $Z_L = (100 + j50) \, \Omega$ through a 50- Ω , 67-cm long lossless transmission line. The phase velocity of the line is $0.7c$, where c is the velocity of light in a vacuum. Find $v(d, t)$ and $i(d, t)$ on the line.

Solution: From the relationship $u_p = \lambda f$, we find the wavelength

$$\lambda = \frac{u_p}{f} = \frac{0.7 \times 3 \times 10^8}{1.05 \times 10^9} = 0.2 \, \text{m},$$

and

$$\beta l = \frac{2\pi}{\lambda} l = \frac{2\pi}{0.2} \times 0.67 = 6.7\pi = 0.7\pi = 126^\circ,$$

where we have subtracted multiples of 2π . The voltage reflection coefficient at the load is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{(100 + j50) - 50}{(100 + j50) + 50} = 0.45e^{j26.6^\circ}.$$

With reference to **Fig. 2-18**, the input impedance of the line given by Eq. (2.76) is

$$\begin{aligned} Z_{in} &= Z_0 \left(\frac{1 + \Gamma_l}{1 - \Gamma_l} \right) \\ &= Z_0 \left(\frac{1 + \Gamma e^{-j2\beta l}}{1 - \Gamma e^{-j2\beta l}} \right) \\ &= 50 \left(\frac{1 + 0.45e^{j26.6^\circ} e^{-j252^\circ}}{1 - 0.45e^{j26.6^\circ} e^{-j252^\circ}} \right) = (21.9 + j17.4) \, \Omega. \end{aligned}$$

Rewriting the expression for the generator voltage with the cosine reference, we have

$$\begin{aligned} v_g(t) &= 10 \sin(\omega t + 30^\circ) \\ &= 10 \cos(90^\circ - \omega t - 30^\circ) \\ &= 10 \cos(\omega t - 60^\circ) \\ &= \Re[10e^{-j60^\circ} e^{j\omega t}] = \Re[\tilde{V}_g e^{j\omega t}] \quad (\text{V}). \end{aligned}$$

Hence, the phasor voltage $\tilde{V}_g$ is given by

$$\tilde{V}_g = 10e^{-j60^\circ} = 10 \angle -60^\circ \quad (\text{V}).$$

Application of Eq. (2.82) gives

$$\begin{aligned} V_0^+ &= \left(\frac{\tilde{V}_g Z_{in}}{Z_g + Z_{in}} \right) \left(\frac{1}{e^{j\beta l} + \Gamma e^{-j\beta l}} \right) \\ &= \left[\frac{10e^{-j60^\circ} (21.9 + j17.4)}{10 + 21.9 + j17.4} \right] \\ &\quad \cdot (e^{j126^\circ} + 0.45e^{j26.6^\circ} e^{-j126^\circ})^{-1} \\ &= 10.2e^{j159^\circ} \quad (\text{V}). \end{aligned}$$

Using Eq. (2.63a) with $z = -d$, the phasor voltage on the line is

$$\begin{aligned} \tilde{V}(d) &= V_0^+ (e^{j\beta d} + \Gamma e^{-j\beta d}) \\ &= 10.2e^{j159^\circ} (e^{j\beta d} + 0.45e^{j26.6^\circ} e^{-j\beta d}), \end{aligned}$$

and the corresponding instantaneous voltage $v(d, t)$ is

$$\begin{aligned} v(d, t) &= \Re[\tilde{V}(d) e^{j\omega t}] \\ &= 10.2 \cos(\omega t + \beta d + 159^\circ) \\ &\quad + 4.55 \cos(\omega t - \beta d + 185.6^\circ) \quad (\text{V}). \end{aligned}$$

Similarly, Eq. (2.63b) leads to

$$\begin{aligned} \tilde{I}(d) &= 0.20e^{j159^\circ} (e^{j\beta d} - 0.45e^{j26.6^\circ} e^{-j\beta d}), \\ i(d, t) &= 0.20 \cos(\omega t + \beta d + 159^\circ) \\ &\quad + 0.091 \cos(\omega t - \beta d + 185.6^\circ) \quad (\text{A}). \end{aligned}$$

等式(2.80) 至方程式(2.81) 然后求解 V_0^+ 导致

$$V_0^+ = \left(\frac{\tilde{V}_g Z_{in}}{Z_g + Z_{in}} \right) \left(\frac{1}{e^{j\beta l} + \Gamma e^{-j\beta l}} \right). \quad (2.82)$$

这就完成了方程式给出的传输线波动方程的求解。(2.21) 和 (2.23) 适用于无损传输线的特殊情况。我们从方程式给出的通用解决方案开始。(2.26), 其中包括四个未知幅度, V_0^+, V_0^-, I_0^+ 和 I_0^- 。然后我们确定 $Z_0 = V_0^+ / I_0^+ = -V_0^- / I_0^-$, 从而仅将未知数减少到两个电压幅度。在负载处应用边界条件后, 我们能够通过 Γ 将 V_0^- 与 V_0^+ 联系起来, 最后, 通过在源处应用边界条件, 我们获得了 V_0^+ 的表达式。

例 2-7: $v(d, t)$ 和 $i(d, t)$ 的完整解

具有串联阻抗 $Z_g = 10 \, \Omega$ 和电压源的 1.05 GHz 发生器电路

$$v_g(t) = 10 \sin(\omega t + 30^\circ) \quad (\text{V})$$

通过50- Ω 、67厘米长的无损传输线连接到负载 $Z_L = (100 + j50) \, \Omega$ 。线的相速度为 $0.7c$, 其中 c 是真空中光速。找到直线上的 $v(d, t)$ 和 $i(d, t)$ 。

解: 根据 $u_p = \lambda f$ 的关系, 我们求出波长

$$\lambda = \frac{u_p}{f} = \frac{0.7 \times 3 \times 10^8}{1.05 \times 10^9} = 0.2 \, \text{m},$$

和

$$\beta l = \frac{2\pi}{\lambda} l = \frac{2\pi}{0.2} \times 0.67 = 6.7\pi = 0.7\pi = 126^\circ,$$

我们减去了 2π 的倍数。负载处的电压反射系数为

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{(100 + j50) - 50}{(100 + j50) + 50} = 0.45e^{j26.6^\circ}.$$

参考图2-18, 线路的输入阻抗由式(1)给出。(2.76) 是

$$\begin{aligned} Z_{in} &= Z_0 \left(\frac{1 + \Gamma_l}{1 - \Gamma_l} \right) \\ &= Z_0 \left(\frac{1 + \Gamma e^{-j2\beta l}}{1 - \Gamma e^{-j2\beta l}} \right) \\ &= 50 \left(\frac{1 + 0.45e^{j26.6^\circ} e^{-j252^\circ}}{1 - 0.45e^{j26.6^\circ} e^{-j252^\circ}} \right) = (21.9 + j17.4) \, \Omega. \end{aligned}$$

用余弦参考重写发电机电压的表达式, 我们有

$$\begin{aligned} v_g(t) &= 10 \sin(\omega t + 30^\circ) \\ &= 10 \cos(90^\circ - \omega t - 30^\circ) \\ &= 10 \cos(\omega t - 60^\circ) \\ &= \Re[10e^{-j60^\circ} e^{j\omega t}] = \Re[\tilde{V}_g e^{j\omega t}] \quad (\text{V}). \end{aligned}$$

因此, 相量电压 $\tilde{V}_g$ 由下式给出

$$\tilde{V}_g = 10e^{-j60^\circ} = 10 \angle -60^\circ \quad (\text{V}).$$

方程的应用(2.82) 给出

$$\begin{aligned} V_0^+ &= \left(\frac{\tilde{V}_g Z_{in}}{Z_g + Z_{in}} \right) \left(\frac{1}{e^{j\beta l} + \Gamma e^{-j\beta l}} \right) \\ &= \left[\frac{10e^{-j60^\circ} (21.9 + j17.4)}{10 + 21.9 + j17.4} \right] \\ &\quad \cdot (e^{j126^\circ} + 0.45e^{j26.6^\circ} e^{-j126^\circ})^{-1} \\ &= 10.2e^{j159^\circ} \quad (\text{V}). \end{aligned}$$

使用方程式(2.63a) 对于 $z = -d$, 线路上的相量电压为

$$\begin{aligned} \tilde{V}(d) &= V_0^+ (e^{j\beta d} + \Gamma e^{-j\beta d}) \\ &= 10.2e^{j159^\circ} (e^{j\beta d} + 0.45e^{j26.6^\circ} e^{-j\beta d}), \end{aligned}$$

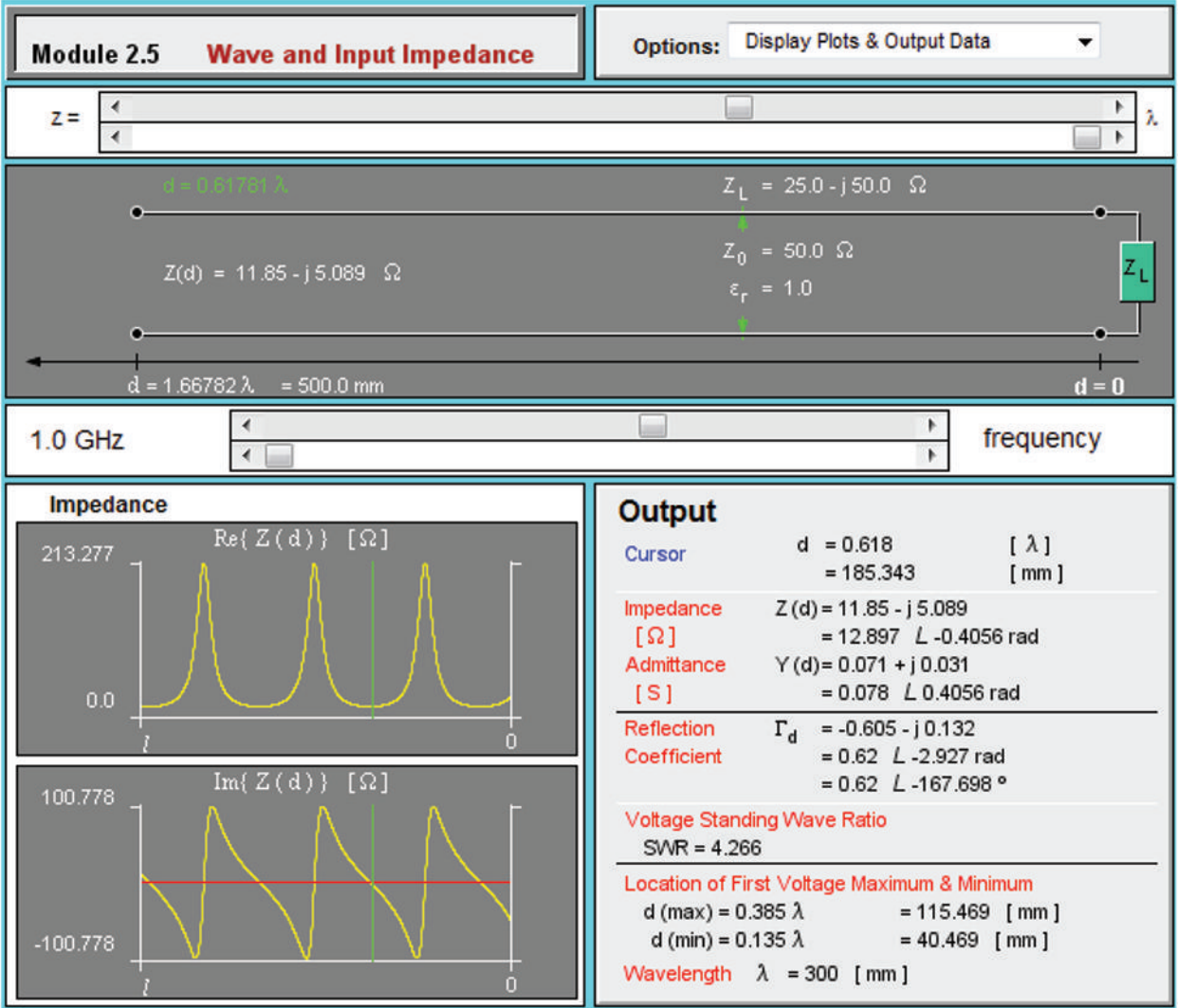
对应的瞬时电压 $v(d, t)$ 为

$$\begin{aligned} v(d, t) &= \Re[\tilde{V}(d) e^{j\omega t}] \\ &= 10.2 \cos(\omega t + \beta d + 159^\circ) \\ &\quad + 4.55 \cos(\omega t - \beta d + 185.6^\circ) \quad (\text{V}). \end{aligned}$$

同样, 等式。(2.63b) 导致

$$\begin{aligned} \tilde{I}(d) &= 0.20e^{j159^\circ} (e^{j\beta d} - 0.45e^{j26.6^\circ} e^{-j\beta d}), \\ i(d, t) &= 0.20 \cos(\omega t + \beta d + 159^\circ) \\ &\quad + 0.091 \cos(\omega t - \beta d + 185.6^\circ) \quad (\text{A}). \end{aligned}$$

Module 2.5 Wave and Input Impedance The wave impedance, $Z(d) = \tilde{V}(d)/\tilde{I}(d)$, exhibits a cyclical pattern as a function of position along the line. This module displays plots of the real and imaginary parts of $Z(d)$, specifies the locations of the voltage maximum and minimum nearest to the load, and provides other related information.



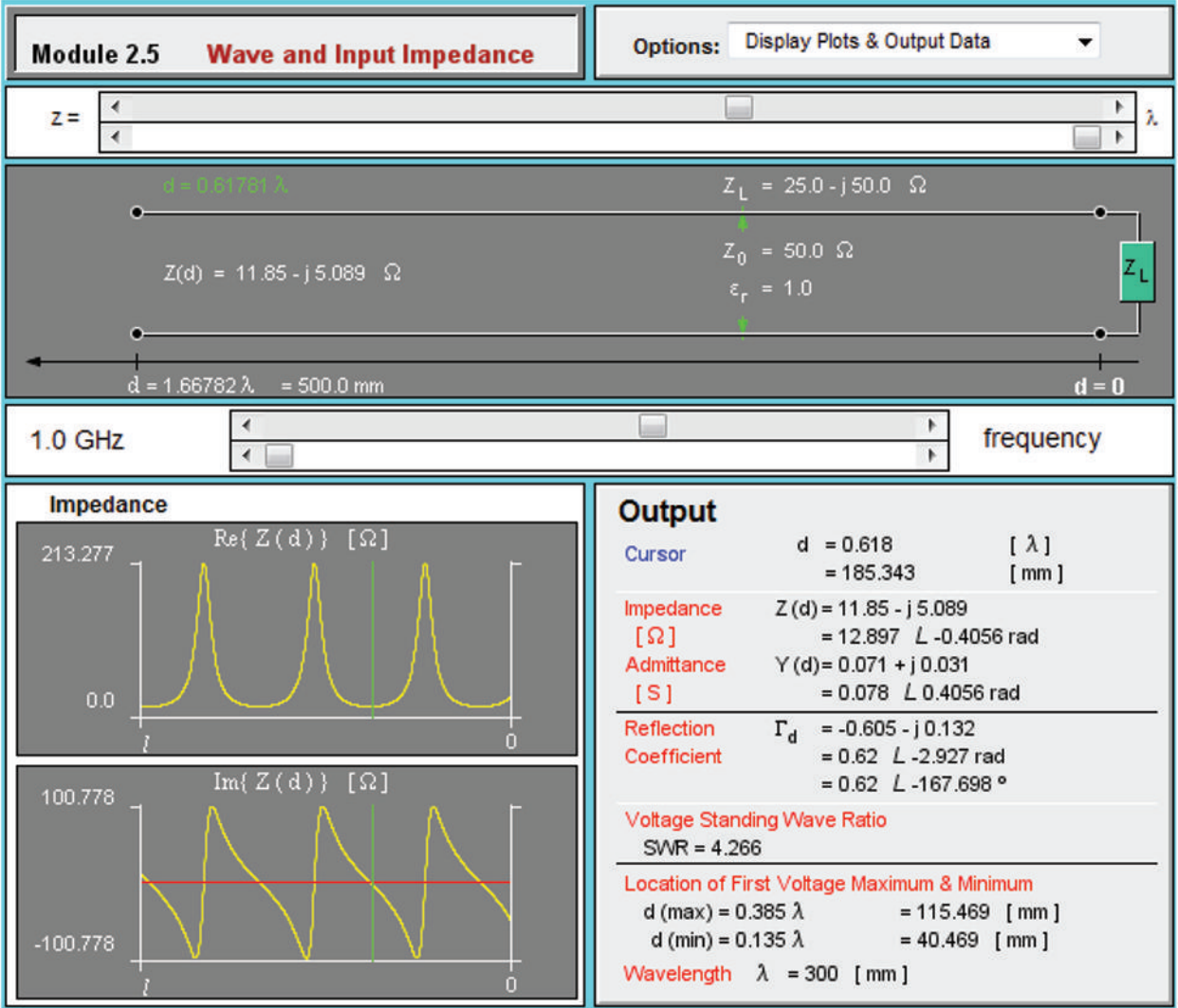
2-8 Special Cases of the Lossless Line

We often encounter situations involving lossless transmission lines with particular terminations or lines whose lengths lead to particularly useful line properties. We now consider some of these special cases.

2-8.1 Short-Circuited Line

The transmission line shown in Fig. 2-19(a) is terminated in a short circuit, $Z_L = 0$. Consequently, the voltage reflection coefficient defined by Eq. (2.59) is $\Gamma = -1$, and the voltage standing-wave ratio given by Eq. (2.73) is $S = \infty$. With $z = -d$ and $\Gamma = -1$ in Eqs. (2.63a) and (2.63b), and $\Gamma = -1$ in Eq. (2.74), the voltage, current, and wave impedance on a short-circuited lossless transmission line are given by

模块 2.5 波和输入阻抗 波阻抗 $Z(d) = \tilde{V}(d)/\tilde{I}(d)$ 呈现出作为沿线位置函数的循环模式。该模块显示 $Z(d)$ 的实部和虚部图，指定最接近负载的电压最大值和最小值的位置，并提供其他相关信息。



2-8 无损线路的特例

我们经常遇到涉及具有特定终端的无损传输线或其长度导致特别有用的线路属性的情况。我们现在考虑其中一些特殊情况。

2-8.1 短路线

图 2-19(a) 所示的传输线以短路终止， $Z_L = 0$ 。因此，电压反射系数由式 (1) 定义。(2.59) 为 $\Gamma = -1$ ，电压驻波比由式(2.59)给出。(2.73) 是 $S = \infty$ 。式中 $z = -d$ 和 $\Gamma = -1$ (2.63a) 和 (2.63b)，以及等式中的 $\Gamma = -1$ (2.74)，短路无损传输线上的电压、电流和波阻抗由下式给出

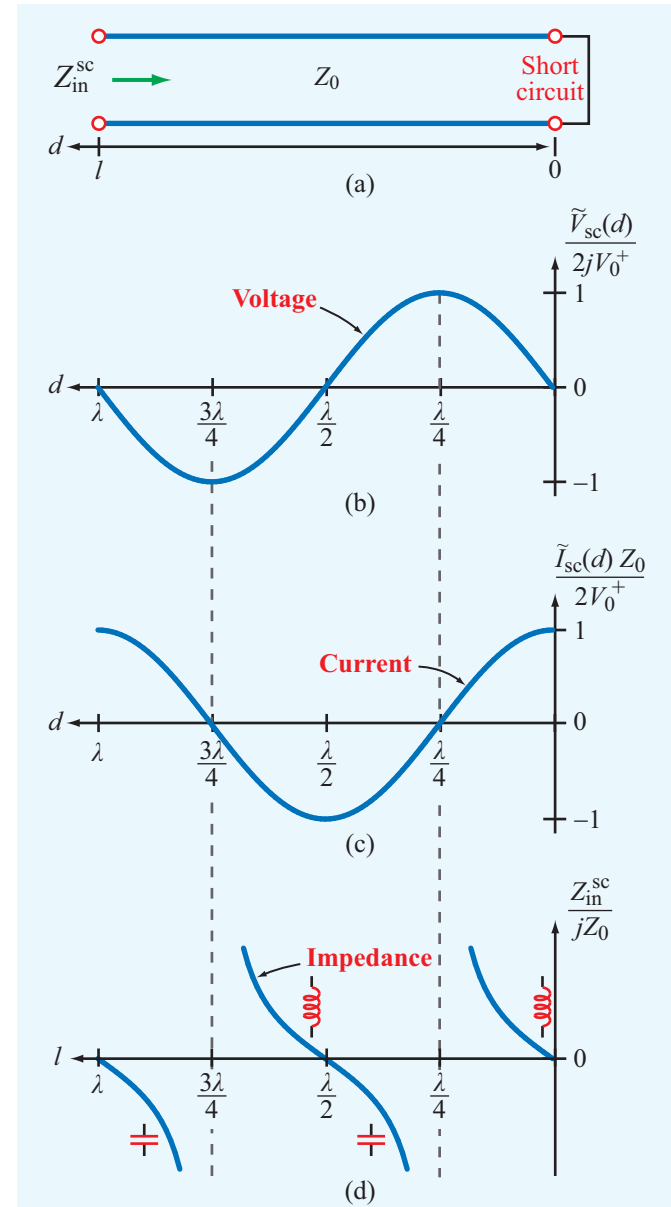


Figure 2-19 Transmission line terminated in a short circuit: (a) schematic representation, (b) normalized voltage on the line, (c) normalized current, and (d) normalized input impedance.

$$\tilde{V}_{sc}(d) = V_0^+ [e^{j\beta d} - e^{-j\beta d}] = 2jV_0^+ \sin \beta d, \quad (2.83a)$$

$$\tilde{I}_{sc}(d) = \frac{V_0^+}{Z_0} [e^{j\beta d} + e^{-j\beta d}] = \frac{2V_0^+}{Z_0} \cos \beta d, \quad (2.83b)$$

$$Z_{sc}(d) = \frac{\tilde{V}_{sc}(d)}{\tilde{I}_{sc}(d)} = jZ_0 \tan \beta d. \quad (2.83c)$$

The voltage $\tilde{V}_{sc}(d)$ is zero at the load ($d = 0$), as it should be for a short circuit, and its amplitude varies as $\sin \beta d$. In contrast, the current $\tilde{I}_{sc}(d)$ is a maximum at the load, and it varies as $\cos \beta d$. Both quantities are displayed in **Fig. 2-19** as a function of d .

Denoting Z_{in}^{sc} as the **input impedance of a short-circuited line** of length l ,

$$Z_{in}^{sc} = \frac{\tilde{V}_{sc}(l)}{\tilde{I}_{sc}(l)} = jZ_0 \tan \beta l. \quad (2.84)$$

A plot of Z_{in}^{sc}/jZ_0 versus l is shown in **Fig. 2-19(d)**. For the short-circuited line, if its length is less than $\lambda/4$, its impedance is equivalent to that of an inductor, and if it is between $\lambda/4$ and $\lambda/2$, it is equivalent to that of a capacitor.

In general, the input impedance Z_{in} of a line terminated in an arbitrary load has a real part, called the **input resistance** R_{in} , and an imaginary part, called the **input reactance** X_{in} :

$$Z_{in} = R_{in} + jX_{in}. \quad (2.85)$$

In the case of the short-circuited lossless line, the input impedance is purely reactive ($R_{in} = 0$). If $\tan \beta l \geq 0$, the line appears inductive to the source, acting like an equivalent inductor L_{eq} whose impedance equals Z_{in}^{sc} . Thus,

$$j\omega L_{eq} = jZ_0 \tan \beta l, \quad \text{if } \tan \beta l \geq 0, \quad (2.86)$$

or

$$L_{eq} = \frac{Z_0 \tan \beta l}{\omega} \quad (\text{H}). \quad (2.87)$$

The minimum line length l that would result in an input impedance Z_{in}^{sc} equivalent to that of an inductor with inductance L_{eq} is

$$l = \frac{1}{\beta} \tan^{-1} \left(\frac{\omega L_{eq}}{Z_0} \right) \quad (\text{m}). \quad (2.88)$$

Similarly, if $\tan \beta l \leq 0$, the input impedance is capacitive, in which case the line acts like an equivalent capacitor with capacitance C_{eq} such that

$$\frac{1}{j\omega C_{eq}} = jZ_0 \tan \beta l, \quad \text{if } \tan \beta l \leq 0, \quad (2.89)$$

or

$$C_{eq} = -\frac{1}{Z_0 \omega \tan \beta l} \quad (\text{F}). \quad (2.90)$$

Since l is a positive number, the shortest length l for which $\tan \beta l \leq 0$ corresponds to the range $\pi/2 \leq \beta l \leq \pi$. Hence, the

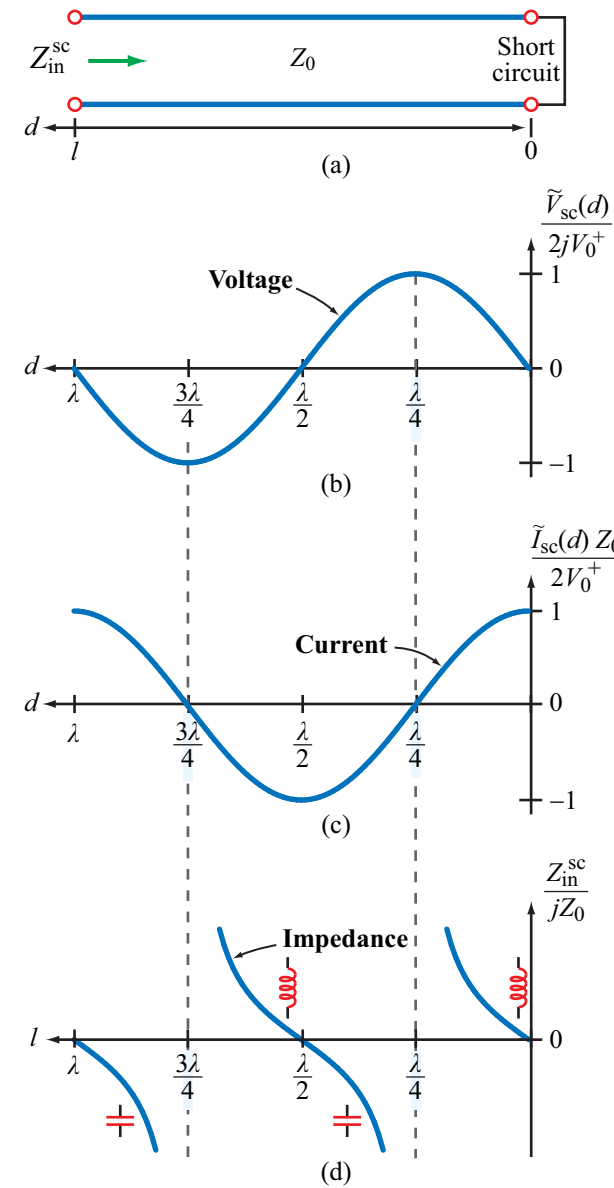


图 2-19 以短路终止的传输线: (a) 示意图, (b) 线路上的标准化电压, (c) 标准化电流, 以及 (d) 标准化输入阻抗。

$$Z_{sc}(d) = \frac{\tilde{V}_{sc}(d)}{\tilde{I}_{sc}(d)} = jZ_0 \tan \beta d. \quad (2.83c)$$

负载处的电压 $\tilde{V}_{sc}(d)$ 为零 ($d = 0$), 短路时应如此, 其幅度随 $\sin \beta d$ 变化。相反, 电流 $\tilde{I}_{sc}(d)$ 是负载时的最大值, 并且随着 $\cos \beta d$ 而变化。这两个量都作为 d 的函数显示在图 2-19 中。

将 Z_{in}^{sc} 表示为长度为 l 的短路线的输入阻抗,

$$Z_{in}^{sc} = \frac{\tilde{V}_{sc}(l)}{\tilde{I}_{sc}(l)} = jZ_0 \tan \beta l. \quad (2.84)$$

Z_{in}^{sc}/jZ_0 与 l 的关系图如图 2-19(d) 所示。对于短路线, 如果其长度小于 $\lambda/4$, 则其阻抗相当于电感, 如果在 $\lambda/4$ 和 $\lambda/2$ 之间, 则相当于电容器的阻抗。

一般来说, 终止于任意负载的线路的输入阻抗 Z_{in} 具有实部 (称为输入电阻 R_{in}) 和虚部 (称为输入电抗 X_{in}) :

$$Z_{in} = R_{in} + jX_{in}. \quad (2.85)$$

在短路无损线路的情况下, 输入阻抗是纯电抗性的 ($R_{in} = 0$)。如果 $\tan \beta l \geq 0$, 则线路对源表现为电感, 其作用类似于阻抗等于 Z_{in}^{sc} 的等效电感器 L_{eq} 。因此,

$$j\omega L_{eq} = jZ_0 \tan \beta l, \quad \text{if } \tan \beta l \geq 0, \quad (2.86)$$

or

$$L_{eq} = \frac{Z_0 \tan \beta l}{\omega} \quad (\text{H}). \quad (2.87)$$

产生与电感为 L_{eq} 的电感器等效的输入阻抗 Z_{in}^{sc} 的最小线路长度 l 为

$$l = \frac{1}{\beta} \tan^{-1} \left(\frac{\omega L_{eq}}{Z_0} \right) \quad (\text{m}). \quad (2.88)$$

类似地, 如果 $\tan \beta l \leq 0$, 则输入阻抗为电容性, 在这种情况下, 线路的作用类似于具有电容 C_{eq} 的等效电容器, 使得

$$\frac{1}{j\omega C_{eq}} = jZ_0 \tan \beta l, \quad \text{if } \tan \beta l \leq 0, \quad (2.89)$$

或者

$$C_{eq} = -\frac{1}{Z_0 \omega \tan \beta l} \quad (\text{F}). \quad (2.90)$$

由于 l 是正数, 因此 $\tan \beta l \leq 0$ 的最短长度 l 对应于范围 $\pi/2 \leq \beta l \leq \pi$ 。因此,

minimum line length l that would result in an input impedance $Z_{\text{in}}^{\text{sc}}$ equivalent to that of a capacitor of capacitance C_{eq} is

$$l = \frac{1}{\beta} \left[\pi - \tan^{-1} \left(\frac{1}{\omega C_{\text{eq}} Z_0} \right) \right] \quad (\text{m}). \quad (2.91)$$

▶ These results imply that, through proper choice of the length of a short-circuited line, we can make them into equivalent capacitors and inductors of any desired reactance. ◀

Such a practice is indeed common in the design of microwave circuits and high-speed integrated circuits, because making an actual capacitor or inductor often is much more difficult than fabricating a microstrip transmission line with a shorted end on a circuit board.

Example 2-8: Equivalent Reactive Elements

Choose the length of a shorted $50\ \Omega$ lossless transmission line (Fig. 2-20) such that its input impedance at 2.25 GHz is identical to that of a capacitor with capacitance $C_{\text{eq}} = 4\ \text{pF}$. The wave velocity on the line is $0.75c$.

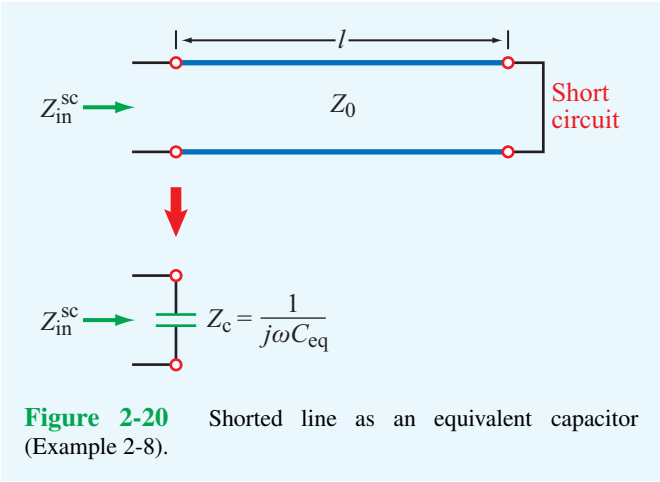


Figure 2-20 Shorted line as an equivalent capacitor (Example 2-8).

Solution: We are given

$$\begin{aligned} u_p &= 0.75c = 0.75 \times 3 \times 10^8 = 2.25 \times 10^8 \text{ m/s}, \\ Z_0 &= 50\ \Omega, \\ f &= 2.25 \text{ GHz} = 2.25 \times 10^9 \text{ Hz}, \\ C_{\text{eq}} &= 4 \text{ pF} = 4 \times 10^{-12} \text{ F}. \end{aligned}$$

The phase constant is

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi f}{u_p} = \frac{2\pi \times 2.25 \times 10^9}{2.25 \times 10^8} = 62.8 \quad (\text{rad/m}).$$

From Eq. (2.89), it follows that

$$\begin{aligned} \tan \beta l &= -\frac{1}{Z_0 \omega C_{\text{eq}}} \\ &= -\frac{1}{50 \times 2\pi \times 2.25 \times 10^9 \times 4 \times 10^{-12}} = -0.354. \end{aligned}$$

The tangent function is negative when its argument is in the second or fourth quadrants. The solution for the second quadrant is

$$\beta l_1 = 2.8 \text{ rad} \quad \text{or} \quad l_1 = \frac{2.8}{\beta} = \frac{2.8}{62.8} = 4.46 \text{ cm},$$

and the solution for the fourth quadrant is

$$\beta l_2 = 5.94 \text{ rad} \quad \text{or} \quad l_2 = \frac{5.94}{62.8} = 9.46 \text{ cm}.$$

We also could have obtained the value of l_1 by applying Eq. (2.91). The length l_2 is greater than l_1 by exactly $\lambda/2$. In fact, any length $l = 4.46 \text{ cm} + n\lambda/2$, where n is a positive integer, also is a solution.

2-8.2 Open-Circuited Line

With $Z_L = \infty$, as illustrated in Fig. 2-21(a), we have $\Gamma = 1$ and $S = \infty$, and the voltage, current, and input impedance are given by

$$\tilde{V}_{\text{oc}}(d) = V_0^+ [e^{j\beta d} + e^{-j\beta d}] = 2V_0^+ \cos \beta d, \quad (2.92a)$$

$$\tilde{I}_{\text{oc}}(d) = \frac{V_0^+}{Z_0} [e^{j\beta d} - e^{-j\beta d}] = \frac{2jV_0^+}{Z_0} \sin \beta d, \quad (2.92b)$$

$$Z_{\text{in}}^{\text{oc}} = \frac{\tilde{V}_{\text{oc}}(l)}{\tilde{I}_{\text{oc}}(l)} = -jZ_0 \cot \beta l. \quad (2.93)$$

Plots of these quantities are displayed in Fig. 2-21 as a function of d for the voltage and current and as a function of l for the input impedance.

2-8.3 Application of Short-Circuit/Open-Circuit Technique

A network analyzer is a radio-frequency (RF) instrument capable of measuring the impedance of any load connected

导致输入阻抗 $Z_{\text{in}}^{\text{sc}}$ 相当于电容 C_{eq} 的电容器的最小线路长度 l 为

$$l = \frac{1}{\beta} \left[\pi - \tan^{-1} \left(\frac{1}{\omega C_{\text{eq}} Z_0} \right) \right] \quad (\text{m}). \quad (2.91)$$

▶ 这些结果意味着，通过正确选择短路线的长度，我们可以将它们制成任何所需电抗的等效电容器和电感器。 ◀

这种做法在微波电路和高速集成电路的设计中确实很常见，因为制造实际的电容器或电感器通常比在电路板上制造具有短路端的微带传输线困难得多。

示例 2-8：等效反应元素

选择短路的 $50\ \Omega$ 无损传输线（图 2-20）的长度，使其在 2.25 GHz 时的输入阻抗与电容 $C_{\text{eq}} = 4\ \text{pF}$ 的电容器的输入阻抗相同。线上的波速为 $0.75c$ 。

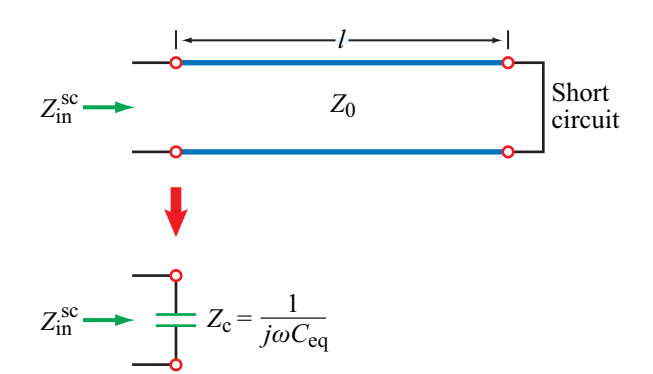


图 2-20 作为等效电容器的短路线（示例 2-8）。

解：我们给定

$$\begin{aligned} u_p &= 0.75c = 0.75 \times 3 \times 10^8 = 2.25 \times 10^8 \text{ m/s}, \\ Z_0 &= 50\ \Omega, \\ f &= 2.25 \text{ GHz} = 2.25 \times 10^9 \text{ Hz}, \\ C_{\text{eq}} &= 4 \text{ pF} = 4 \times 10^{-12} \text{ F}. \end{aligned}$$

相位常数为

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi f}{u_p} = \frac{2\pi \times 2.25 \times 10^9}{2.25 \times 10^8} = 62.8 \quad (\text{弧度/米}).$$

从方程。 (2.89) ， 由此可知

$$\begin{aligned} \tan \beta l &= -\frac{1}{Z_0 \omega C_{\text{eq}}} \\ &= -\frac{1}{50 \times 2\pi \times 2.25 \times 10^9 \times 4 \times 10^{-12}} = -0.354. \end{aligned}$$

当参数位于第二或第四象限时，正切函数为负。第二象限的解为

$$\beta l_1 = 2.8 \text{ rad} \quad \text{or} \quad l_1 = \frac{2.8}{\beta} = \frac{2.8}{62.8} = 4.46 \text{ cm},$$

第四象限的解为

$$\beta l_2 = 5.94 \text{ rad} \quad \text{or} \quad l_2 = \frac{5.94}{62.8} = 9.46 \text{ cm}.$$

我们还可以通过应用等式获得 l_1 的值。 (2.91) 。长度 l_2 比 l_1 正好大 $\lambda/2$ 。事实上，任意长度 $l = 4.46 \text{ cm} + n\lambda/2$ ，其中 n 为正整数，也是一个解。

2-8.2 开路线路

对于 $Z_L = \infty$ ，如图 2-21(a) 所示，我们有 $\Gamma = 1$ 和 $S = \infty$ ，电压、电流和输入阻抗由下式给出

$$\tilde{V}_{\text{oc}}(d) = V_0^+ [e^{j\beta d} + e^{-j\beta d}] = 2V_0^+ \cos \beta d, \quad (2.92a)$$

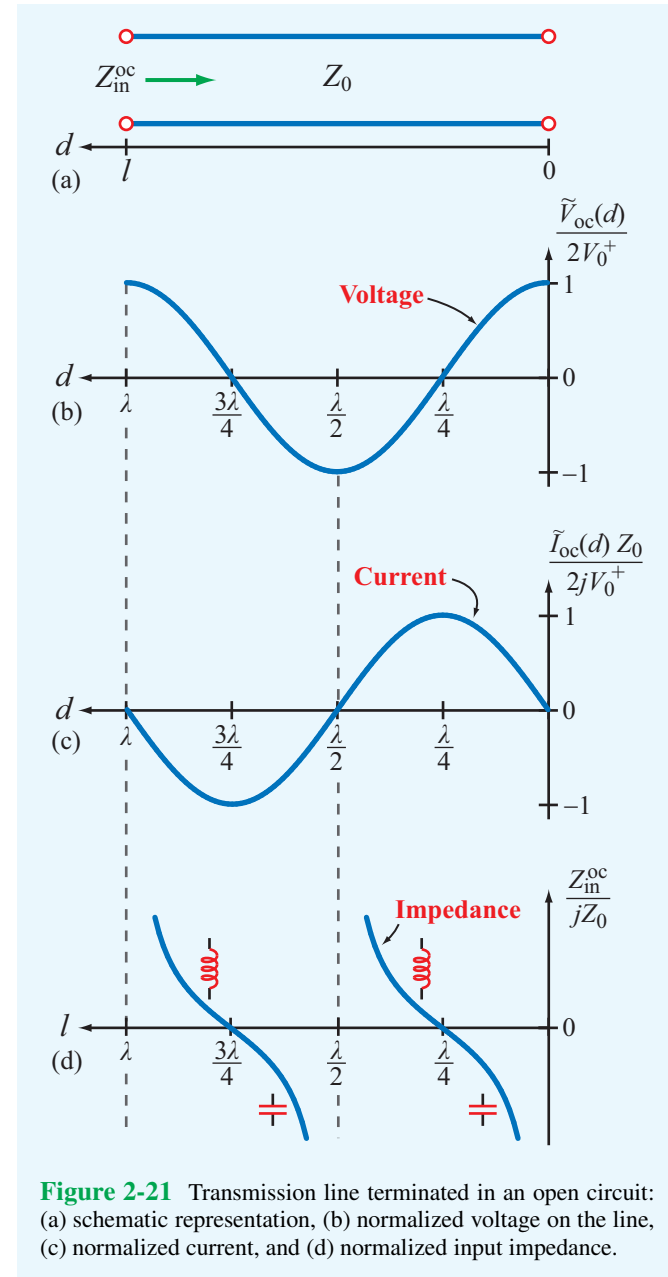
$$\tilde{I}_{\text{oc}}(d) = \frac{V_0^+}{Z_0} [e^{j\beta d} - e^{-j\beta d}] = \frac{2jV_0^+}{Z_0} \sin \beta d, \quad (2.92b)$$

$$Z_{\text{in}}^{\text{oc}} = \frac{\tilde{V}_{\text{oc}}(l)}{\tilde{I}_{\text{oc}}(l)} = -jZ_0 \cot \beta l. \quad (2.93)$$

这些量的图显示在图 2-21 中，作为电压和电流的 d 函数以及输入阻抗的 l 函数。

2-8.3 短路/开路技术的应用

网络分析仪是一种射频 (RF) 仪器，能够测量任何连接负载的阻抗



to its input terminal. When used to measure (1) Z_{in}^{sc} , which is the input impedance of a lossless line terminated in a short circuit, and (2) Z_{in}^{oc} , which is the input impedance of the line when terminated in an open circuit, the combination of the two measurements can be used to determine the characteristic impedance of the line Z_0 and its phase constant β . Indeed, the product of the expressions given by Eqs. (2.84) and (2.93)

gives

$$Z_0 = \sqrt[+]{Z_{in}^{sc} Z_{in}^{oc}}, \quad (2.94)$$

and the ratio of the same expressions leads to

$$\tan \beta l = \sqrt{\frac{-Z_{in}^{sc}}{Z_{in}^{oc}}}. \quad (2.95)$$

Because of the π phase ambiguity associated with the tangent function, the length l should be less than or equal to $\lambda/2$ to provide an unambiguous result.

Example 2-9: Measuring Z_0 and β

Find Z_0 and β of a 57-cm long lossless transmission line whose input impedance was measured as $Z_{in}^{sc} = j40.42 \, \Omega$ when terminated in a short circuit and as $Z_{in}^{oc} = -j121.24 \, \Omega$ when terminated in an open circuit. From other measurements, we know that the line is between 3 and 3.25 wavelengths long.

Solution: From Eqs. (2.94) and (2.95),

$$Z_0 = \sqrt[+]{Z_{in}^{sc} Z_{in}^{oc}} = \sqrt{(j40.42)(-j121.24)} = 70 \, \Omega,$$

$$\tan \beta l = \sqrt{\frac{-Z_{in}^{sc}}{Z_{in}^{oc}}} = \sqrt{\frac{1}{3}}.$$

Since l is between 3λ and 3.25λ , $\beta l = (2\pi l/\lambda)$ is between 6π radians and $(13\pi/2)$ radians. This places βl in the first quadrant (0 to $\pi/2$) radians. Hence, the only acceptable solution for $\tan \beta l = \sqrt{1/3}$ is $\beta l = \pi/6$ radians. This value, however, does not include the 2π multiples associated with the integer λ multiples of l . Hence, the true value of βl is

$$\beta l = 6\pi + \frac{\pi}{6} = 19.4 \quad (\text{rad}),$$

in which case

$$\beta = \frac{19.4}{0.57} = 34 \quad (\text{rad/m}).$$

2-8.4 Lines of Length $l = n\lambda/2$

If $l = n\lambda/2$, where n is an integer,

$$\tan \beta l = \tan [(2\pi/\lambda)(n\lambda/2)] = \tan n\pi = 0.$$

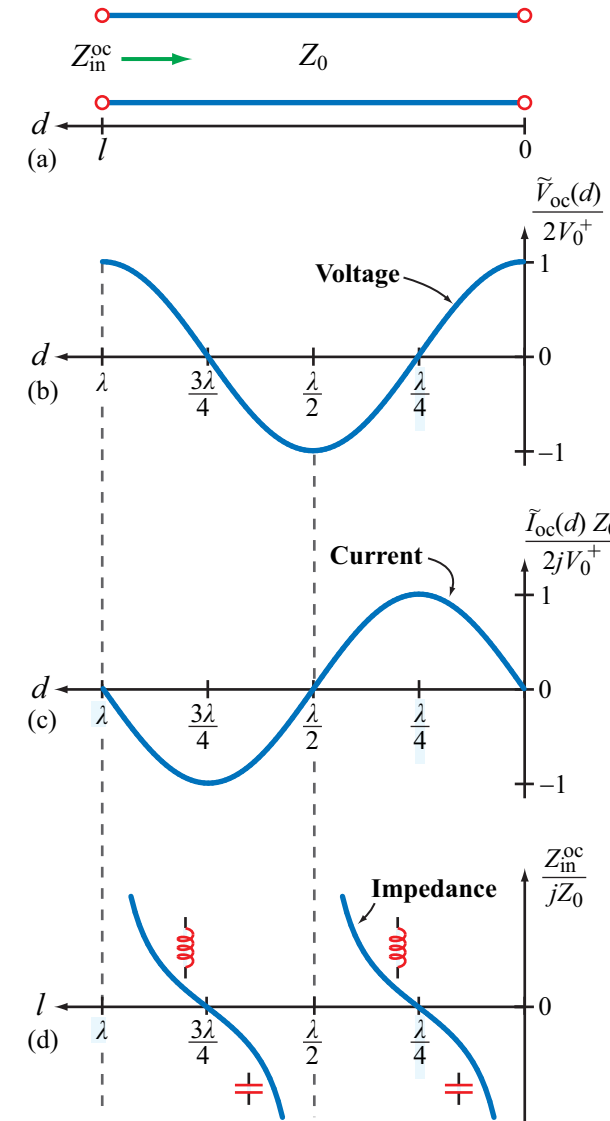


图 2-21 以开路端接的传输线: (a) 示意图, (b) 线路上的标准化电压, (c) 标准化电流, 以及 (d) 标准化输入阻抗。

到其输入端子。当用于测量 (1) Z_{in}^{sc} (以短路方式端接的无损线路的输入阻抗) 和 (2) Z_{in}^{oc} (以开路方式端接时线路的输入阻抗) 时, 两个测量值的组合可用于确定线路 Z_0 的特性阻抗及其相位常数 β 。事实上, 方程给出的表达式的乘积。 (2.84) 和 (2.93)

给出

$$Z_0 = \sqrt[+]{Z_{in}^{sc} Z_{in}^{oc}}, \quad (2.94)$$

并且相同表达式的比率导致

$$\tan \beta l = \sqrt{\frac{-Z_{in}^{sc}}{Z_{in}^{oc}}}. \quad (2.95)$$

由于与正切函数相关的 π 相位模糊性, 长度 l 应小于或等于 $\lambda/2$ 以提供明确的结果。

例 2-9: 测量 Z_0 和 β

求 57 厘米长的无损传输线的 Z_0 和 β , 其输入阻抗在短路端接时测量为 $Z_{in}^{sc} = j40.42 \, \Omega$, 在开路端接时测量为 $Z_{in}^{oc} = -j121.24 \, \Omega$ 。从其他测量中, 我们知道该线的长度在 3 到 3.25 个波长之间。

解决方案: 从方程。 (2.94) 和 (2.95),

$$Z_0 = \sqrt[+]{Z_{in}^{sc} Z_{in}^{oc}} = \sqrt{(j40.42)(-j121.24)} = 70 \, \Omega,$$

$$\tan \beta l = \sqrt{\frac{-Z_{in}^{sc}}{Z_{in}^{oc}}} = \sqrt{\frac{1}{3}}.$$

由于 l 介于 3λ 和 3.25λ 之间, 因此 $\beta l = (2\pi l/\lambda)$ 介于 6π 弧度和 $(13\pi/2)$ 弧度之间。这会将 βl 置于第一象限 (0 到 $\pi/2$) 弧度内。因此, $\tan \beta l = \sqrt{1/3}$ 唯一可接受的解决方案是 $\beta l = \pi/6$ 弧度。然而, 该值不包括与 l 的整数 λ 倍数相关的 2π 倍数。因此, βl 的真实值为

$$\beta l = 6\pi + \frac{\pi}{6} = 19.4 \quad (\text{rad}),$$

在这种情况下

$$\beta = \frac{19.4}{0.57} = 34 \quad (\text{rad/m}).$$

2-8.4 长度为 $l = n\lambda/2$ 的线

如果 $l = n\lambda/2$, 其中 n 是整数,

$$\tan \beta l = \tan [(2\pi/\lambda)(n\lambda/2)] = \tan n\pi = 0.$$

Consequently, Eq. (2.79) reduces to

$$Z_{\text{in}} = Z_{\text{L}}, \quad \text{for } l = n\lambda/2,$$
(2.96)

which means that a half-wavelength line (or any integer multiple of $\lambda/2$) does not modify the load impedance.

2-8.5 Quarter-Wavelength Transformer

Another case of interest is when the length of the line is a quarter-wavelength (or $\lambda/4 + n\lambda/2$, where $n = 0$ or a positive integer), corresponding to $\beta l = (2\pi/\lambda)(\lambda/4) = \pi/2$. From Eq. (2.79), the input impedance becomes

$$Z_{\text{in}} = \frac{Z_0^2}{Z_{\text{L}}}, \quad \text{for } l = \lambda/4 + n\lambda/2.$$
(2.97)

The utility of such a quarter-wave transformer is illustrated by Example 2-10.

Example 2-10: $\lambda/4$ Transformer

A $50\ \Omega$ lossless transmission line is to be matched to a resistive load impedance with $Z_{\text{L}} = 100\ \Omega$ via a quarter-wave section as shown in Fig. 2-22, thereby eliminating reflections along the feedline. Find the required characteristic impedance of the quarter-wave transformer.

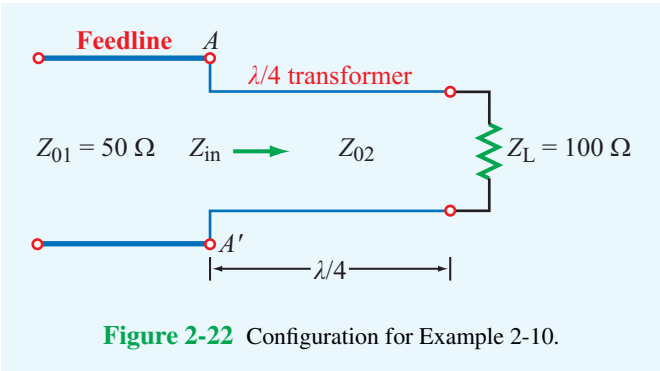


Figure 2-22 Configuration for Example 2-10.

Solution: To eliminate reflections at terminal AA' , the input impedance Z_{in} looking into the quarter-wave line should be equal to Z_{01} , which is the characteristic impedance of the feedline. Thus, $Z_{\text{in}} = 50\ \Omega$. From Eq. (2.97),

$$Z_{\text{in}} = \frac{Z_{02}^2}{Z_{\text{L}}},$$

or

$$Z_{02} = \sqrt{Z_{\text{in}} Z_{\text{L}}} = \sqrt{50 \times 100} = 70.7\ \Omega.$$

Whereas this eliminates reflections on the feedline, it does not eliminate them on the $\lambda/4$ line. However, since the lines are lossless, all of the power incident on AA' will end up getting transferred into the load Z_{L} .

► In this example, Z_{L} is purely resistive. To apply the $\lambda/4$ transformer technique to match a transmission line to a load with a complex impedance, a more elaborate procedure is required (Section 2-11). ◀

2-8.6 Matched Transmission Line: $Z_{\text{L}} = Z_0$

For a matched lossless transmission line with $Z_{\text{L}} = Z_0$, (1) the input impedance $Z_{\text{in}} = Z_0$ for all locations d on the line, (2) $\Gamma = 0$, and (3) all of the incident power is delivered to the load, regardless of the line length l . A summary of the properties of standing waves is given in Table 2-4.

Concept Question 2-10: What is the difference between the characteristic impedance Z_0 and the input impedance Z_{in} ? When are they the same?

Concept Question 2-11: What is a quarter-wave transformer? How can it be used?

Concept Question 2-12: A lossless transmission line of length l is terminated in a short circuit. If $l < \lambda/4$, is the input impedance inductive or capacitive?

Concept Question 2-13: What is the input impedance of an infinitely long line?

Concept Question 2-14: If the input impedance of a lossless line is inductive when terminated in a short circuit, will it be inductive or capacitive when the line is terminated in an open circuit?

Exercise 2-14: A $50\ \Omega$ lossless transmission line uses an insulating material with $\epsilon_{\text{r}} = 2.25$. When terminated in an open circuit, how long should the line be for its input impedance to be equivalent to a $10\ \text{pF}$ capacitor at $50\ \text{MHz}$?

Answer: $l = 9.92\ \text{cm}$. (See EM.)

因此，等式。(2.79) 减少为

$$Z_{\text{in}} = Z_{\text{L}}, \quad \text{for } l = n\lambda/2,$$
(2.96)

这意味着半波长线（或 $\lambda/2$ 的任何整数倍）不会改变负载阻抗。

2-8.5 四分之一波长变压器

另一种有趣的情况是当线的长度为四分之一波长（或 $\lambda/4 + n\lambda/2$ ，其中 $n = 0$ 或正整数）时，对应于 $\beta l = (2\pi/\lambda)(\lambda/4) = \pi/2$ 。从方程。(2.79)，输入阻抗变为

$$Z_{\text{in}} = \frac{Z_0^2}{Z_{\text{L}}}, \quad \text{for } l = \lambda/4 + n\lambda/2.$$
(2.97)

例 2-10 说明了这种四分之一波长变压器的实用性。

例 2-10: $\lambda/4$ 变压器

$50\ \Omega$ 无损传输线将通过四分之一波长部分与 $Z_{\text{L}} = 100\ \Omega$ 的电阻负载阻抗匹配，如图 2-22 所示，从而消除沿馈线的反射。找到四分之一波长变压器所需的特性阻抗。

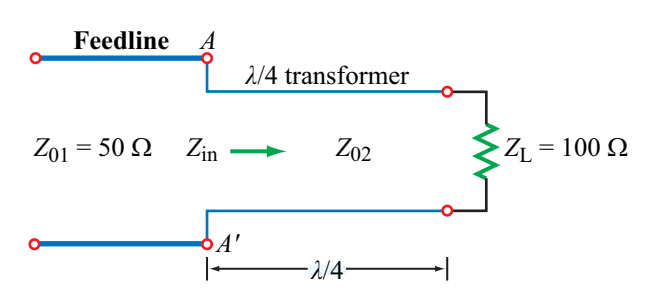


图 2-22 示例 2-10 的配置。

解决方案：为了消除端子 AA' 处的反射，四分之一波长线路的输入阻抗 Z_{in} 应等于 Z_{01} ，即馈线的特性阻抗。因此， $Z_{\text{in}} = 50\ \Omega$ 。从方程。(2.97),

$$Z_{\text{in}} = \frac{Z_{02}^2}{Z_{\text{L}}},$$

或者

$$Z_{02} = \sqrt{Z_{\text{in}} Z_{\text{L}}} = \sqrt{50 \times 100} = 70.7\ \Omega.$$

虽然这消除了馈线上的反射，但它并没有消除 $\lambda/4$ 线上的反射。然而，由于线路是无损的，所有入射到 AA' 上的功率最终都会转移到负载 Z_{L} 中。

► 在此示例中， Z_{L} 是纯电阻性的。要应用 $\lambda/4$ 变压器技术将传输线与具有复阻抗的负载相匹配，需要更复杂的过程（第 2-11 节）。◀

2-8.6 匹配传输线: $Z_{\text{L}} = Z_0$

对于具有 $Z_{\text{L}} = Z_0$ 的匹配无损传输线，(1) 线路上所有位置 d 的输入阻抗 $Z_{\text{in}} = Z_0$ ，(2) $\Gamma = 0$ ，以及 (3) 无论线路长度 l ，所有入射功率都传递到负载。表 2-4 总结了驻波的特性。

概念问题2-10: 特性阻抗 Z_0 和输入阻抗 Z_{in} 有什么区别？他们什么时候一样了？

概念问题 2-11: 什么是四分之一波变压器？如何使用？

概念问题 2-12: 长度为 l 的无损传输线以短路终止。如果 $l < \lambda/4$ ，输入阻抗是感性的还是容性的？

概念问题 2-13: 无限长线路的输入阻抗是多少？

概念问题 2-14: 如果无损线路的输入阻抗在短路端接时是感性的，那么当线路在开路端接时它是感性的还是容性的？

练习 2-14: $50\ \Omega$ 无损传输线使用 $\epsilon_{\text{r}} = 2.25$ 的绝缘材料。当以开路端接时，线路应有多长才能使其输入阻抗相当于 $50\ \text{MHz}$ 下的 $10\ \text{pF}$ 电容器？

答案: $l = 9.92\ \text{厘米}$ 。（参见^{EM}。）

Table 2-4 Properties of standing waves on a lossless transmission line.

Voltage maximum	$ \tilde{V} _{\max} = V_0^+ [1 + \Gamma]$
Voltage minimum	$ \tilde{V} _{\min} = V_0^+ [1 - \Gamma]$
Positions of voltage maxima (also positions of current minima)	$d_{\max} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}, \quad n = 0, 1, 2, \dots$
Position of first maximum (also position of first current minimum)	$d_{\max} = \begin{cases} \frac{\theta_r \lambda}{4\pi}, & \text{if } 0 \leq \theta_r \leq \pi \\ \frac{\theta_r \lambda}{4\pi} + \frac{\lambda}{2}, & \text{if } -\pi \leq \theta_r \leq 0 \end{cases}$
Positions of voltage minima (also positions of current maxima)	$d_{\min} = \frac{\theta_r \lambda}{4\pi} + \frac{(2n+1)\lambda}{4}, \quad n = 0, 1, 2, \dots$
Position of first minimum (also position of first current maximum)	$d_{\min} = \frac{\lambda}{4} \left(1 + \frac{\theta_r}{\pi} \right)$
Input impedance	$Z_{\text{in}} = Z_0 \left(\frac{z_L + j \tan \beta l}{1 + j z_L \tan \beta l} \right) = Z_0 \left(\frac{1 + \Gamma_l}{1 - \Gamma_l} \right)$
Positions at which Z_{in} is real	at voltage maxima and minima
Z_{in} at voltage maxima	$Z_{\text{in}} = Z_0 \left(\frac{1 + \Gamma }{1 - \Gamma } \right)$
Z_{in} at voltage minima	$Z_{\text{in}} = Z_0 \left(\frac{1 - \Gamma }{1 + \Gamma } \right)$
Z_{in} of short-circuited line	$Z_{\text{in}}^{\text{sc}} = jZ_0 \tan \beta l$
Z_{in} of open-circuited line	$Z_{\text{in}}^{\text{oc}} = -jZ_0 \cot \beta l$
Z_{in} of line of length $l = n\lambda/2$	$Z_{\text{in}} = Z_L, \quad n = 0, 1, 2, \dots$
Z_{in} of line of length $l = \lambda/4 + n\lambda/2$	$Z_{\text{in}} = Z_0^2/Z_L, \quad n = 0, 1, 2, \dots$
Z_{in} of matched line	$Z_{\text{in}} = Z_0$
$ V_0^+ $ = amplitude of incident wave; $\Gamma = \Gamma e^{j\theta_r}$ with $-\pi < \theta_r < \pi$; θ_r in radians; $\Gamma_l = \Gamma e^{-j2\beta l}$.	

Exercise 2-15: A 300 Ω feedline is to be connected to a 3-m long, 150 Ω line terminated in a 150 Ω resistor. Both lines are lossless and use air as the insulating material, and the operating frequency is 50 MHz. Determine (a) the input impedance of the 3-m long line, (b) the voltage standing-wave ratio on the feedline, and (c) the characteristic impedance of a quarter-wave transformer were it to be used between the two lines in order to achieve $S = 1$ on the feedline. (See [EM](#).)

Answer: (a) $Z_{\text{in}} = 150 \, \Omega$, (b) $S = 2$, (c) $Z_0 = 212.1 \, \Omega$.

Exercise 2-16: Through multiple trials, it was determined that a load with unknown impedance Z_L can be perfectly matched to a feedline with $Z_{\text{in}} = 50 \, \Omega$ by using a $\lambda/4$ transformer section with characteristic impedance of 60 Ω. What is Z_L ?

Answer: $Z_L = 72 \, \Omega$.

表2-4 无损传输线上的驻波特性。

Voltage maximum	$ \tilde{V} _{\max} = V_0^+ [1 + \Gamma]$
Voltage minimum	$ \tilde{V} _{\min} = V_0^+ [1 - \Gamma]$
Positions of voltage maxima (also positions of current minima)	$d_{\max} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}, \quad n = 0, 1, 2, \dots$
Position of first maximum (also position of first current minimum)	$d_{\max} = \begin{cases} \frac{\theta_r \lambda}{4\pi}, & \text{if } 0 \leq \theta_r \leq \pi \\ \frac{\theta_r \lambda}{4\pi} + \frac{\lambda}{2}, & \text{if } -\pi \leq \theta_r \leq 0 \end{cases}$
Positions of voltage minima (also positions of current maxima)	$d_{\min} = \frac{\theta_r \lambda}{4\pi} + \frac{(2n+1)\lambda}{4}, \quad n = 0, 1, 2, \dots$
Position of first minimum (also position of first current maximum)	$d_{\min} = \frac{\lambda}{4} \left(1 + \frac{\theta_r}{\pi} \right)$
Input impedance	$Z_{\text{in}} = Z_0 \left(\frac{z_L + j \tan \beta l}{1 + j z_L \tan \beta l} \right) = Z_0 \left(\frac{1 + \Gamma_l}{1 - \Gamma_l} \right)$
Positions at which Z_{in} is real	at voltage maxima and minima
Z_{in} at voltage maxima	$Z_{\text{in}} = Z_0 \left(\frac{1 + \Gamma }{1 - \Gamma } \right)$
Z_{in} at voltage minima	$Z_{\text{in}} = Z_0 \left(\frac{1 - \Gamma }{1 + \Gamma } \right)$
Z_{in} of short-circuited line	$Z_{\text{in}}^{\text{sc}} = jZ_0 \tan \beta l$
Z_{in} of open-circuited line	$Z_{\text{in}}^{\text{oc}} = -jZ_0 \cot \beta l$
Z_{in} of line of length $l = n\lambda/2$	$Z_{\text{in}} = Z_L, \quad n = 0, 1, 2, \dots$
Z_{in} of line of length $l = \lambda/4 + n\lambda/2$	$Z_{\text{in}} = Z_0^2/Z_L, \quad n = 0, 1, 2, \dots$
Z_{in} of matched line	$Z_{\text{in}} = Z_0$
$ V_0^+ $ = amplitude of incident wave; $\Gamma = \Gamma e^{j\theta_r}$ with $-\pi < \theta_r < \pi$; θ_r in radians; $\Gamma_l = \Gamma e^{-j2\beta l}$.	

练习 2-15：一条 300 Ω 馈线将连接到一条 3 米长、150 Ω 线路，该线路端接一个 150 Ω 电阻器。两条线路都是无损的，并使用空气作为绝缘材料，工作频率为 50 MHz。确定 (a) 3 米长线路的输入阻抗，(b) 馈线上的电压驻波比，以及 (c) 在两条线路之间使用四分之一波长变压器的特性阻抗，以便在馈线上实现 $S = 1$ 。（参见^{EM}。）

答案：(a) $Z_{\text{in}} = 150 \, \Omega$, (b) $S = 2$, (c) $Z_0 = 212.1 \, \Omega$ 。

练习 2-16：通过多次试验，确定阻抗 Z_L 未知的负载可以通过使用特性阻抗为 60 Ω 的 $\lambda/4$ 变压器部分与 $Z_{\text{in}} = 50 \, \Omega$ 的馈线完美匹配。 Z_L 是什么？

答案： $Z_L = 72 \, \Omega$ 。

Technology Brief 3: Microwave Ovens

Percy Spencer, while working for Raytheon in the 1940s on the design and construction of **magnetrons** for radar, observed that a chocolate bar that had unintentionally been exposed to microwaves had melted in his pocket. The process of cooking by microwave was patented in 1946 and by the 1970s, microwave ovens had become standard household items.

Microwave Absorption

A microwave is an **electromagnetic wave** whose frequency lies in the 300 MHz–300 GHz range (see **Fig. 1-16**). When a material containing water is exposed to microwaves, the water molecule reacts by rotating itself in order to align its own **electric dipole** along the direction of the oscillating electric field of the microwave. The rapid vibration motion creates heat in the material, resulting in the conversion of microwave energy into thermal energy. The absorption coefficient of water, $\alpha(f)$, exhibits a microwave spectrum that depends on the temperature of the water and the concentration of dissolved

salts and sugars present in it. If the frequency f is chosen such that $\alpha(f)$ is high, the water-containing material absorbs much of the microwave energy passing through it and converts it to heat. However, this also means that most of the energy is absorbed by a thin surface layer of the material with not much energy remaining to heat deeper layers. The penetration depth δ_p of a material, defined as $\delta_p = 1/2\alpha$, is a measure of how deep the power carried by an EM wave can penetrate into the material. Approximately 95% of the microwave energy incident upon a material is absorbed by the surface layer of thickness $3\delta_p$. **Figure TF3-1** displays calculated spectra of δ_p for pure water and two materials with different water contents.

► The frequency most commonly used in microwave ovens is 2.45 GHz. The magnitude of δ_p at 2.45 GHz varies between ~2 cm for pure water and 8 cm for a material with a water content of only 20%. ◀

This is a practical range for cooking food in a microwave oven; at much lower frequencies, the food is not a good absorber of energy (in addition to the fact that the design

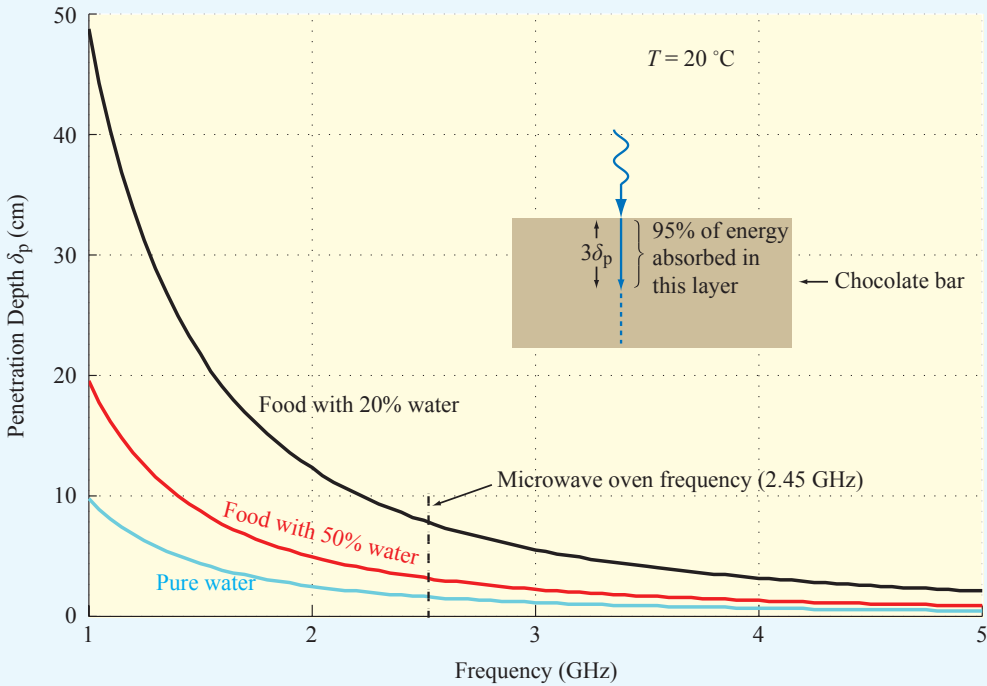


Figure TF3-1 Penetration depth as a function of frequency (1–5 GHz) for pure water and two foods with different water contents.

技术简介 3：微波炉

20 世纪 40 年代，珀西·斯宾塞 (Percy Spencer) 在雷神公司 (Raytheon) 从事雷达磁控管的设计和建造工作时，发现他口袋里的一块巧克力棒在无意中接触到微波后融化了。微波炉烹饪工艺于 1946 年获得专利，到 20 世纪 70 年代，微波炉已成为标准家庭用品。

微波吸收

微波是一种频率在 300 MHz 至 300 GHz 范围内的电磁波（见图 1-16）。当含水材料暴露于微波时，水分子通过自身旋转而发生反应，以便沿着微波振荡电场的方向排列其自身的电偶极子。快速振动运动在材料中产生热量，从而将微波能转化为热能。水的吸收系数 $\alpha(f)$ 呈现出取决于水温和溶解物浓度的微波光谱。

其中含有盐和糖。如果选择频率 f 使得 $\alpha(f)$ 较高，则含水材料吸收大量穿过它的微波能量并将其转化为热量。然而，这也意味着大部分能量被材料的薄表面层吸收，没有太多能量剩余来加热更深的层。材料的穿透深度 δ_p 定义为 $\delta_p = 1/2\alpha$ ，是电磁波携带的功率可以穿透材料多深的量度。入射到材料上的微波能量大约 95% 被厚度为 $3\delta_p$ 的表面层吸收。图 TF3-1 显示了纯水和两种不同含水量材料的 δ_p 计算光谱。

► 微波炉最常用的频率是 2.45 GHz。2.45 GHz 时， δ_p 的幅度在纯水的 ~2 cm 和含水量仅为 20% 的材料的 8 cm 之间变化。◀

这是一个实用的微波炉烹饪食物的系列；在低得多的频率下，食物并不是很好的能量吸收剂（此外，设计

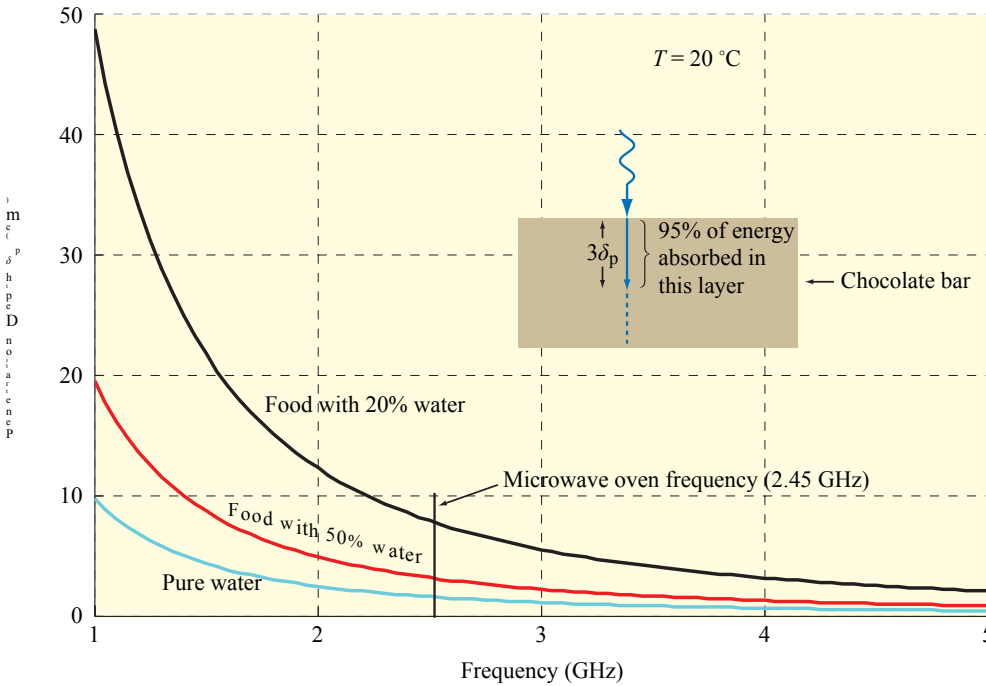


图 TF3-1 纯水和两种不同含水量食品的渗透深度与频率 (1–5 GHz) 的函数关系。

of the magnetron and the oven cavity become problematic), and at much higher frequencies, the microwave energy cooks the food very unevenly (mostly the surface layer). Whereas microwaves are readily absorbed by water, fats, and sugars, they can penetrate through most ceramics, glass, or plastics without loss of energy, thereby imparting little or no heat to those materials.

Oven Operation

To generate high-power microwaves (~700 watts), the microwave oven uses a **magnetron tube** (Fig. TF3-2), which requires the application of a voltage on the order of 4000 volts. The typical household voltage of 115 volts is increased to the required voltage level through a **high-voltage transformer**. The microwave energy generated by the magnetron is transferred into a cooking chamber designed to contain the microwaves within it through the use of metal surfaces and safety Interlock switches.

► Microwaves are reflected by metal surfaces, so they can bounce around the interior of the chamber, be absorbed by the food, but not escape to the outside. ◀

If the oven door is made of a glass panel, a **metal screen** or a layer of conductive mesh is attached to it to ensure the necessary shielding; microwaves cannot pass through the metal screen if the mesh width is much smaller than the wavelength of the microwave ($\lambda \approx 12\text{ cm}$ at 2.5 GHz). In the chamber, the microwave energy establishes a **standing-wave pattern**, which leads to an uneven distribution. This is mitigated by using a rotating **metal stirrer** that disperses the microwave energy to different parts of the chamber.

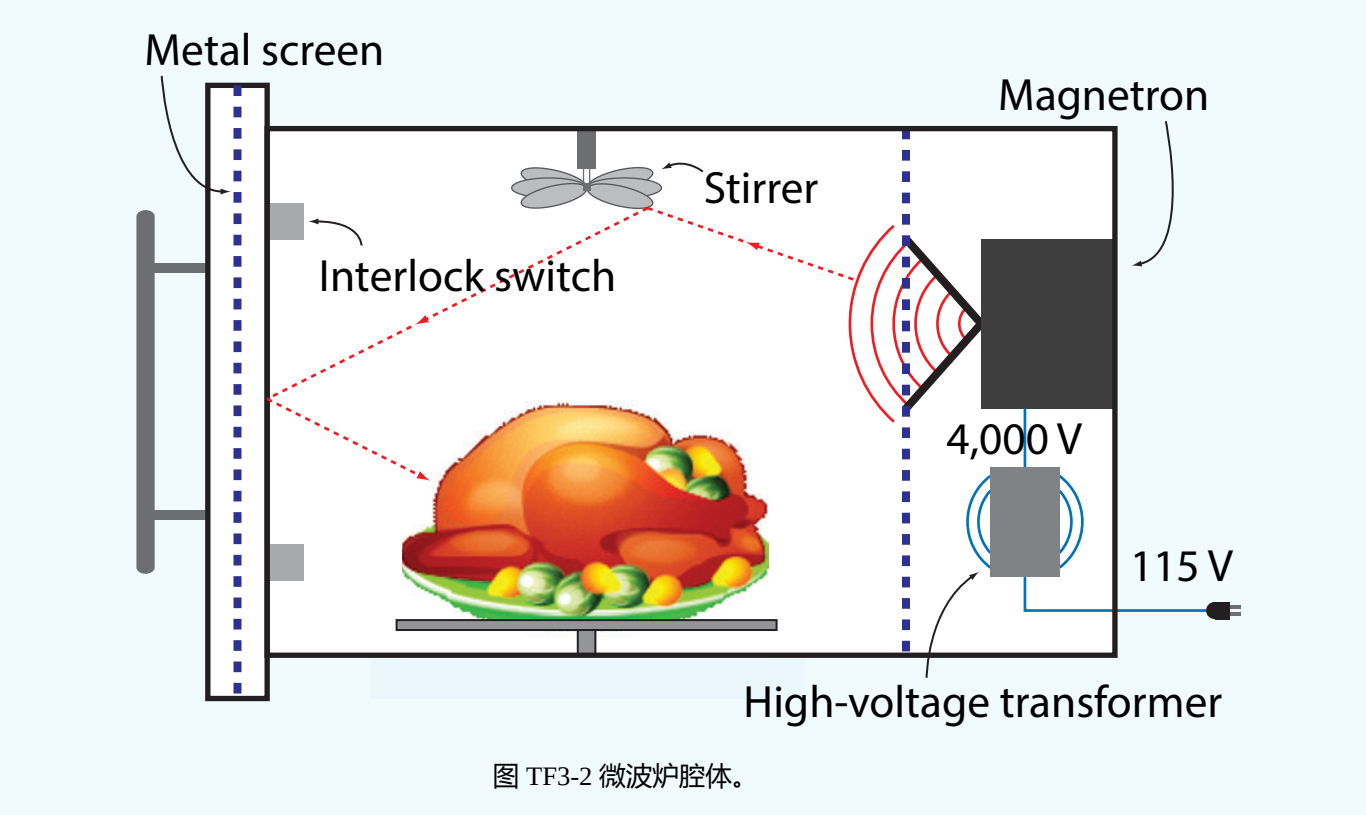
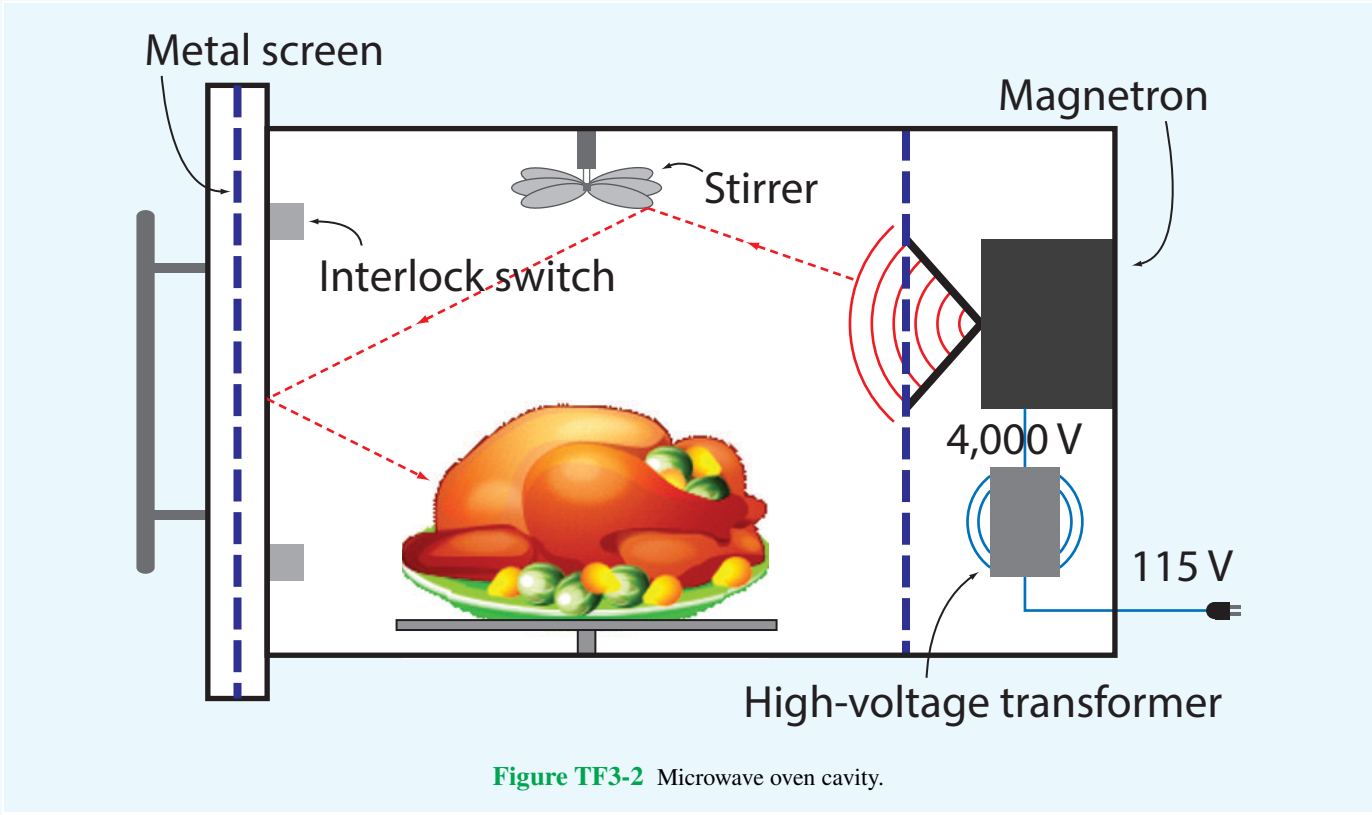
磁控管和烤箱腔体的内部结构会出现问题), 并且在更高的频率下, 微波能量对食物的烹饪非常不均匀 (主要是表面层)。虽然微波很容易被水、脂肪和糖吸收, 但它们可以穿透大多数陶瓷、玻璃或塑料而不会损失能量, 从而向这些材料传递很少或不传递热量。

► 微波会被金属表面反射, 因此它们可以在室内部反弹, 被食物吸收, 但不会逸出到外部。 ◀

烤箱操作

为了产生高功率微波 (~700瓦), 微波炉使用磁控管 (图TF3-2), 需要施加约4000伏的电压。典型的家庭电压为 115 伏, 通过高压变压器增加到所需的电压水平。磁控管产生的微波能量被转移到烹饪室中, 该烹饪室设计为通过使用金属表面和安全联锁开关将微波容纳在其中。

如果烤箱门由玻璃面板制成, 则在其上附有金属丝网或一层导电网, 以确保必要的屏蔽; 如果网孔宽度远小于微波的波长 (2.5 GHz 时 $\lambda \approx 12\text{ cm}$), 微波就无法穿过金属筛网。在腔室中, 微波能量形成驻波模式, 从而导致分布不均匀。通过使用旋转金属搅拌器将微波能量分散到腔室的不同部分可以减轻这种情况。



2-9 Power Flow on a Lossless Transmission Line

Our discussion thus far has focused on the voltage and current attributes of waves propagating on a transmission line. Now we examine the flow of power carried by the incident and reflected waves. We begin by reintroducing Eqs. (2.63a) and (2.63b) with $z = -d$:

$$\tilde{V}(d) = V_0^+(e^{j\beta d} + \Gamma e^{-j\beta d}), \quad (2.98a)$$

$$\tilde{I}(d) = \frac{V_0^+}{Z_0}(e^{j\beta d} - \Gamma e^{-j\beta d}). \quad (2.98b)$$

In these expressions, the first terms represent the incident-wave voltage and current, and the terms involving Γ represent the reflected-wave voltage and current. The time-domain expressions for the voltage and current at location d from the load are obtained by transforming Eq. (2.98) to the time domain:

$$\begin{aligned} v(d, t) &= \Re[\tilde{V}e^{j\omega t}] \\ &= \Re[|V_0^+|e^{j\phi^+}(e^{j\beta d} + |\Gamma|e^{j\theta_r}e^{-j\beta d})e^{j\omega t}] \\ &= |V_0^+|[\cos(\omega t + \beta d + \phi^+) \\ &\quad + |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)], \end{aligned} \quad (2.99a)$$

$$\begin{aligned} i(d, t) &= \frac{|V_0^+|}{Z_0}[\cos(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)], \end{aligned} \quad (2.99b)$$

where we used the relations $V_0^+ = |V_0^+|e^{j\phi^+}$ and $\Gamma = |\Gamma|e^{j\theta_r}$ that were introduced earlier as Eqs. (2.31a) and (2.62), respectively.

2-9.1 Instantaneous Power

The instantaneous power carried by the transmission line is equal to the product of $v(d, t)$ and $i(d, t)$:

$$\begin{aligned} P(d, t) &= v(d, t)i(d, t) \\ &= |V_0^+|[\cos(\omega t + \beta d + \phi^+) \\ &\quad + |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)] \\ &\quad \times \frac{|V_0^+|}{Z_0}[\cos(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)] \\ &= \frac{|V_0^+|^2}{Z_0}[\cos^2(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|^2\cos^2(\omega t - \beta d + \phi^+ + \theta_r)] \quad (\text{W}). \end{aligned} \quad (2.100)$$

Per our earlier discussion in connection with Eq. (1.31), if the signs preceding ωt and βd in the argument of the cosine term are both positive or both negative, then the cosine term represents a wave traveling in the negative d direction. Since d points from the load to the generator, the first term in Eq. (2.100) represents the **instantaneous incident power** traveling towards the load. This is the power that would be delivered to the load in the absence of wave reflection (when $\Gamma = 0$). Because βd is preceded by a minus sign in the argument of the cosine of the second term in Eq. (2.100), that term represents the **instantaneous reflected power** traveling in the $+d$ direction—away from the load. Accordingly, we label these two power components

$$P^i(d, t) = \frac{|V_0^+|^2}{Z_0}\cos^2(\omega t + \beta d + \phi^+) \quad (\text{W}), \quad (2.101a)$$

$$P^r(d, t) = -|\Gamma|^2\frac{|V_0^+|^2}{Z_0}\cos^2(\omega t - \beta d + \phi^+ + \theta_r) \quad (\text{W}). \quad (2.101b)$$

Using the trigonometric identity

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x),$$

the expressions in Eq. (2.101) can be rewritten as

$$P^i(d, t) = \frac{|V_0^+|^2}{2Z_0}[1 + \cos(2\omega t + 2\beta d + 2\phi^+)], \quad (2.102a)$$

$$P^r(d, t) = -|\Gamma|^2\frac{|V_0^+|^2}{2Z_0}[1 + \cos(2\omega t - 2\beta d + 2\phi^+ + 2\theta_r)]. \quad (2.102b)$$

We note that in each case, the instantaneous power consists of a dc term (not varying in time) and an ac term that oscillates at an angular frequency of 2ω .

► The power oscillates at twice the rate of the voltage or current. ◀

2-9.2 Time-Average Power

From a practical standpoint, we usually are more interested in the time-average power flowing along the transmission line, $P_{\text{av}}(d)$, than in the instantaneous power $P(d, t)$. To compute $P_{\text{av}}(d)$, we can use a time-domain approach or a computationally simpler phasor-domain approach. For completeness, we consider both.

2-9 无损传输线上的功率流

到目前为止，我们的讨论主要集中在传输线上传播的波的电压和电流属性。现在我们研究入射波和反射波所携带的能量流。我们首先重新引入方程。(2.63a)和(2.63b)与 $z = -d$:

$$\tilde{V}(d) = V_0^+(e^{j\beta d} + \Gamma e^{-j\beta d}), \quad (2.98a)$$

$$\tilde{I}(d) = \frac{V_0^+}{Z_0}(e^{j\beta d} - \Gamma e^{-j\beta d}). \quad (2.98b)$$

在这些表达式中，第一项代表入射波电压和电流，涉及 Γ 的项代表反射波电压和电流。通过对式 (1) 进行变换，得到负载 d 处电压和电流的时域表达式。(2.98) 到 时域:

$$\begin{aligned} v(d, t) &= \Re[\tilde{V}e^{j\omega t}] \\ &= \Re[|V_0^+|e^{j\phi^+}(e^{j\beta d} + |\Gamma|e^{j\theta_r}e^{-j\beta d})e^{j\omega t}] \\ &= |V_0^+|[\cos(\omega t + \beta d + \phi^+) \\ &\quad + |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)], \end{aligned} \quad (2.99a)$$

$$\begin{aligned} i(d, t) &= \frac{|V_0^+|}{Z_0}[\cos(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)], \end{aligned} \quad (2.99b)$$

其中我们使用了之前作为等式引入的关系 $V_0^+ = |V_0^+|e^{j\phi^+}$ 和 $\Gamma = |\Gamma|e^{j\theta_r}$ 。分别为 (2.31a) 和 (2.62)。

2-9.1 瞬时功率

传输线承载的瞬时功率等于 $v(d, t)$ 和 $i(d, t)$ 的乘积:

$$\begin{aligned} P(d, t) &= v(d, t)i(d, t) \\ &= |V_0^+|[\cos(\omega t + \beta d + \phi^+) \\ &\quad + |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)] \\ &\quad \times \frac{|V_0^+|}{Z_0}[\cos(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|\cos(\omega t - \beta d + \phi^+ + \theta_r)] \\ &= \frac{|V_0^+|^2}{Z_0}[\cos^2(\omega t + \beta d + \phi^+) \\ &\quad - |\Gamma|^2\cos^2(\omega t - \beta d + \phi^+ + \theta_r)] \quad (\text{W}). \end{aligned} \quad (2.100)$$

根据我们之前关于方程式的讨论。(1.31)，如果余弦项的自变量中 ωt 和 βd 之前的符号均为正或均为负，则余弦项表示沿 $-d$ 方向传播的波。由于 d 从负载指向发电机，因此方程中的第一项。(2.100) 表示流向负载的瞬时入射功率。这是在没有波反射的情况下（当 $\Gamma = 0$ 时）传递到负载的功率。因为在方程第二项的余弦参数中， βd 前面有一个负号。(2.100)，该术语表示沿 $+d$ 方向（远离负载）传播的瞬时反射功率。因此，我们将这两个功率组件标记为

$$P^i(d, t) = \frac{|V_0^+|^2}{Z_0}\cos^2(\omega t + \beta d + \phi^+) \quad (\text{W}), \quad (2.101a)$$

$$P^r(d, t) = -|\Gamma|^2\frac{|V_0^+|^2}{Z_0}\cos^2(\omega t - \beta d + \phi^+ + \theta_r) \quad (\text{W}). \quad (2.101b)$$

使用三角恒等式

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x),$$

方程中的表达式(2.101) 可以重写为

$$P^i(d, t) = \frac{|V_0^+|^2}{2Z_0}[1 + \cos(2\omega t + 2\beta d + 2\phi^+)], \quad (2.102a)$$

$$P^r(d, t) = -|\Gamma|^2\frac{|V_0^+|^2}{2Z_0}[1 + \cos(2\omega t - 2\beta d + 2\phi^+ + 2\theta_r)]. \quad (2.102b)$$

我们注意到，在每种情况下，瞬时功率由直流项（不随时间变化）和以 2ω 角频率振荡的交流项组成。

► 功率以电压或电流速率的两倍振荡。 ◀

2-9.2 时间平均功率

从实际角度来看，我们通常对沿传输线流动的时间平均功率 $P_{\text{av}}(d)$ 更感兴趣，而不是瞬时功率 $P(d, t)$ 。为了计算 $P_{\text{av}}(d)$ ，我们可以使用时域方法或计算更简单的相量域方法。为了完整起见，我们同时考虑两者。

Time-Domain Approach

The time-average power is equal to the instantaneous power averaged over one time period $T = 1/f = 2\pi/\omega$. For the incident wave, its time-average power is

$$P_{av}^i(d) = \frac{1}{T} \int_0^T P^i(d,t) dt = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} P^i(d,t) dt. \quad (2.103)$$

Upon inserting Eq. (2.102a) into Eq. (2.103) and performing the integration, we obtain

$$P_{av}^i = \frac{|V_0^+|^2}{2Z_0} \quad (\text{W}), \quad (2.104)$$

which is identical with the dc term of $P^i(d,t)$ given by Eq. (2.102a). A similar treatment for the reflected wave gives

$$P_{av}^r = -|\Gamma|^2 \frac{|V_0^+|^2}{2Z_0} = -|\Gamma|^2 P_{av}^i. \quad (2.105)$$

► The average reflected power is equal to the average incident power, diminished by a multiplicative factor of $|\Gamma|^2$. ◀

Note that the expressions for P_{av}^i and P_{av}^r are independent of d , which means that the time-average powers carried by the incident and reflected waves do not change as they travel along the transmission line. This is as expected, because the transmission line is lossless.

The **net average power flowing towards (and then absorbed by) the load** shown in Fig. 2-23 is

$$P_{av} = P_{av}^i + P_{av}^r = \frac{|V_0^+|^2}{2Z_0} [1 - |\Gamma|^2] \quad (\text{W}). \quad (2.106)$$

Phasor-Domain Approach

For any propagating wave with voltage and current phasors $\tilde{V}$ and $\tilde{I}$, a useful formula for computing the time-average power is

$$P_{av} = \frac{1}{2} \Re [\tilde{V} \cdot \tilde{I}^*], \quad (2.107)$$

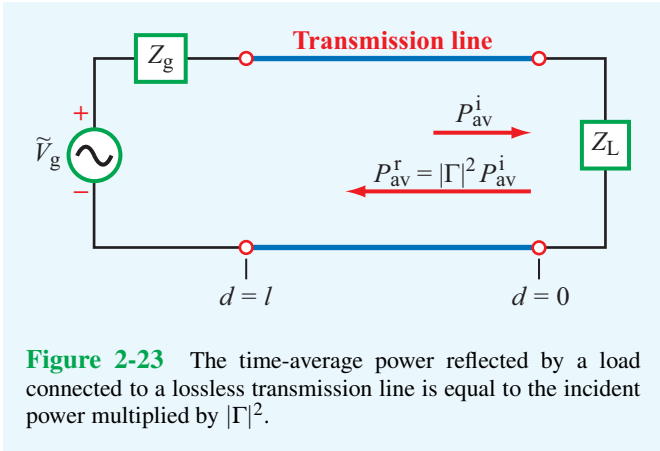


Figure 2-23 The time-average power reflected by a load connected to a lossless transmission line is equal to the incident power multiplied by $|\Gamma|^2$.

where $\tilde{I}^*$ is the complex conjugate of $\tilde{I}$. Application of this formula to Eqs. (2.98a) and (2.98b) gives

$$\begin{aligned} P_{av} &= \frac{1}{2} \Re \left[V_0^+ (e^{j\beta d} + \Gamma e^{-j\beta d}) \frac{V_0^{+*}}{Z_0} (e^{-j\beta d} - \Gamma^* e^{j\beta d}) \right] \\ &= \frac{1}{2} \Re \left[\frac{|V_0^+|^2}{Z_0} (1 - |\Gamma|^2 + \Gamma e^{-j2\beta d} - \Gamma^* e^{j2\beta d}) \right] \\ &= \frac{|V_0^+|^2}{2Z_0} \left\{ [1 - |\Gamma|^2] + \Re [\Gamma |e^{-j(2\beta d - \theta_r)} - \Gamma^* e^{j(2\beta d - \theta_r)}] \right\} \\ &= \frac{|V_0^+|^2}{2Z_0} \left\{ [1 - |\Gamma|^2] + |\Gamma| [\cos(2\beta d - \theta_r) - \cos(2\beta d - \theta_r)] \right\} \\ &= \frac{|V_0^+|^2}{2Z_0} [1 - |\Gamma|^2], \end{aligned} \quad (2.108)$$

which is identical to Eq. (2.106).

Exercise 2-17: For a $50 \, \Omega$ lossless transmission line terminated in a load impedance $Z_L = (100 + j50) \, \Omega$, determine the fraction of the average incident power reflected by the load.
Answer: 20%. (See [EM](#).)

Exercise 2-18: For the line of Exercise 2-17, what is the magnitude of the average reflected power if $|V_0^+| = 1 \, \text{V}$?
Answer: $P_{av}^r = 2 \, (\text{mW})$. (See [EM](#).)

Concept Question 2-15: According to Eq. (2.102b), the instantaneous value of the reflected power depends on the phase of the reflection coefficient θ_r , but the average reflected power given by Eq. (2.105) does not. Explain.

时域方法

时间平均功率等于一个时间段 $T = 1/f = 2\pi/\omega$ 内平均的瞬时功率。对于入射波，其时均功率为

$$P_{av}^i(d) = \frac{1}{T} \int_0^T P^i(d,t) dt = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} P^i(d,t) dt. \quad (2.103)$$

插入方程后，(2.102a) 代入方程。(2.103) 并进行积分，我们得到

$$P_{av}^i = \frac{|V_0^+|^2}{2Z_0} \quad (\text{W}), \quad (2.104)$$

这与式 (1) 给出的 $P^i(d,t)$ 的直流项相同。(2.102a) 对反射波的类似处理给出

$$P_{av}^r = -|\Gamma|^2 \frac{|V_0^+|^2}{2Z_0} = -|\Gamma|^2 P_{av}^i. \quad (2.105)$$

► 平均反射功率等于平均入射功率减去乘法因子 $|\Gamma|^2$ 。
◀

请注意， P_{av}^i 和 P_{av}^r 的表达式与 d 无关，这意味着入射波和反射波所携带的时间平均功率在沿传输线传播时不会改变。这是符合预期的，因为传输线是无损的。

流向负载（然后被负载吸收）的净平均功率如图 2-23 所示：

$$P_{av} = P_{av}^i + P_{av}^r = \frac{|V_0^+|^2}{2Z_0} [1 - |\Gamma|^2] \quad (\text{W}). \quad (2.106)$$

相量域方法

对于任何具有电压和电流相量 $\tilde{V}$ 和 $\tilde{I}$ 的传播波，计算时间平均功率的有用公式为

$$P_{av} = \frac{1}{2} \Re [\tilde{V} \cdot \tilde{I}^*], \quad (2.107)$$

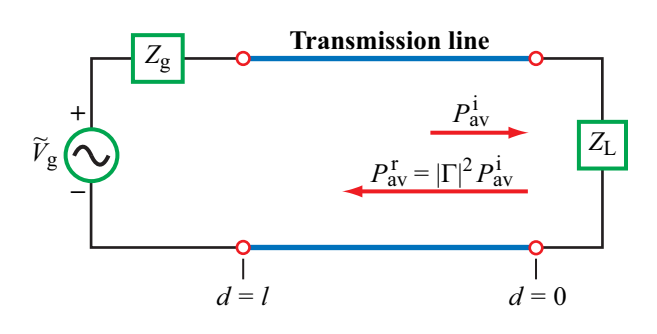


图2-23 连接到无损传输线的负载反射的时间平均功率等于入射功率乘以 $|\Gamma|^2$ 。

其中 $\tilde{I}^*$ 是 $\tilde{I}$ 的复共轭。将此公式应用于方程式。(2.98a) 和 (2.98b) 给出

$$\begin{aligned} P_{av} &= \frac{1}{2} \Re \left[V_0^+ (e^{j\beta d} + \Gamma e^{-j\beta d}) \frac{V_0^{+*}}{Z_0} (e^{-j\beta d} - \Gamma^* e^{j\beta d}) \right] \\ &= \frac{1}{2} \Re \left[\frac{|V_0^+|^2}{Z_0} (1 - |\Gamma|^2 + \Gamma e^{-j2\beta d} - \Gamma^* e^{j2\beta d}) \right] \\ &= \frac{|V_0^+|^2}{2Z_0} \left\{ [1 - |\Gamma|^2] + \Re [\Gamma |e^{-j(2\beta d - \theta_r)} - \Gamma^* e^{j(2\beta d - \theta_r)}] \right\} \\ &= \frac{|V_0^+|^2}{2Z_0} \left\{ [1 - |\Gamma|^2] + |\Gamma| [\cos(2\beta d - \theta_r) - \cos(2\beta d - \theta_r)] \right\} \\ &= \frac{|V_0^+|^2}{2Z_0} [1 - |\Gamma|^2], \end{aligned} \quad (2.108)$$

这与等式相同。(2.106)。

练习 2-17: 对于终止于负载阻抗 $Z_L = (100 + j50) \, \Omega$ 的 $50 \, \Omega$ 无损传输线，确定负载反射的平均入射功率的分数。

答案: 20%。(参见 [EM](#)。)

练习 2-18: 对于练习 2-17 的直线，如果 $|V_0^+| = 1 \, \text{V}$ ，平均反射功率的大小是多少？

Answer: $P_{av}^r = 2 \, (\text{mW})$. (See [EM](#).)

概念问题 2-15: 根据方程式(2.102b)中，反射功率的瞬时值取决于反射系数 θ_r 的相位，但平均反射功率由式(2.102b)给出。(2.105) 则不然。解释。

Concept Question 2-16: What is the average power delivered by a lossless transmission line to a reactive load?

Concept Question 2-17: What fraction of the incident power is delivered to a matched load?

Concept Question 2-18: Verify that

$$\frac{1}{T} \int_0^T \cos^2 \left(\frac{2\pi t}{T} + \beta d + \phi \right) dt = \frac{1}{2},$$

regardless of the values of d and ϕ , so long as neither is a function of t .

2-10 The Smith Chart

The **Smith chart**, developed by P. H. Smith in 1939, is a widely used graphical tool for analyzing and designing transmission-line circuits. Even though it was originally intended to facilitate calculations involving complex impedances, the Smith chart has become an important avenue for comparing and characterizing the performance of microwave circuits. As the material in this and the next section demonstrates, use of the Smith chart not only avoids tedious manipulations of complex numbers, but it also allows an engineer to design impedance-matching circuits with relative ease.

2-10.1 Parametric Equations

The reflection coefficient Γ is, in general, a complex quantity composed of a magnitude $|\Gamma|$ and a phase angle θ_Γ or, equivalently, a real part Γ_r and an imaginary part Γ_i ,

$$\Gamma = |\Gamma|e^{j\theta_\Gamma} = \Gamma_r + j\Gamma_i, \quad (2.109)$$

where

$$\Gamma_r = |\Gamma| \cos \theta_\Gamma, \quad (2.110a)$$

$$\Gamma_i = |\Gamma| \sin \theta_\Gamma. \quad (2.110b)$$

The Smith chart lies in the complex Γ plane. In **Fig. 2-24**, point A represents a reflection coefficient $\Gamma_A = 0.3 + j0.4$ or, equivalently,

$$|\Gamma_A| = [(0.3)^2 + (0.4)^2]^{1/2} = 0.5$$

and

$$\theta_{r_A} = \tan^{-1}(0.4/0.3) = 53^\circ.$$

Similarly, point B represents $\Gamma_B = -0.5 - j0.2$, or $|\Gamma_B| = 0.54$ and $\theta_{r_B} = 202^\circ$ or, equivalently,

$$\theta_{r_B} = (360^\circ - 202^\circ) = -158^\circ.$$

► When both Γ_r and Γ_i are negative, θ_Γ is in the third quadrant in the Γ_r – Γ_i plane. Thus, when using $\theta_\Gamma = \tan^{-1}(\Gamma_i/\Gamma_r)$ to compute θ_Γ , it may be necessary to add or subtract 180° to obtain the correct value of θ_Γ . ◀

The **unit circle** shown in **Fig. 2-24** corresponds to $|\Gamma| = 1$. Because $|\Gamma| \leq 1$ for a transmission line terminated with a passive load, only that part of the Γ_r – Γ_i plane that lies within the unit circle is useful to us; hence, future drawings will be limited to the domain contained within the unit circle.

Impedances on a Smith chart are represented by their values normalized to the line’s characteristic impedance, Z_0 . From

$$\begin{aligned} \Gamma &= \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} \\ &= \frac{z_L - 1}{z_L + 1}, \end{aligned} \quad (2.111)$$

the inverse relation is

$$z_L = \frac{1 + \Gamma}{1 - \Gamma}. \quad (2.112)$$

The normalized load impedance z_L is, in general, a complex quantity composed of a **normalized load resistance** r_L and a **normalized load reactance** x_L :

$$z_L = r_L + jx_L. \quad (2.113)$$

Using Eqs. (2.109) and (2.113) in Eq. (2.112), we have

$$r_L + jx_L = \frac{(1 + \Gamma_r) + j\Gamma_i}{(1 - \Gamma_r) - j\Gamma_i}, \quad (2.114)$$

which can be manipulated to obtain explicit expressions for each of r_L and x_L in terms of Γ_r and Γ_i . This is accomplished by multiplying the numerator and denominator of the right-hand side of Eq. (2.114) by the complex conjugate of the denominator and then separating the result into real and imaginary parts. These steps lead to

$$r_L = \frac{1 - \Gamma_r^2 - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2}, \quad (2.115a)$$

$$x_L = \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}. \quad (2.115b)$$

概念问题 2-16：无损传输线向无功负载传送的平均功率是多少？

概念问题 2-17：入射功率的哪一部分传输至匹配负载？

概念问题 2-18：验证

$$\frac{1}{T} \int_0^T \cos^2 \left(\frac{2\pi t}{T} + \beta d + \phi \right) dt = \frac{1}{2},$$

无论 d 和 ϕ 的值如何，只要两者都不是 t 的函数。

2-10 史密斯圆图

史密斯圆图由 P. H. Smith 于 1939 年开发，是一种广泛使用的用于分析和设计传输线电路的图形工具。尽管史密斯圆图最初的目的是促进涉及复数阻抗的计算，但它已成为比较和表征微波电路性能的重要途径。正如本节和下一节中的材料所示，史密斯圆图的使用不仅避免了对复数的繁琐操作，而且还允许工程师相对轻松地设计阻抗匹配电路。

2-10.1 参数方程

一般来说，反射系数 Γ 是一个由幅度 $|\Gamma|$ 和相位角 θ_Γ 组成的复数，或者等效地，由实部 Γ_r 和虚部 Γ_i 组成，

$$\Gamma = |\Gamma|e^{j\theta_\Gamma} = \Gamma_r + j\Gamma_i, \quad (2.109)$$

在哪里

$$\Gamma_r = |\Gamma| \cos \theta_\Gamma, \quad (2.110a)$$

$$\Gamma_i = |\Gamma| \sin \theta_\Gamma. \quad (2.110b)$$

史密斯圆图位于复数 Γ 平面中。在图 2-24 中，点 A 代表反射系数 $\Gamma_A = 0.3 + j0.4$ 或等效地，

$$|\Gamma_A| = [(0.3)^2 + (0.4)^2]^{1/2} = 0.5$$

和

$$\theta_{r_A} = \tan^{-1}(0.4/0.3) = 53^\circ.$$

类似地，点 B 表示 $\Gamma_B = -0.5 - j0.2$ ，或 $|\Gamma_B| = 0.54$ 和 $\theta_{r_B} = 202^\circ$ ，或者等效地，

$$\theta_{r_B} = (360^\circ - 202^\circ) = -158^\circ.$$

► 当 Γ_r 和 Γ_i 均为负数时， θ_Γ 位于 Γ_r – Γ_i 平面的第三象限。因此，当使用 $\theta_\Gamma = \tan^{-1}(\Gamma_i/\Gamma_r)$ 计算 θ_Γ 时，可能需要加上或减去 180° 才能获得正确的 θ_Γ 值。◀

图 2-24 所示的单位圆对应于 $|\Gamma| = 1$ 。因为 $|\Gamma| \leq 1$ 对于以无源负载端接的传输线来说，只有位于单位圆内的 Γ_r – Γ_i 平面的部分对我们有用；因此，未来的绘图将仅限于单位圆内包含的域。

史密斯圆图上的阻抗由其归一化为线路特征阻抗 Z_0 的值表示。从

$$\begin{aligned} \Gamma &= \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} \\ &= \frac{z_L - 1}{z_L + 1}, \end{aligned} \quad (2.111)$$

反比关系是

$$z_L = \frac{1 + \Gamma}{1 - \Gamma}. \quad (2.112)$$

归一化负载阻抗 z_L 通常是由归一化负载电阻 r_L 和归一化负载电抗 x_L 组成的复数：

$$z_L = r_L + jx_L. \quad (2.113)$$

使用方程式。式 (2.109) 和 (2.113) (2.112)，我们有

$$r_L + jx_L = \frac{(1 + \Gamma_r) + j\Gamma_i}{(1 - \Gamma_r) - j\Gamma_i}, \quad (2.114)$$

可以对其进行操作以获得 r_L 和 x_L 中每个以 Γ_r 和 Γ_i 表示的显式表达式。这是通过将等式右侧的分子和分母相乘来实现的。(2.114) 通过分母的复共轭，然后将结果分为实部和虚部。这些步骤导致

$$r_L = \frac{1 - \Gamma_r^2 - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2}, \quad (2.115a)$$

$$x_L = \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}. \quad (2.115b)$$

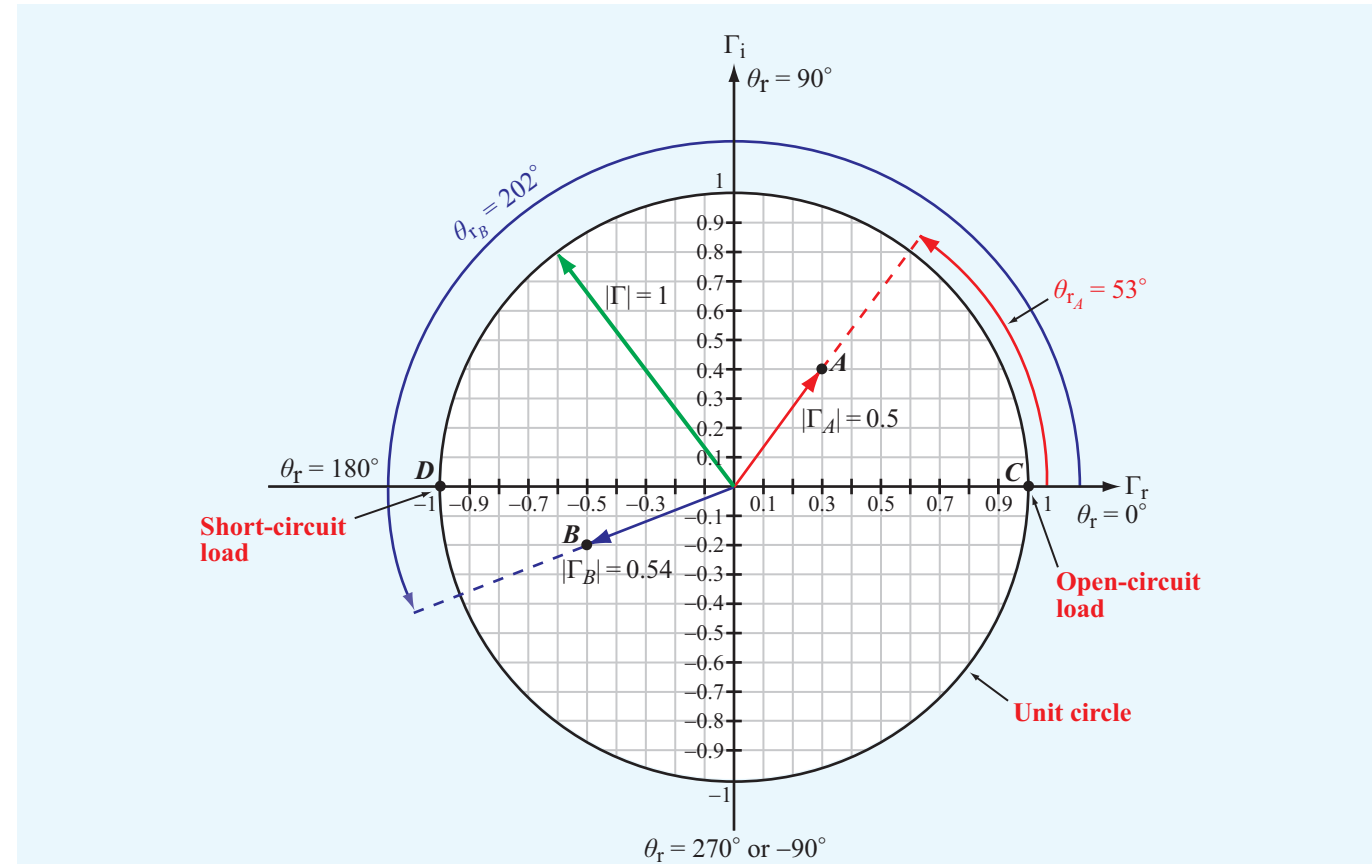


Figure 2-24 The complex Γ plane. Point A is at $\Gamma_A = 0.3 + j0.4 = 0.5e^{j53^\circ}$, and point B is at $\Gamma_B = -0.5 - j0.2 = 0.54e^{j202^\circ}$. The unit circle corresponds to $|\Gamma| = 1$. At point C, $\Gamma = 1$, corresponding to an open-circuit load, and at point D, $\Gamma = -1$, corresponding to a short circuit.

Equation (2.115a) implies that there exist many combinations of values for Γ_r and Γ_i that yield the same value for the normalized load resistance r_L . For example, $(\Gamma_r, \Gamma_i) = (0.33, 0)$ gives $r_L = 2$, as does $(\Gamma_r, \Gamma_i) = (0.5, 0.29)$, as well as an infinite number of other combinations. In fact, if we were to plot in the Γ_r - Γ_i plane all possible combinations of Γ_r and Γ_i corresponding to $r_L = 2$, we would obtain the circle labeled $r_L = 2$ in **Fig. 2-25**. Similar circles can be obtained for other values of r_L . After some algebraic manipulations, Eq. (2.115a) can be rearranged into the **parametric equation** for the circle in the Γ_r - Γ_i plane corresponding to a given value of r_L :

$$\left(\Gamma_r - \frac{r_L}{1+r_L}\right)^2 + \Gamma_i^2 = \left(\frac{1}{1+r_L}\right)^2. \quad (2.116)$$

The standard equation for a circle in the x - y plane with center at (x_0, y_0) and radius a is

$$(x - x_0)^2 + (y - y_0)^2 = a^2. \quad (2.117)$$

Comparison of Eq. (2.116) with Eq. (2.117) shows that the r_L circle is centered at $\Gamma_r = r_L/(1+r_L)$ and $\Gamma_i = 0$, and its radius is $1/(1+r_L)$. It therefore follows that all r_L -circles pass through the point $(\Gamma_r, \Gamma_i) = (1, 0)$. The largest circle shown in **Fig. 2-25** is for $r_L = 0$, which also is the unit circle corresponding to $|\Gamma| = 1$. This is to be expected, because when $r_L = 0$, $|\Gamma| = 1$ regardless of the magnitude of x_L .

A similar manipulation of the expression for x_L given by Eq. (2.115b) leads to

$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L}\right)^2 = \left(\frac{1}{x_L}\right)^2, \quad (2.118)$$

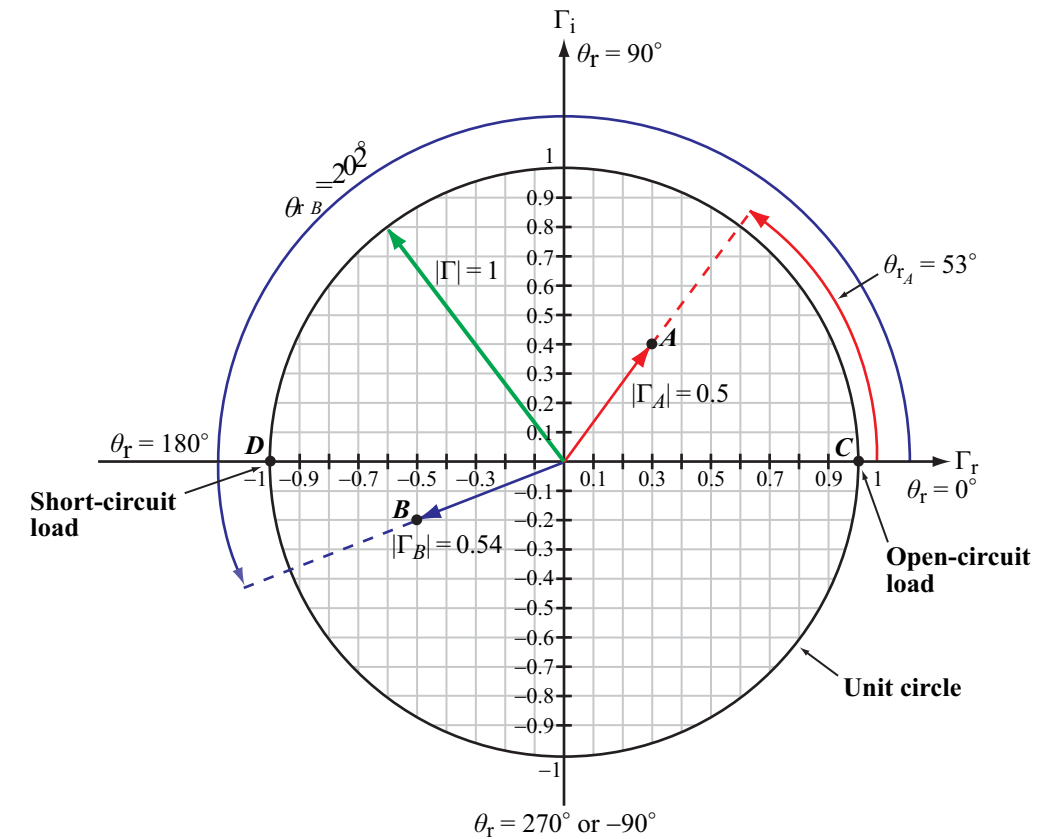


图 2-24 复数 Γ 平面。A 点位于 $\Gamma_A = 0.3 + j0.4 = 0.5e^{j53^\circ}$ 处，B 点位于 $\Gamma_B = -0.5 - j0.2 = 0.54e^{j202^\circ}$ 处。单位圆对应 $|\Gamma| = 1$ 。在 C 点， $\Gamma = 1$ ，对应开路负载，在 D 点， $\Gamma = -1$ ，对应短路负载。

方程 (2.115a) 意味着存在许多 Γ_r 和 Γ_i 值的组合，它们产生相同的归一化负载电阻 r_L 值。例如， $(\Gamma_r, \Gamma_i) = (0.33, 0)$ 给出 $r_L = 2$ ， $(\Gamma_r, \Gamma_i) = (0.5, 0.29)$ 以及无数其他组合也是如此。事实上，如果我们在 Γ_r - Γ_i 平面上绘制对应于 $r_L = 2$ 的 Γ_r 和 Γ_i 的所有可能组合，我们将得到图 2-25 中标记为 $r_L = 2$ 的圆。对于 r_L 的其他值也可以获得类似的圆。经过一些代数运算后，方程 (2.115a) 可以重新整理成对应于给定 r_L 值的 Γ_r - Γ_i 平面中圆的参数方程：

$$\left(\Gamma_r - \frac{r_L}{1+r_L}\right)^2 + \Gamma_i^2 = \left(\frac{1}{1+r_L}\right)^2. \quad (2.116)$$

x - y 平面上以 (x_0, y_0) 为圆心、半径为 a 的圆的标准方程为

$$(x - x_0)^2 + (y - y_0)^2 = a^2. \quad (2.117)$$

等式的比较(2.116)与等式 (2.117) 表明 r_L 圆以 $\Gamma_r = r_L/(1+r_L)$ 和 $\Gamma_i = 0$ 为圆心，半径为 $1/(1+r_L)$ 。因此，所有 r_L 圆都经过点 $(\Gamma_r, \Gamma_i) = (1, 0)$ 。图2-25所示的最大圆是 $r_L = 0$ 时的，它也是对应于 $|\Gamma| = 1$ 的单位圆。这是可以预料的，因为当 $r_L = 0$ 时，无论 x_L 的大小如何， $|\Gamma|$ 都是 1。

对等式给出的 x_L 表达式进行类似的操作。(2.115b) 导致

$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L}\right)^2 = \left(\frac{1}{x_L}\right)^2, \quad (2.118)$$

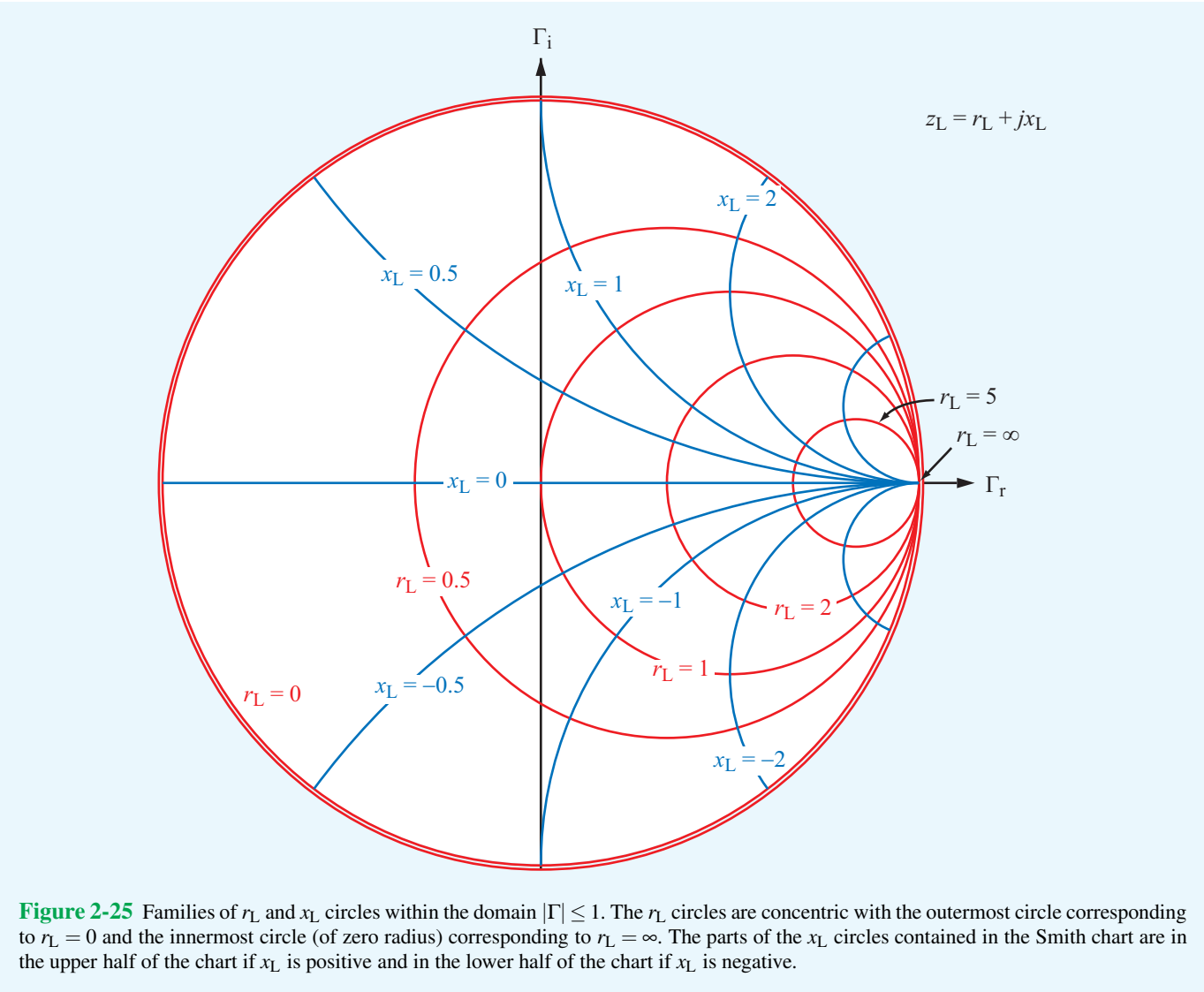


Figure 2-25 Families of r_L and x_L circles within the domain $|\Gamma| \leq 1$. The r_L circles are concentric with the outermost circle corresponding to $r_L = 0$ and the innermost circle (of zero radius) corresponding to $r_L = \infty$. The parts of the x_L circles contained in the Smith chart are in the upper half of the chart if x_L is positive and in the lower half of the chart if x_L is negative.

which is the equation of a circle of radius $(1/x_L)$ centered at $(\Gamma_r, \Gamma_i) = (1, 1/x_L)$. The x_L circles in the Γ_r - Γ_i plane are quite different from those for constant r_L . To start with, the normalized reactance x_L may assume both positive and negative values, whereas the normalized resistance cannot be negative (negative resistances cannot be realized in passive circuits). Hence, Eq. (2.118) represents two families of circles: one for positive values of x_L and another for negative ones. Furthermore, as shown in **Fig. 2-25**, only part of a given circle falls within the bounds of the $|\Gamma| = 1$ unit circle.

The families of circles of the two parametric equations given by Eqs. (2.116) and (2.118) plotted for selected values of r_L and x_L constitute the Smith chart shown in **Fig. 2-26**. The

Smith chart provides a graphical evaluation of Eqs. (2.115a and b) and their inverses.

► Each point in the Smith chart conveys the values of two interrelated quantities, namely z_L and Γ . The intersection of the r_L and x_L circles at that point defines $z_L = r_L + jx_L$. Simultaneously, the location of the point defines $|\Gamma|$ and θ_r . ◀

For example, point P in **Fig. 2-26** represents a normalized load impedance $z_L = 2 - j1$, which corresponds to a voltage reflection coefficient $\Gamma = 0.45 \exp(-j26.6^\circ)$. The magnitude

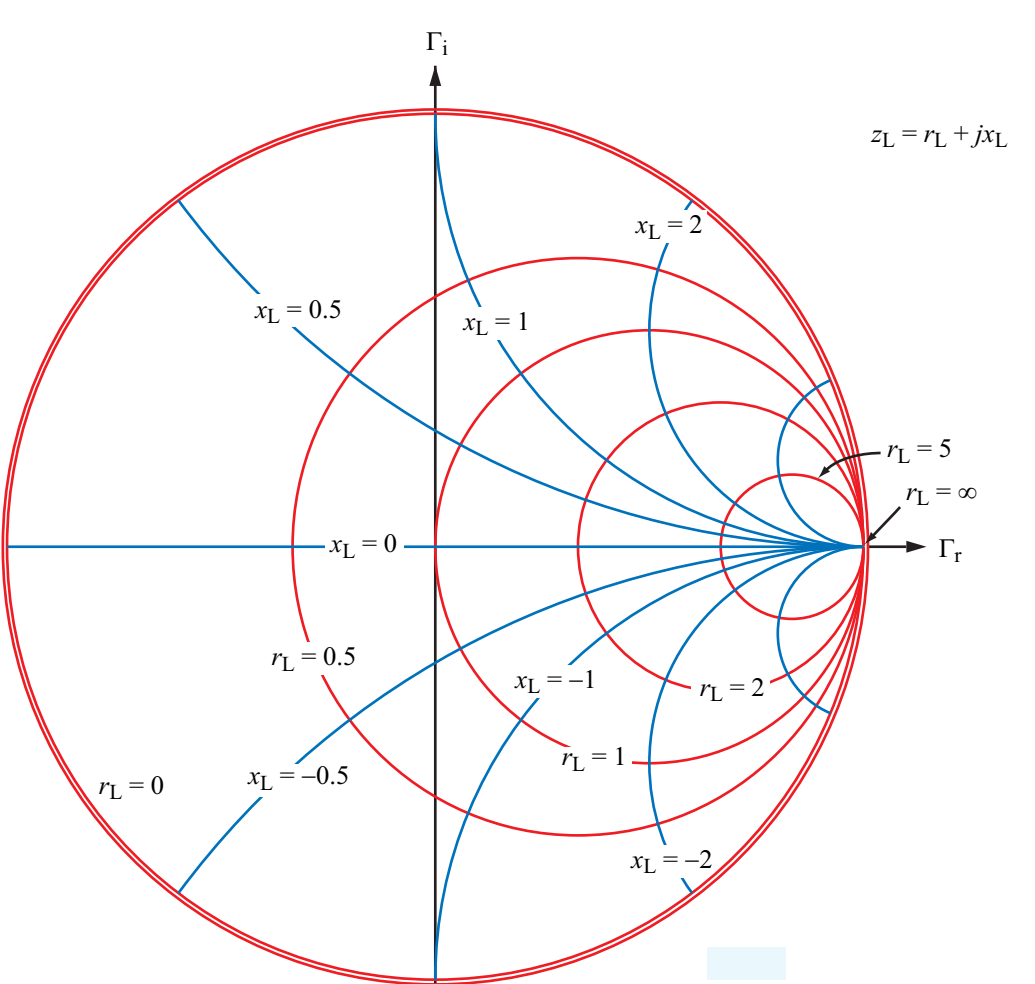


图 2-25 $|\Gamma| \leq 1$ 域内的 r_L 和 x_L 圆族 1. r_L 圆与对应于 $r_L = 0$ 的最外圆和对应于 $r_L = \infty$ 的最内圆（半径为零）同心。如果 x_L 为正，则史密斯圆图中包含的 x_L 圆的部分位于图表的上半部分；如果 x_L 为负，则位于图表的下半部分。

这是以 $(\Gamma_r, \Gamma_i) = (1, 1/x_L)$ 为中心、半径为 $(1/x_L)$ 的圆方程。 Γ_r - Γ_i 平面中的 x_L 圆与常数 r_L 的圆有很大不同。首先，归一化电抗 x_L 可以取正值和负值，而归一化电阻不能为负值（负电阻不能在无源电路中实现）。因此，等式 (2.118) 表示两个圆族：一个代表 x_L 的正值，另一个代表负值。此外，如图2-25所示，只有给定圆的一部分落在 $|\Gamma| = 1$ 单位圆的边界内。

方程给出的两个参数方程的圆族。针对 r_L 和 x_L 的选定值绘制的 (2.116) 和 (2.118) 构成了图 2-26 所示的史密斯圆图。这

史密斯圆图提供了方程式的图形评估。（2.115a 和 b）及其逆。

► 史密斯圆图中的每个点都表示两个相互关联的量的值，即 z_L 和 Γ 。 r_L 和 x_L 圆在该点的交点定义了 $z_L = r_L + jx_L$ 。同时，该点的位置定义了 $|\Gamma|$ 和 θ_r 。 ◀

例如，图2-26中的点P表示归一化负载阻抗 $z_L = 2 - j1$ ，其对应于电压反射系数 $\Gamma = 0.45 \exp(-j26.6^\circ)$ 。震级

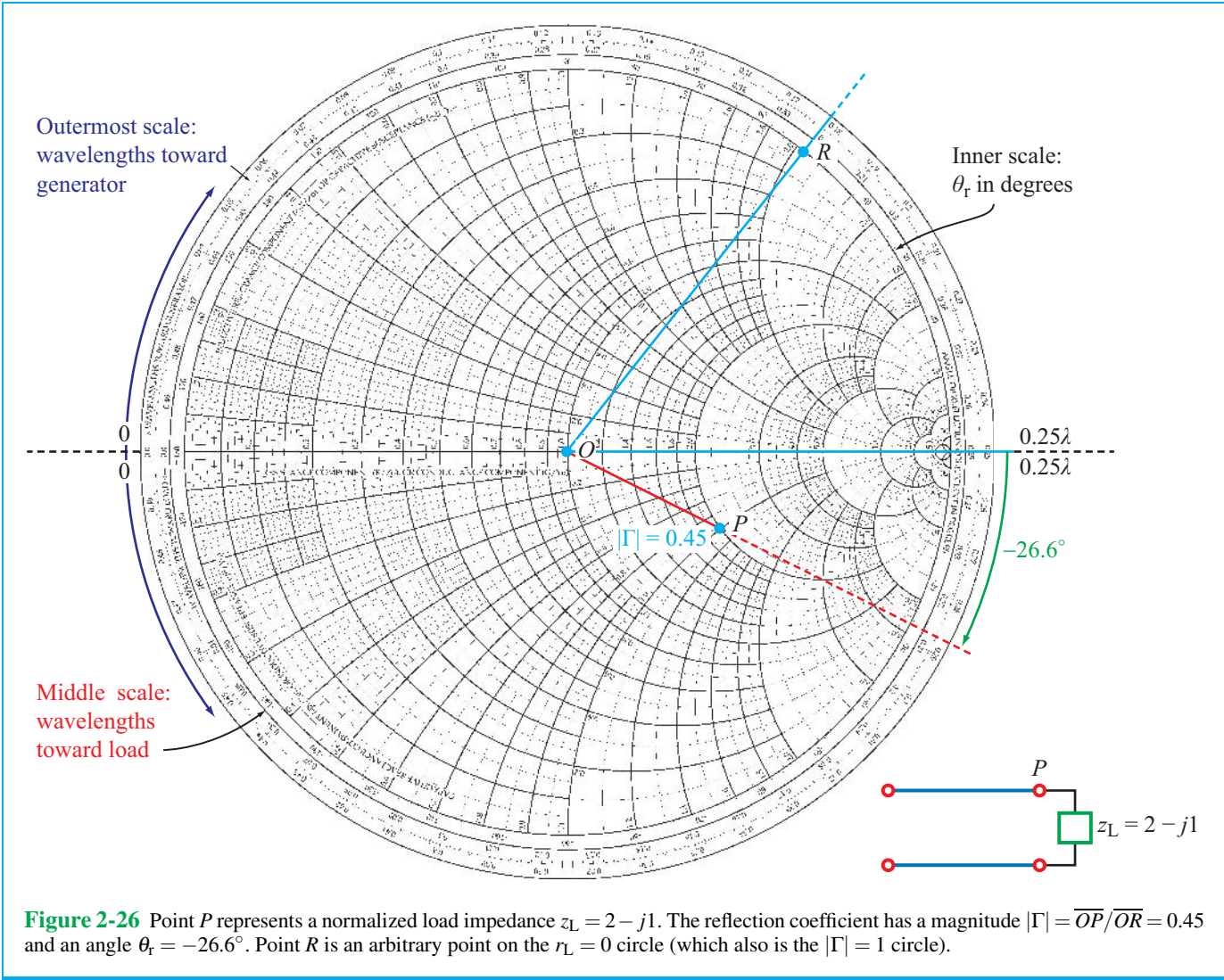
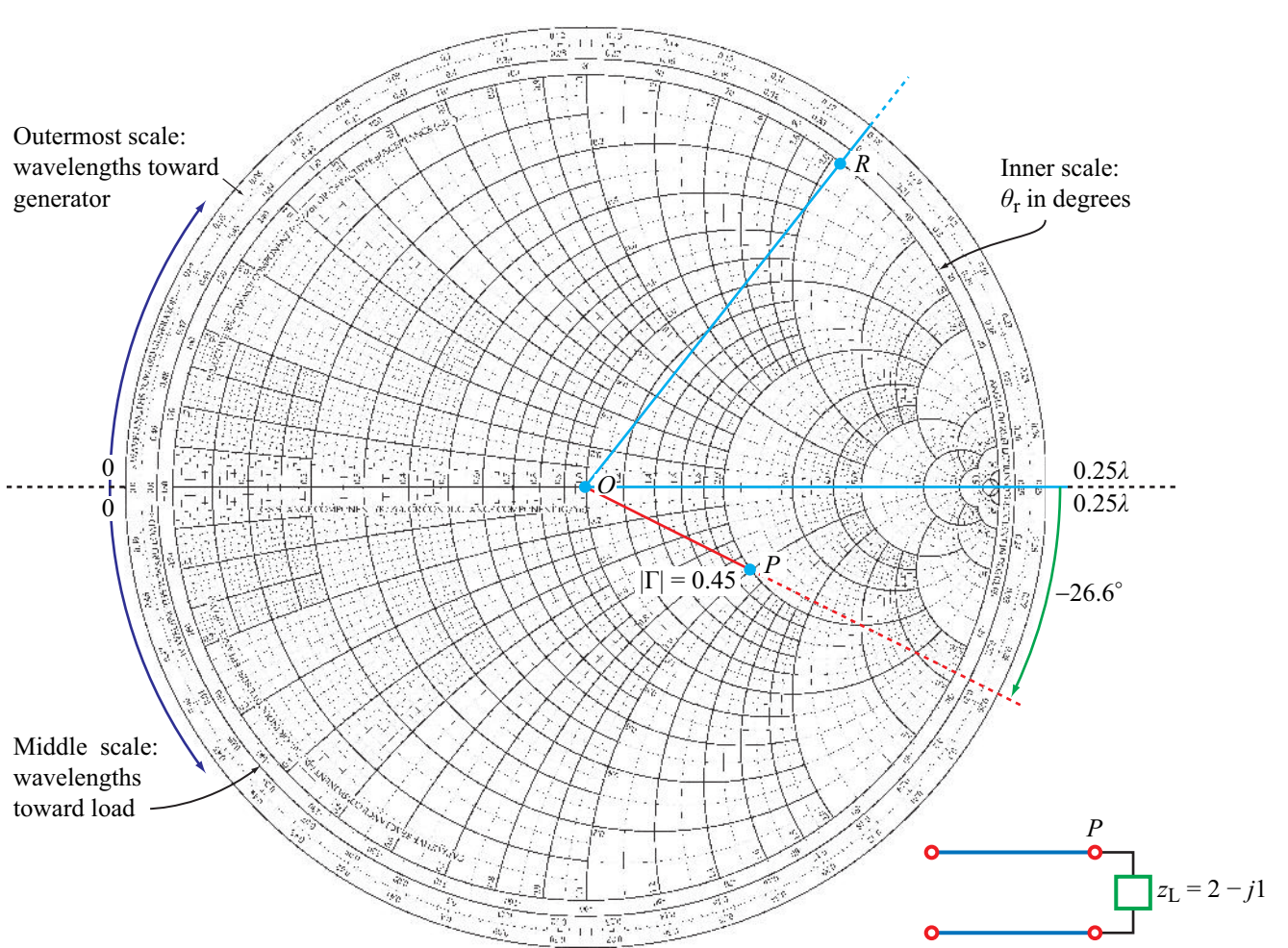


Figure 2-26 Point P represents a normalized load impedance $z_L = 2 - j1$. The reflection coefficient has a magnitude $|\Gamma| = \overline{OP}/\overline{OR} = 0.45$ and an angle $\theta_r = -26.6^\circ$. Point R is an arbitrary point on the $r_L = 0$ circle (which also is the $|\Gamma| = 1$ circle).

$|\Gamma| = 0.45$ is obtained by dividing the length of the line between the center of the Smith chart and point P (designated $\overline{OP}$ in Fig. 2-26), by $\overline{OR}$, which is the length of the line between the center of the Smith chart and the edge of the unit circle (the radius of the unit circle corresponds to $|\Gamma| = 1$). The perimeter of the Smith chart contains three concentric scales. The innermost scale is labeled **angle of reflection coefficient in degrees**. This is the scale for θ_r . As indicated in Fig. 2-26, $\theta_r = -26.6^\circ$ (-0.46 rad) for point P . The meanings and uses of the other two scales are discussed next.

Exercise 2-19: Use the Smith chart to find the values of Γ corresponding to the following normalized load impedances: (a) $z_L = 2 + j0$, (b) $z_L = 1 - j1$, (c) $z_L = 0.5 - j2$, (d) $z_L = -j3$, (e) $z_L = 0$ (short circuit), (f) $z_L = \infty$ (open circuit), and (g) $z_L = 1$ (matched load).

Answer: (a) $\Gamma = 0.33$, (b) $\Gamma = 0.45 \angle -63.4^\circ$, (c) $\Gamma = 0.83 \angle -50.9^\circ$, (d) $\Gamma = 1 \angle -36.9^\circ$, (e) $\Gamma = -1$, (f) $\Gamma = 1$, (g) $\Gamma = 0$. (See EN.)

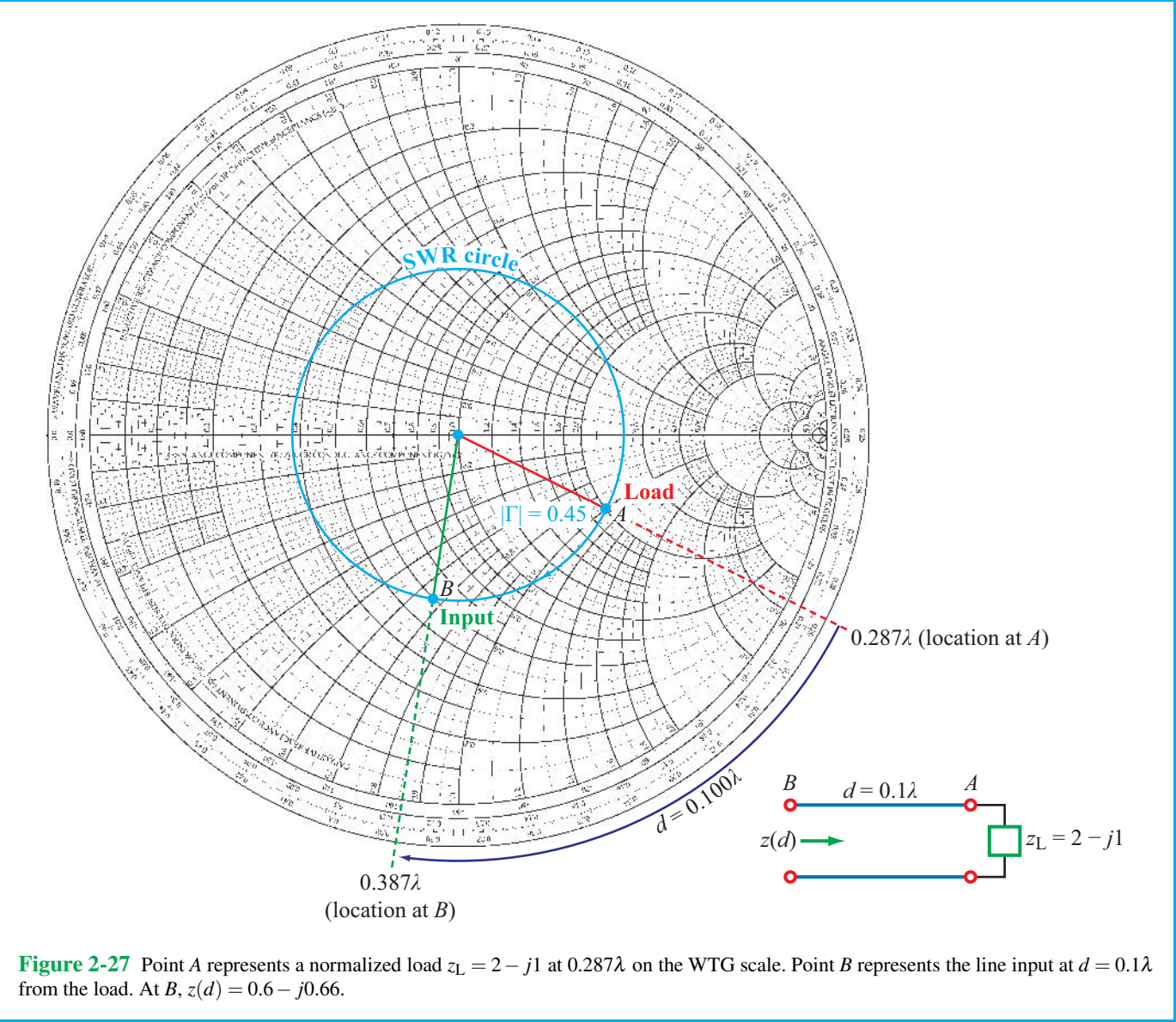


菲图2-26 P点表示归一化负载阻抗 $z_L = 2 - j1$ 。反射系数的大小为 $|\Gamma| = \overline{OP}/\overline{OR} = 0.45$ 一个角 $\theta_r = -26.6^\circ$ 。点 R 是 $r_L = 0$ 圆 (也是 $|\Gamma| = 1$ 圆) 上的任意点。

$|\Gamma| = 0.45$ 是用史密斯圆图中心与 P 点 (图 2-26 中指定的 OP) 之间的连线长度除以 OR 得到的, OR 是史密斯圆图中心与单位圆边缘之间的连线长度 (单位圆的半径对应于 $|\Gamma| = 1$)。史密斯圆图的周长包含三个同心刻度。最里面的刻度标有反射角系数 (以度为单位)。这是 θ_r 的比例。如图2-26所示, P 点为 $\theta_r = -26.6^\circ$ (-0.46 rad)。接下来讨论其他两个尺度的含义和用途。

练习 2-19: 使用史密斯圆图找出与以下标准化负载阻抗相对应的 Γ 值: (a) $z_L = 2 + j0$, (b) $z_L = 1 - j1$, (c) $z_L = 0.5 - j2$, (d) $z_L = -j3$, (e) $z_L = 0$ (短路), (f) $z_L = \infty$ (开路), (g) $z_L = 1$ (匹配负载)。

答案: (a) $\Gamma = 0.33$, (b) $\Gamma = 0.45 \angle -63.4^\circ$, (c) $\Gamma = 0.83 \angle -50.9^\circ$, (d) $\Gamma = 1 \angle -36.9^\circ$, (e) $\Gamma = -1$, (f) $\Gamma = 1$, (g) $\Gamma = 0$ 。(参见 EN.)



2-10.2 SWR Circle

Consider point A in the Smith chart of Fig. 2-27. At point A, the normalized load impedance is $z_L = 2 - j1$. The magnitude of the corresponding reflection coefficient is

$$|\Gamma| = \left| \frac{z_L - 1}{z_L + 1} \right| = \left| \frac{2 - j1 - 1}{2 - j1 + 1} \right| = \left| \frac{1 - j1}{3 - j1} \right| = \frac{\sqrt{2}}{\sqrt{10}} = 0.45.$$

Let us construct a circle centered at $(\Gamma_r, \Gamma_i) = (0, 0)$ and passing through point A. Every point on this circle has the same value for $|\Gamma|$, namely 0.45. This **constant- $|\Gamma|$ circle** is also a

constant-SWR circle. This follows from the relation between the voltage standing-wave ratio (SWR) and $|\Gamma|$, namely

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \quad (2.119)$$

► A constant value of $|\Gamma|$ corresponds to a constant value of S , and vice versa. ◀

The utility of the SWR circle will become evident shortly.

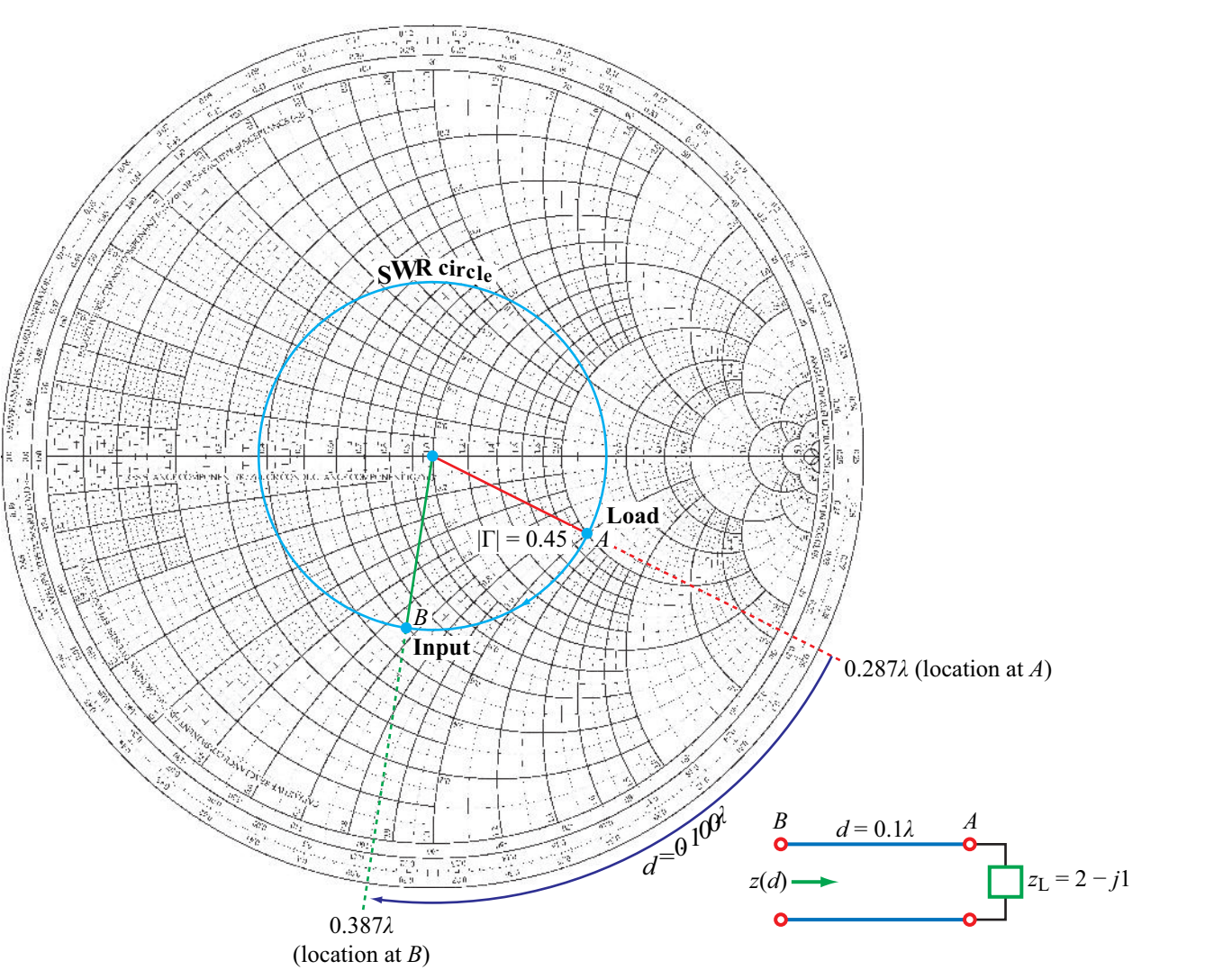


图 2-27 点 A 表示 WTG 规模上 0.287λ 处的归一化载荷 $z_L = 2 - j1$ 。点 B 表示来自负载的 $d = 0.1\lambda$ 处的线路输入。在 B 处, $z(d) = 0.6 - j0.66$ 。

2-10.2 驻波比圆

考虑图 2-27 史密斯圆图中的 A 点。在 A 点, 归一化负载阻抗为 $z_L = 2 - j1$ 。相应的反射系数的大小为

$$|\Gamma| = \left| \frac{z_L - 1}{z_L + 1} \right| = \left| \frac{2 - j1 - 1}{2 - j1 + 1} \right| = \left| \frac{1 - j1}{3 - j1} \right| = \frac{\sqrt{2}}{\sqrt{10}} = 0.45.$$

让我们构造一个以 $(\Gamma_r, \Gamma_i) = (0, 0)$ 为圆心并经过点 A 的圆。该圆上的每个点都有相同的 $|\Gamma|$ 值, 即 0.45。这个常数- $|\Gamma|$ 圆也是

恒定驻波比圆。这是根据电压驻波比 (SWR) 与 $|\Gamma|$ 之间的关系得出的, 即

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \quad (2.119)$$

► $|\Gamma|$ 的常数对应于 S 的常数, 反之亦然。◀

SWR 环的效用很快就会显现出来。

2-10.3 Wave Impedance

From Eq. (2.74), the *normalized wave impedance* looking toward the load at a distance d from the load is

$$z(d) = \frac{Z(d)}{Z_0} = \frac{1 + \Gamma_d}{1 - \Gamma_d}, \tag{2.120}$$

where

$$\Gamma_d = \Gamma e^{-j2\beta d} = |\Gamma|e^{j(\theta_r - 2\beta d)} \tag{2.121}$$

is the *phase-shifted* voltage reflection coefficient. The form of Eq. (2.120) is identical with that for z_L given by Eq. (2.112):

$$z_L = \frac{1 + \Gamma}{1 - \Gamma}. \tag{2.122}$$

This similarity in form suggests that, if Γ is transformed into Γ_d , z_L gets transformed into $z(d)$. On the Smith chart, the transformation from Γ to Γ_d is achieved by maintaining $|\Gamma|$ constant and *decreasing its phase* θ_r by $2\beta d$, which corresponds to a *clockwise rotation* (on the Smith chart) over an angle of $2\beta d$ radians. A complete rotation around the Smith chart corresponds to a phase change of 2π in Γ . The length d corresponding to this phase change satisfies

$$2\beta d = 2 \frac{2\pi}{\lambda} d = 2\pi, \tag{2.123}$$

from which it follows that $d = \lambda/2$.

► The outermost scale around the perimeter of the Smith chart (Fig. 2-26), called the *wavelengths toward generator* (WTG) scale, has been constructed to denote movement on the transmission line toward the generator, in units of the wavelength λ . That is, d is measured in wavelengths, and one complete rotation corresponds to $d = \lambda/2$. ◀

In some transmission-line problems, it may be necessary to move from some point on the transmission line toward a point closer to the load, in which case the phase of Γ must be increased, which corresponds to rotation in the counterclockwise direction. For convenience, the Smith chart contains a third scale around its perimeter (in between the θ_r scale and the WTG scale) for accommodating such an operation. It is called the *wavelengths toward load* (WTL) scale.

Exercise 2-20: Use the Smith chart to find the normalized input impedance of a lossless line of length l terminated in a normalized load impedance z_L for each of the following combinations: (a) $l = 0.25\lambda$, $z_L = 1 + j0$; (b) $l = 0.5\lambda$, $z_L = 1 + j1$; (c) $l = 0.3\lambda$, $z_L = 1 - j1$; (d) $l = 1.2\lambda$, $z_L = 0.5 - j0.5$; (e) $l = 0.1\lambda$, $z_L = 0$ (short circuit); (f) $l = 0.4\lambda$, $z_L = j3$; and (g) $l = 0.2\lambda$, $z_L = \infty$ (open circuit).

Illustration:
Using the Smith Chart to Find $Z(d)$

To illustrate how the Smith chart is used to find $Z(d)$, consider a $50\ \Omega$ lossless transmission line terminated in a load impedance $Z_L = (100 - j50)\ \Omega$. Our objective is to find $Z(d)$ at a distance $d = 0.1\lambda$ from the load.

1. The normalized load impedance is

$$z_L = \frac{Z_L}{Z_0} = \frac{100 - j50}{50} = 2 - j1 \quad (\text{point A in Fig. 2-27}).$$

2. Point A is located at 0.287λ on the WTG scale.

3. Using a compass, we construct the SWR circle with its center at the center of the Smith chart and radius passing through A.

4. As was stated earlier, to transform z_L to $z(d)$, we need to maintain $|\Gamma|$ constant, which means staying on the SWR circle, while decreasing the phase of Γ by $2\beta d$ radians. This is equivalent to moving a distance $d = 0.1\lambda$ toward the generator on the WTG scale. Since point A is located at 0.287λ on the WTG scale, $z(d)$ is found by moving to location $0.287\lambda + 0.1\lambda = 0.387\lambda$ on the WTG scale. A radial line through this new position on the WTG scale intersects the SWR circle at point B.

5. Point B represents $z(d)$, whose value is

$$z(d) = 0.6 - j0.66.$$

To obtain $Z(d)$, we unnormalize $z(d)$ by multiplying it by $Z_0 = 50\ \Omega$:

$$Z(d) = (0.6 - j0.66) \times 50 = (30 - j33)\ \Omega.$$

This result can be verified analytically using Eq. (2.120). The points between points A and B on the SWR circle represent different locations along the transmission line.

If a line is of length l , its *input impedance* is $Z_{in} = Z_0 z(l)$ with $z(l)$ determined by rotating a distance l from the load along the WTG scale.

Answer: (a) $z_{in} = 1 + j0$, (b) $z_{in} = 1 + j1$, (c) $z_{in} = 0.76 + j0.84$, (d) $z_{in} = 0.59 + j0.66$, (e) $z_{in} = 0 + j0.73$, (f) $z_{in} = 0 + j0.72$, (g) $z_{in} = 0 - j0.32$. (See EM.)

2-10.3 波阻抗

从方程。 (2.74), 在距负载距离 d 处看向负载的归一化波阻抗为

$$z(d) = \frac{Z(d)}{Z_0} = \frac{1 + \Gamma_d}{1 - \Gamma_d}, \tag{2.120}$$

在哪里

$$\Gamma_d = \Gamma e^{-j2\beta d} = |\Gamma|e^{j(\theta_r - 2\beta d)} \tag{2.121}$$

是相移电压反射系数。方程的形式(2.120) 与式 (2.120) 给出的 z_L 相同。 (2.112) :

$$z_L = \frac{1 + \Gamma}{1 - \Gamma}. \tag{2.122}$$

这种形式上的相似性表明, 如果 Γ 转换为 Γ_d , 则 z_L 也会转换为 $z(d)$ 。在史密斯圆图上, 从 Γ 到 Γ_d 的变换是通过保持 $|\Gamma|$ 恒定并将其相位 θ_r 减少 $2\beta d$ 来实现的, 这对应于顺时针旋转 (在史密斯圆图上) $2\beta d$ 弧度的角度。围绕史密斯圆图的完整旋转对应于 Γ 中 2π 的相变。对应于该相变的长度 d 满足

$$2\beta d = 2 \frac{2\pi}{\lambda} d = 2\pi, \tag{2.123}$$

由此得出 $d = \lambda/2$ 。

► 史密斯圆图周边的最外层刻度 (图 2-26) 称为朝向发生器的波长 (WTG) 刻度, 用于表示传输线上朝向发生器的移动, 以波长 λ 为单位。也就是说, d 以波长为单位进行测量, 完整旋转一圈对应于 $d = \lambda/2$ 。
◀

在某些传输线问题中, 可能需要从传输线上的某个点向更靠近负载的点移动, 在这种情况下, 必须增加 Γ 的相位, 这对应于逆时针方向的旋转。为了方便起见, 史密斯圆图在其周边包含第三个标度 (在 θ_r 标度和 WTG 标度之间) 以适应此类操作。它称为负载波长 (WTL) 标度。

练习 2-20: 使用史密斯圆图计算以下每种组合的长度为 l 且终止于归一化负载阻抗 z_L 的无损线路的归一化输入阻抗: (a) $l = 0.25\lambda$, $z_L = 1 + j0$; (b) $l = 0.5\lambda$, $z_L = 1 + j1$; (c) $l = 0.3\lambda$, $z_L = 1 - j1$; (d) $l = 1.2\lambda$, $z_L = 0.5 - j0.5$; (e) $l = 0.1\lambda$, $z_L = 0$ (短路); (f) $l = 0.4\lambda$, $z_L = j3$; (g) $l = 0.2\lambda$, $z_L = \infty$ (开路)。

插图: 使用史密斯圆图
查找 $Z(d)$

为了说明如何使用史密斯圆图来查找 $Z(d)$, 请考虑终止于负载阻抗 $Z_L = (100 - j50)\ \Omega$ 的 $50\ \Omega$ 无损传输线。我们的目标是在距负载距离 $d = 0.1\lambda$ 处找到 $Z(d)$ 。

1. 归一化负载阻抗为

$$z_L = \frac{Z_L}{Z_0} = \frac{100 - j50}{50} = 2 - j1 \quad (\text{point A in Fig. 2-27}).$$

2. A点位于WTG标度上的0.287λ处。

3. 使用圆规, 以史密斯圆图的中心为圆心, 以 A 为半径构建 SWR 圆。

4. 如前所述, 要将 z_L 转换为 $z(d)$, 我们需要保持 $|\Gamma|$ 恒定, 这意味着保持在 SWR 圆上, 同时将 Γ 的相位减小 $2\beta d$ 弧度。这相当于在 WTG 尺度上向发电机移动距离 $d = 0.1\lambda$ 。由于点 A 位于 WTG 尺度上的 0.287λ , 因此通过移动到 WTG 尺度上的位置 $0.287\lambda + 0.1\lambda = 0.387\lambda$ 即可找到 $z(d)$ 。通过 WTG 标尺上这个新位置的径向线与 SWR 圆相交于 B 点。

5、B点代表 $z(d)$, 其值为

$$z(d) = 0.6 - j0.66.$$

为了获得 $Z(d)$, 我们将 $z(d)$ 乘以 $Z_0 = 50\ \Omega$ 来对其进行非标准化:

$$Z(d) = (0.6 - j0.66) \times 50 = (30 - j33)\ \Omega.$$

这个结果可以使用方程进行分析验证。 (2.120)。SWR 圆上 A 点和 B 点之间的点代表传输线上的不同位置。

如果线路长度为 l , 则其输入阻抗为 $Z_{in} = Z_0 z(l)$, 其中 $z(l)$ 是通过沿 WTG 标度从负载旋转距离 l 来确定的。

答案: (a) $z_{in} = 1 + j0$, (b) $z_{in} = 1 + j1$, (c) $z_{in} = 0.76 + j0.84$, (d) $z_{in} = 0.59 + j0.66$, (e) $z_{in} = 0 + j0.73$, (f) $z_{in} = 0 + j0.72$, (g) $z_{in} = 0 - j0.32$ 。(参见EM。)

2-10.4 SWR, Voltage Maxima, and Minima

Consider a load with $z_L = 2 + j1$. **Figure 2-28** shows a Smith chart with a SWR circle drawn through point A, representing z_L . The SWR circle intersects the real (Γ_r) axis at two points, labeled $P_{\max}$ and $P_{\min}$. At both points, $\Gamma_i = 0$ and $\Gamma = \Gamma_r$. Also, on the real axis, the imaginary part of the load impedance $x_L = 0$. From the definition of Γ ,

$$\Gamma = \frac{z_L - 1}{z_L + 1}, \tag{2.124}$$

It follows that points $P_{\max}$ and $P_{\min}$ correspond to

$$\Gamma = \Gamma_r = \frac{r_0 - 1}{r_0 + 1} \quad (\text{for } \Gamma_i = 0), \tag{2.125}$$

where r_0 is the value of r_L where the SWR circle intersects the Γ_r axis. Point $P_{\min}$ corresponds to $r_0 < 1$, and $P_{\max}$ corresponds to $r_0 > 1$. Rewriting Eq. (2.119) for $|\Gamma|$ in terms of S , we have

$$|\Gamma| = \frac{S - 1}{S + 1}. \tag{2.126}$$

For point $P_{\max}$, $|\Gamma| = \Gamma_r$; hence

$$\Gamma_r = \frac{S - 1}{S + 1}. \tag{2.127}$$

The similarity in form of Eqs. (2.125) and (2.127) suggests that S equals the value of the normalized resistance r_0 . By definition, $S \geq 1$, and at point $P_{\max}$, $r_0 > 1$, which further satisfies the similarity condition. In **Fig. 2-28**, $r_0 = 2.6$ at $P_{\max}$; hence $S = 2.6$.

► S is numerically equal to the value of r_0 at $P_{\max}$, which is the point at which the SWR circle intersects the real Γ axis to the right of the chart's center. ◄

Points $P_{\min}$ and $P_{\max}$ also represent locations on the line where the magnitude of the voltage $|\tilde{V}|$ is a minimum and a maximum, respectively. This is easily demonstrated by considering Eq. (2.121) for Γ_d . At point $P_{\max}$, the total phase of Γ_d , that is, $(\theta_r - 2\beta d)$, equals zero or $-2n\pi$ (with n being a positive integer), which is the condition corresponding to $|\tilde{V}|_{\max}$, as indicated by Eq. (2.69). Similarly, at $P_{\min}$ the total phase of Γ_d equals $-(2n + 1)\pi$, which is the condition for $|\tilde{V}|_{\min}$. Thus, for the transmission line represented by the SWR circle shown in **Fig. 2-28**, the distance between the load and the nearest voltage maximum is $d_{\max} = 0.037\lambda$, which is obtained by moving clockwise from the load at point A to point $P_{\max}$, and the distance to the nearest voltage minimum is $d_{\min} = 0.287\lambda$, corresponding to the clockwise rotation from

A to $P_{\min}$. Since the location of $|\tilde{V}|_{\max}$ corresponds to that of $|\tilde{I}|_{\min}$ and the location of $|\tilde{V}|_{\min}$ corresponds to that of $|\tilde{I}|_{\max}$, the Smith chart provides a convenient way to determine the distances from the load to all maxima and minima on the line (recall that the standing-wave pattern has a repetition period of $\lambda/2$).

2-10.5 Impedance to Admittance Transformations

In solving certain types of transmission line problems, it is often more convenient to work with admittances than with impedances. Any impedance Z is in general a complex quantity consisting of a resistance R and a reactance X :

$$Z = R + jX \quad (\Omega). \tag{2.128}$$

The **admittance** Y is the reciprocal of Z :

$$Y = \frac{1}{Z} = \frac{1}{R + jX} = \frac{R - jX}{R^2 + X^2} \quad (\text{S}). \tag{2.129}$$

The real part of Y is called the **conductance** G , and the imaginary part of Y is called the **susceptance** B . That is,

$$Y = G + jB \quad (\text{S}). \tag{2.130}$$

Comparison of Eq. (2.130) with Eq. (2.129) reveals that

$$G = \frac{R}{R^2 + X^2} \quad (\text{S}), \tag{2.131a}$$

$$B = \frac{-X}{R^2 + X^2} \quad (\text{S}). \tag{2.131b}$$

A **normalized impedance** z is defined as the ratio of Z to the characteristic impedance of the line, Z_0 . The same concept applies to the definition of the **normalized admittance** y ; that is,

$$y = \frac{Y}{Y_0} = \frac{G}{Y_0} + j\frac{B}{Y_0} = g + jb \quad (\text{dimensionless}), \tag{2.132}$$

where $Y_0 = 1/Z_0$ is the **characteristic admittance of the line** and

$$g = \frac{G}{Y_0} = GZ_0 \quad (\text{dimensionless}), \tag{2.133a}$$

$$b = \frac{B}{Y_0} = BZ_0 \quad (\text{dimensionless}). \tag{2.133b}$$

The lowercase quantities g and b represent the **normalized conductance** and **normalized susceptance** of y , respectively. Of course, the normalized admittance y is the reciprocal of the normalized impedance z ,

$$y = \frac{Y}{Y_0} = \frac{Z_0}{Z} = \frac{1}{z}. \tag{2.134}$$

2-10.4 SWR、电压最大值和最小值

考虑 $z_L = 2 + j1$ 的负载。图 2-28 显示了史密斯圆图，其中通过点 A 绘制了 SWR 圆，表示 z_L 。SWR 圆与实轴 (Γ_r) 相交于两个点，标记为 $P_{\max}$ 和 $P_{\min}$ 。在这两点上， $\Gamma_i = 0$ 和 $\Gamma = \Gamma_r$ 。另外，在实轴上，负载阻抗 $x_L =$ 的虚部为 0。根据 Γ 的定义，

$$\Gamma = \frac{z_L - 1}{z_L + 1}, \tag{2.124}$$

由此可知点 $P_{\max}$ 和 $P_{\min}$ 对应于

$$\Gamma = \Gamma_r = \frac{r_0 - 1}{r_0 + 1} \quad (\text{for } \Gamma_i = 0), \tag{2.125}$$

其中 r_0 是 SWR 圆与 Γ_r 轴相交处的 r_L 值。点 $P_{\min}$ 对应于 $r_0 < 1$ ，点 $P_{\max}$ 对应于 $r_0 > 1$ 。(2.119) 对于 S 而言的 $|\Gamma|$ ，我们有

$$|\Gamma| = \frac{S - 1}{S + 1}. \tag{2.126}$$

对于点 $P_{\max}$ ， $|\Gamma| = \Gamma_r$ ；因此

$$\Gamma_r = \frac{S - 1}{S + 1}. \tag{2.127}$$

方程形式的相似性。(2.125) 和 (2.127) 表明 S 等于归一化电阻 r_0 的值。根据定义， $S \geq 1$ ，在点 $P_{\max}$ 处， $r_0 > 1$ ，进一步满足相似性条件。图 2-28 中， $P_{\max}$ 处 $r_0 =$ 为 2.6；因此 $S = 2.6$ 。

► S 在数值上等于 r 的值₀在 $P_{\max}$ ，这是 SWR 圆与真实圆相交的点 Γ 轴位于图表中心右侧。◄

点 $P_{\min}$ 和 $P_{\max}$ 还表示线上电压 $|\tilde{V}|$ 的幅度分别为最小值和最大值的位置。通过考虑等式可以很容易地证明这一点。(2.121) 对于 Γ_d 。在 $P_{\max}$ 点， Γ_d 的总相位，即 $(\theta_r - 2\beta d)$ 等于 0 或 $-2n\pi$ (，其中 n 为正整数)，这是与 $|\tilde{V}|_{\max}$ 对应的条件，如式 (1) 所示。(2.69)。类似地，在 $P_{\min}$ 处， Γ_d 的总相位等于 $-(2n + 1)\pi$ ，这是 $|\tilde{V}|_{\min}$ 的条件。因此，对于图 2-28 所示的 SWR 圆表示的传输线，负载与最近电压最大值之间的距离为 $d_{\max} = 0.037\lambda$ ，这是从 A 点的负载顺时针移动到点 $P_{\max}$ 得到的，而到最近电压最小值的距离为 $d_{\min} = 0.287\lambda$ ，对应于从 A 点顺时针旋转

A 到 $P_{\min}$ 。由于 $|\tilde{V}|_{\max}$ 的位置对应于 $|\tilde{I}|_{\min}$ 的位置，而 $|\tilde{V}|_{\min}$ 的位置对应于 $|\tilde{I}|_{\max}$ 的位置，因此史密斯圆图提供了一种方便的方法来确定从负载到线路上所有最大值和最小值的距离（回想一下，驻波模式的重复周期为 $\lambda/2$ ）。

2-10.5 阻抗到导纳的变换

在解决某些类型的传输线问题时，使用导纳通常比使用阻抗更方便。一般而言，任何阻抗 Z 都是由电阻 R 和电抗 X 组成的复数：

$$Z = R + jX \quad (\Omega). \tag{2.128}$$

导纳 Y 是 Z 的倒数：

$$Y = \frac{1}{Z} = \frac{1}{R + jX} = \frac{R - jX}{R^2 + X^2} \quad (\text{S}). \tag{2.129}$$

Y 的实部称为电导 G ， Y 的虚部称为电纳 B 。即：

$$Y = G + jB \quad (\text{S}). \tag{2.130}$$

等式的比较 (2.130) 与等式。(2.129) 表明

$$G = \frac{R}{R^2 + X^2} \quad (\text{S}), \tag{2.131a}$$

$$B = \frac{-X}{R^2 + X^2} \quad (\text{S}). \tag{2.131b}$$

归一化阻抗 z 定义为 Z 与线路特性阻抗 Z_0 的比率。相同的概念适用于归一化导纳 y 的定义；那是，

$$y = \frac{Y}{Y_0} = \frac{G}{Y_0} + j\frac{B}{Y_0} = g + jb \quad (\text{无量纲}), \tag{2.132}$$

其中 $Y_0 = 1/Z_0$ 是线路的特征导纳，

$$g = \frac{G}{Y_0} = GZ_0 \quad (\text{dimensionless}), \tag{2.133a}$$

$$b = \frac{B}{Y_0} = BZ_0 \quad (\text{dimensionless}). \tag{2.133b}$$

小写量 g 和 b 分别代表 y 的归一化电导和归一化电纳。当然，归一化导纳 y 是归一化阻抗 z 的倒数，

$$y = \frac{Y}{Y_0} = \frac{Z_0}{Z} = \frac{1}{z}. \tag{2.134}$$

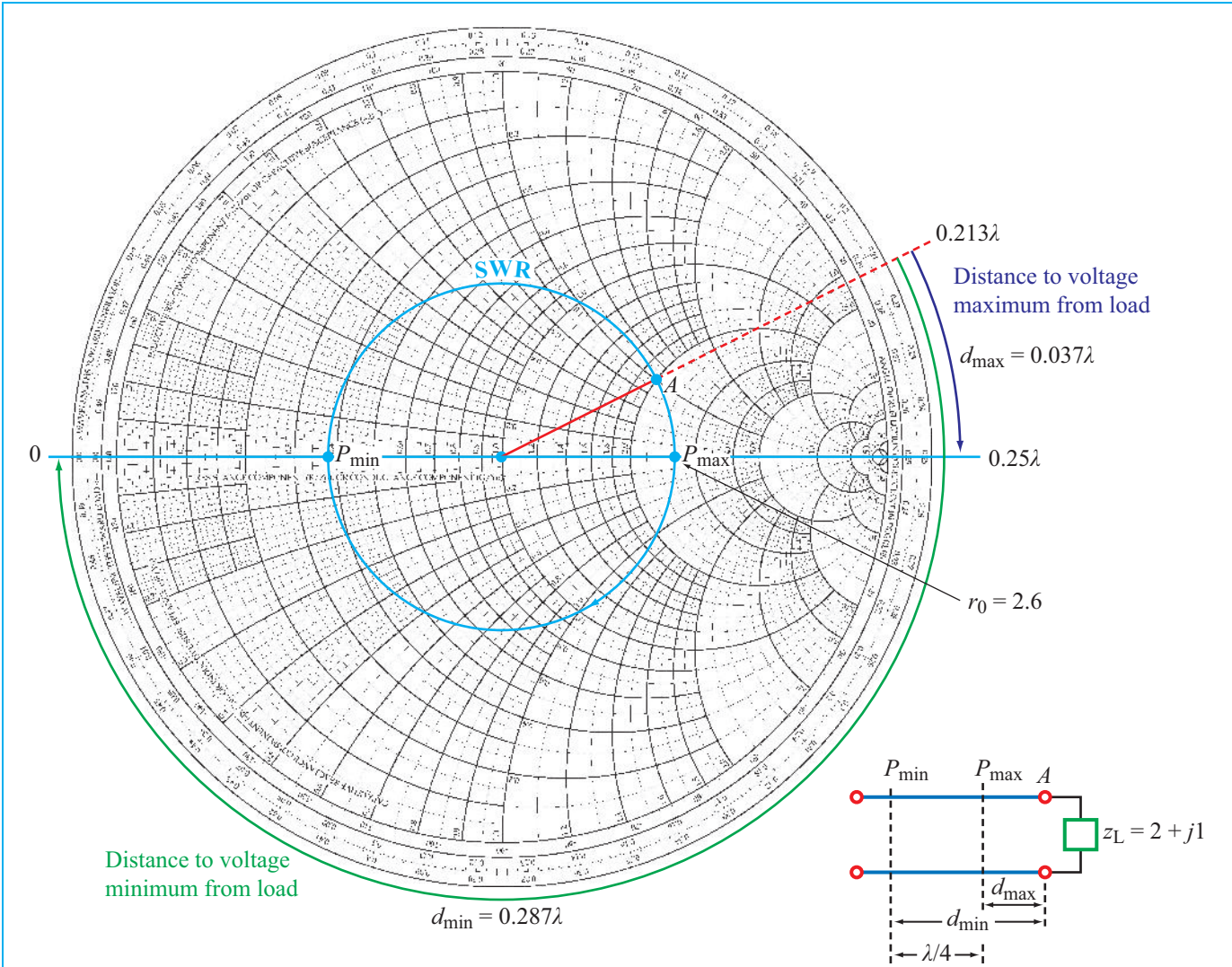


Figure 2-28 Point A represents a normalized load with $z_L = 2 + j1$. The standing-wave ratio is $S = 2.6$ (at $P_{\max}$), the distance between the load and the first voltage maximum is $d_{\max} = (0.25 - 0.213)\lambda = 0.037\lambda$, and the distance between the load and the first voltage minimum is $d_{\min} = (0.037 + 0.25)\lambda = 0.287\lambda$.

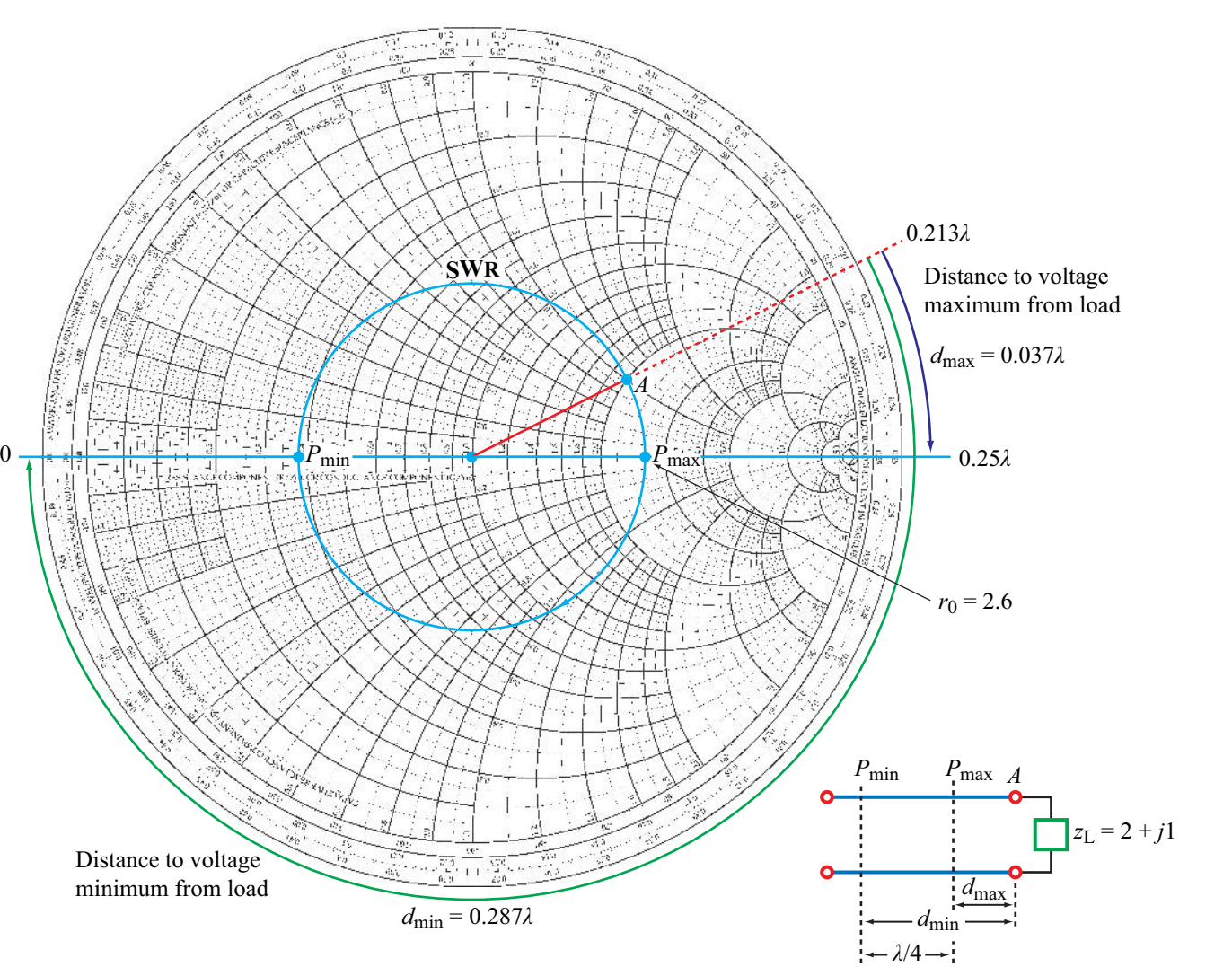


图 2-28 点 A 表示 $z_L = 2 + j1$ 的标准化载荷。驻波比为 $S = 2.6$ (在 $P_{\max}$ 处)，负载与第一电压最大值之间的距离为 $d_{\max} = (0.25 - 0.213)\lambda = 0.037\lambda$ ，负载与第一电压最小值之间的距离为 $d_{\min} = (0.037 + 0.25)\lambda = 0.287\lambda$ 。

Accordingly, using Eq. (2.122), the normalized load admittance y_L is given by

$$y_L = \frac{1}{z_L} = \frac{1 - \Gamma}{1 + \Gamma} \quad (\text{dimensionless}). \quad (2.135)$$

Now let us consider the normalized wave impedance $z(d)$ at a distance $d = \lambda/4$ from the load. Using Eq. (2.120) with

$$2\beta d = 4\pi d/\lambda = 4\pi\lambda/4\lambda = \pi \text{ gives}$$

$$z(d = \lambda/4) = \frac{1 + \Gamma e^{-j\pi}}{1 - \Gamma e^{-j\pi}} = \frac{1 - \Gamma}{1 + \Gamma} = y_L. \quad (2.136)$$

► Rotation by $\lambda/4$ on the SWR circle transforms z into y , and vice versa. ◀

In Fig. 2-29, the points representing z_L and y_L are diametrically opposite to each other on the SWR circle. In fact, such a

因此，使用等式 (2.122)，归一化负载导纳 y_L 由下式给出

$$y_L = \frac{1}{z_L} = \frac{1 - \Gamma}{1 + \Gamma} \quad (\text{dimensionless}). \quad (2.135)$$

现在让我们考虑距负载距离 $d = \lambda/4$ 处的归一化波阻抗 $z(d)$ 。使用方程式 (2.120) 与

$$2\beta d = 4\pi d/\lambda = 4\pi\lambda/4\lambda = \pi \text{ 给出}$$

$$z(d = \lambda/4) = \frac{1 + \Gamma e^{-j\pi}}{1 - \Gamma e^{-j\pi}} = \frac{1 - \Gamma}{1 + \Gamma} = y_L. \quad (2.136)$$

► 在 SWR 圆上旋转 $\lambda/4$ 将 z 转换为 y ，反之亦然。◀

在图 2-29 中，代表 z_L 和 y_L 的点在 SWR 圆上彼此相反。事实上，这样一个

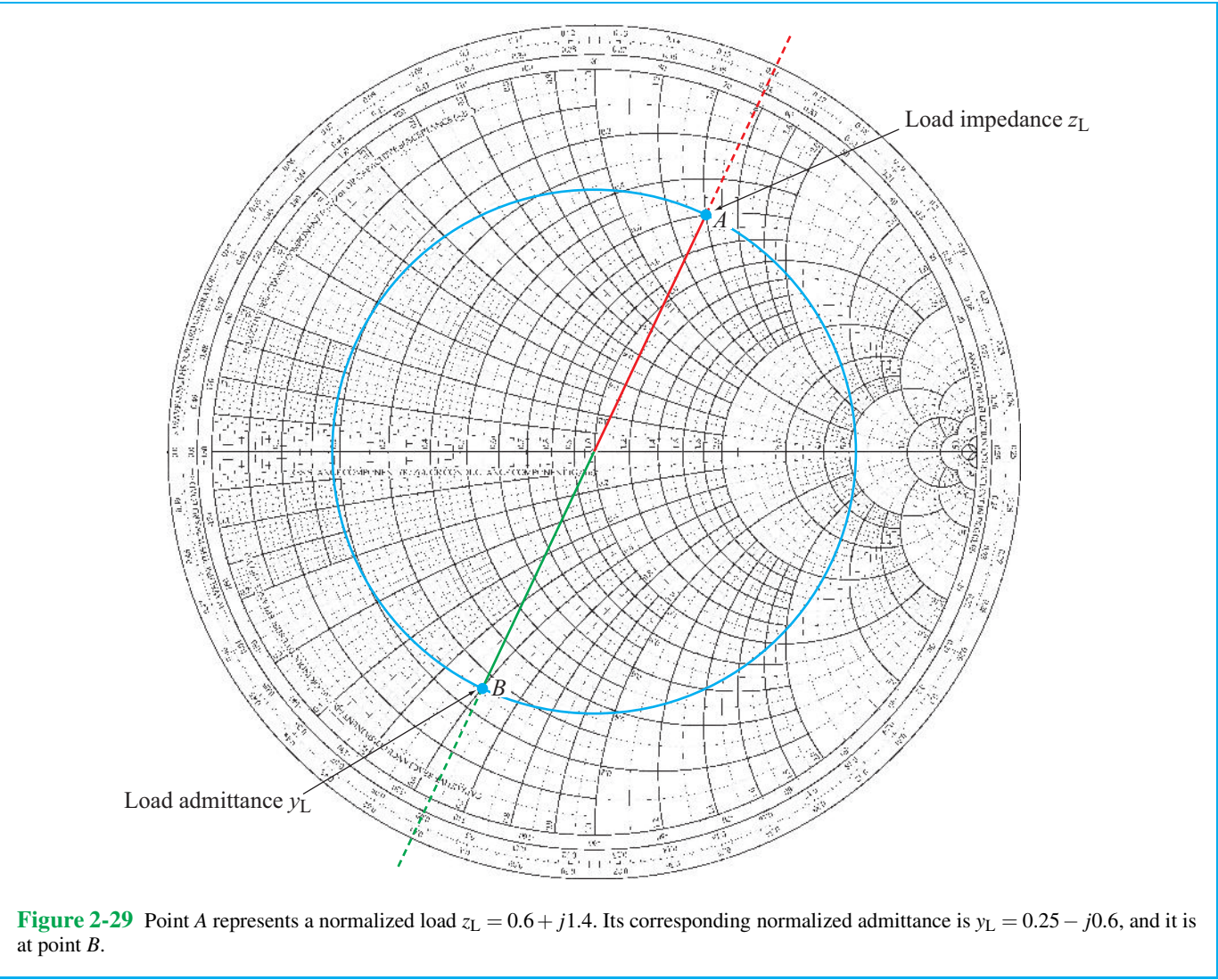


Figure 2-29 Point A represents a normalized load $z_L = 0.6 + j1.4$. Its corresponding normalized admittance is $y_L = 0.25 - j0.6$, and it is at point B.

transformation on the Smith chart can be used to determine any normalized admittance from its corresponding normalized impedance, and vice versa.

The Smith chart can be used with normalized impedances or with normalized admittances. As an impedance chart, the Smith chart consists of r_L and x_L circles: the resistance and reactance of a normalized load impedance z_L , respectively.

► When used as an admittance chart, the r_L circles become g_L circles and the x_L circles become b_L circles, where g_L and b_L are the conductance and susceptance of the normalized load admittance y_L , respectively. ◀

Example 2-11: Smith Chart Calculations

A $50\ \Omega$ lossless transmission line of length 3.3λ is terminated by a load impedance $Z_L = (25 + j50)\ \Omega$. Use the Smith chart to find (a) the voltage reflection coefficient, (b) the voltage standing-wave ratio, (c) the distances of the first voltage maximum and first voltage minimum from the load, (d) the input impedance of the line, and (e) the input admittance of the line.

Solution: (a) The normalized load impedance is

$$z_L = \frac{Z_L}{Z_0} = \frac{25 + j50}{50} = 0.5 + j1,$$

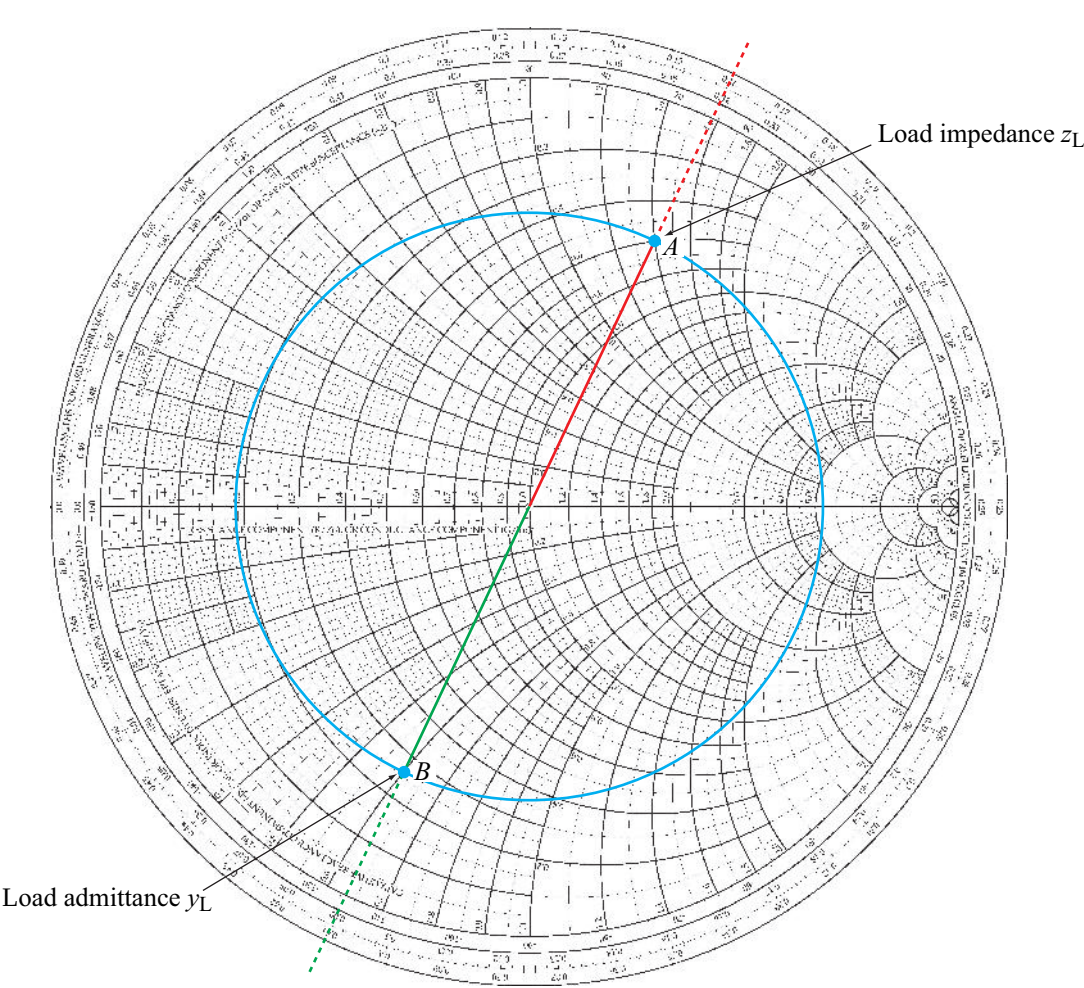


图2-29 A点表示归一化载荷 $z_L = 0.6 + j1.4$ 。其对应的归一化导纳为 $y_L = 0.25 - j0.6$ ，位于B点。

史密斯圆图上的变换可用于根据其相应的归一化阻抗确定任何归一化导纳，反之亦然。

史密斯圆图可与归一化阻抗或归一化导纳一起使用。作为阻抗图，史密斯圆图由 r_L 和 x_L 圆组成：分别是归一化负载阻抗 z_L 的电阻和电抗。

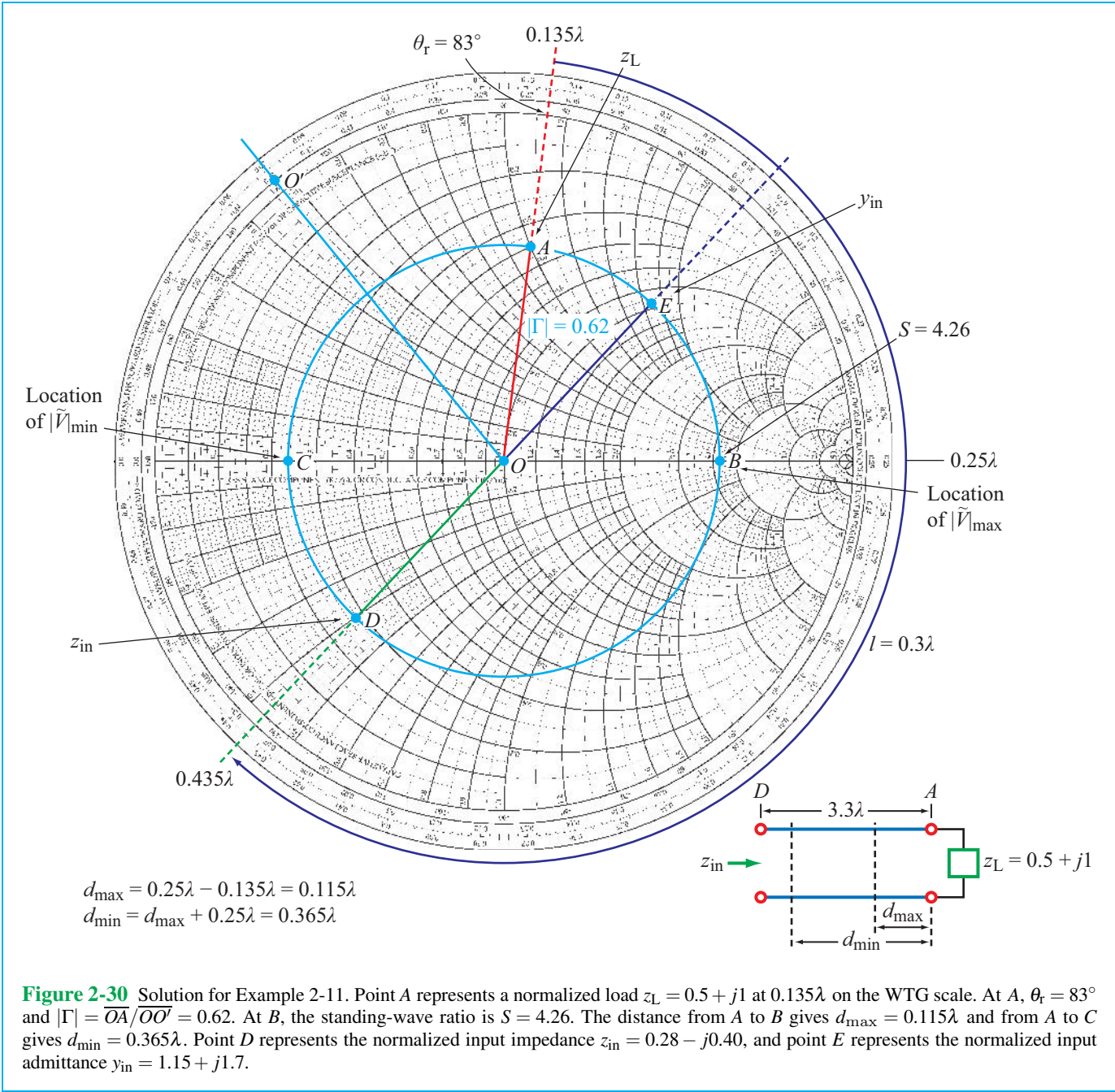
► 当用作导纳图时， r_L 圆变为 g_L 圆， x_L 圆变为 b_L 圆，其中 g_L 和 b_L 分别是归一化负载导纳 y_L 的电导和电纳。 ◀

示例 2-11：史密斯圆图计算

长度为 3.3λ 的 $50\ \Omega$ 无损传输线由负载阻抗 $Z_L = (25 + j50)\ \Omega$ 终止。使用史密斯圆图求 (a) 电压反射系数，(b) 电压驻波比，(c) 第一电压最大值和第一电压最小值与负载的距离，(d) 线路的输入阻抗，以及 (e) 线路的输入导纳。

解： (a) 归一化负载阻抗为

$$z_L = \frac{Z_L}{Z_0} = \frac{25 + j50}{50} = 0.5 + j1,$$



which is marked as point A on the Smith chart in **Fig. 2-30**. A radial line is drawn from the center of the chart at point O through point A to the outer perimeter of the chart. The line crosses the scale labeled “angle of reflection coefficient in degrees” at $\theta_r = 83^\circ$. Next, measurements are made to

determine lengths $\overline{OA}$ and $\overline{OO'}$, of the lines between O and A and between points O and O', respectively, where O' is an arbitrary point on the $r_L = 0$ circle. The length $\overline{OO'}$ is equal to the radius of the $|\Gamma| = 1$ circle. The magnitude of Γ is then

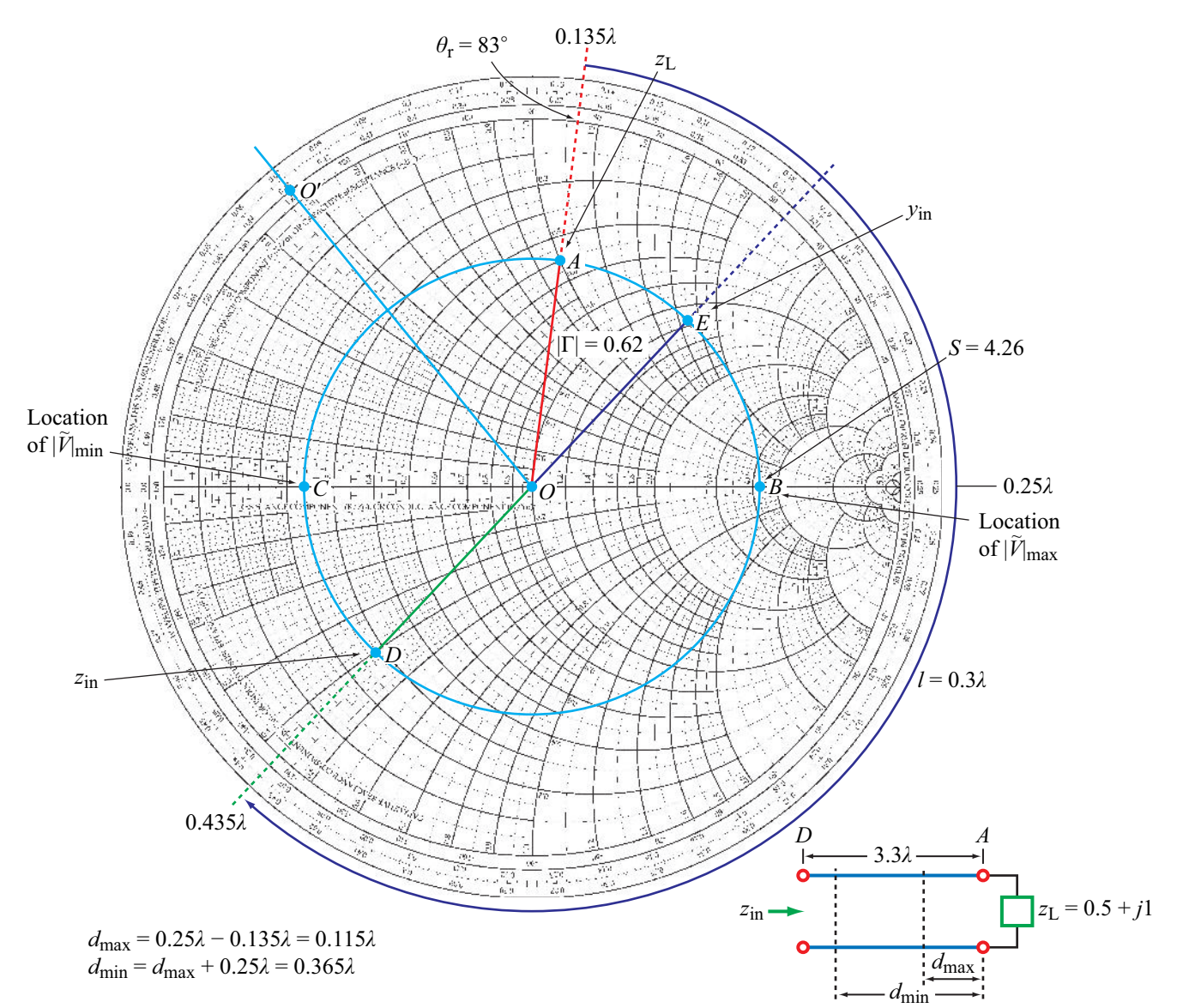


图 2-30 例 2-11 的解决方案。A 点代表 WTG 规模上 0.135λ 处的归一化载荷 $z_L = 0.5 + j1$ 。在 A 处， $\theta_r = 83^\circ$ 和 $|\Gamma| = OA/OO' = 0.62$ 。在 B 处，驻波比为 $S = 4.26$ 。从 A 到 B 的距离为 $d_{max} = 0.115\lambda$ ，从 A 到 C 的距离为 $d_{min} = 0.365\lambda$ 。D 点表示归一化输入阻抗 $z_{in} = 0.28 - j0.40$ ，E 点表示归一化输入导纳 $y_{in} = 1.15 + j1.7$ 。

在图 2-30 的史密斯圆图上标记为 A 点。从图表中心的 O 点经过 A 点到图表的外周界绘制一条径向线。该线在 $\theta_r = 83^\circ$ 处与标有“反射系数角度”的刻度相交。接下来，进行测量

分别确定 O 和 A 之间以及点 O 和 O' 之间的直线的长度 OA 和 OO'，其中 O' 是 $r_L = 0$ 圆上的任意点。长度 OO' 等于 $|\Gamma| = 1$ 圆的半径。Γ 的大小为

obtained from $|\Gamma| = \overline{OA}/\overline{OO'} = 0.62$. Hence,

$$\Gamma = 0.62\angle 83^\circ.$$
(2.137)

(b) The SWR circle passing through point A crosses the Γ_r axis at points B and C . The value of r_L at point B is 4.26, from which it follows that

$$S = 4.26.$$

(c) The first voltage maximum is at point B on the SWR circle, which is at location 0.25λ on the WTG scale. The load, represented by point A , is at 0.135λ on the WTG scale. Hence, the distance between the load and the first voltage maximum is

$$d_{\max} = (0.25 - 0.135)\lambda = 0.115\lambda.$$

The first voltage minimum is at point C . Moving on the WTG scale between points A and C gives

$$d_{\min} = (0.5 - 0.135)\lambda = 0.365\lambda,$$

which is 0.25λ past $d_{\max}$.

(d) The line is 3.3λ long; subtracting multiples of 0.5λ leaves 0.3λ . From the load at 0.135λ on the WTG scale, the input of the line is at $(0.135 + 0.3)\lambda = 0.435\lambda$. This is labeled as point D on the SWR circle, and the normalized impedance is

$$z_{\text{in}} = 0.28 - j0.40,$$

which yields

$$Z_{\text{in}} = z_{\text{in}}Z_0 = (0.28 - j0.40)50 = (14 - j20) \, \Omega.$$

(e) The normalized input admittance y_{in} is found by moving 0.25λ on the Smith chart to the image point of z_{in} across the circle, labeled point E on the SWR circle. The coordinates of point E give

$$y_{\text{in}} = 1.15 + j1.7,$$

and the corresponding input admittance is

$$Y_{\text{in}} = y_{\text{in}}Y_0 = \frac{y_{\text{in}}}{Z_0} = \frac{1.15 + j1.7}{50} = (0.023 + j0.034) \, \text{S}.$$

Example 2-12:
Determining Z_L Using the Smith Chart

This problem is similar to Example 2-6, except that now we demonstrate its solution using the Smith chart.

Given that the voltage standing-wave ratio $S = 3$ on a $50 \, \Omega$ line, that the first voltage minimum occurs at 5 cm from

the load, and that the distance between successive minima is 20 cm, find the load impedance.

Solution: The distance between successive minima equals $\lambda/2$. Hence, $\lambda = 40$ cm. In wavelength units, the first voltage minimum is at

$$d_{\min} = \frac{5}{40} = 0.125\lambda.$$

Point A on the Smith chart in **Fig. 2-31** corresponds to $S = 3$. Using a compass, the constant S circle is drawn through point A . Point B corresponds to locations of voltage minima. Upon moving 0.125λ from point B toward the load on the WTL scale (counterclockwise), we arrive at point C , which represents the location of the load. The normalized load impedance at point C is

$$z_L = 0.6 - j0.8.$$

Multiplying by $Z_0 = 50 \, \Omega$, we obtain

$$Z_L = 50(0.6 - j0.8) = (30 - j40) \, \Omega.$$

Concept Question 2-19: The outer perimeter of the Smith chart represents what value of $|\Gamma|$? Which point on the Smith chart represents a matched load?

Concept Question 2-20: What is an SWR circle? What quantities are constant for all points on an SWR circle?

Concept Question 2-21: What line length corresponds to one complete rotation around the Smith chart? Why?

Concept Question 2-22: Which points on the SWR circle correspond to locations of voltage maxima and minima on the line and why?

Concept Question 2-23: Given a normalized impedance z_L , how do you use the Smith chart to find the corresponding normalized admittance $y_L = 1/z_L$?

2-11 Impedance Matching

A transmission line usually connects a generator circuit at one end to a load at the other. The load may be an antenna, a computer terminal, or any circuit with an equivalent input impedance Z_L .

从 $|\Gamma| = \text{OA}/\text{OO}' = 0.62$ 获得。因此,

$$\Gamma = 0.62\angle 83^\circ.$$
(2.137)

(b) 经过 A 点的 SWR 圆在 B 点和 C 点与 Γ_r 轴相交。 B 点的 r_L 值为 4.26, 由此得出

$$S = 4.26.$$

(c) 第一个电压最大值位于 SWR 圆上的 B 点, 即 WTG 标度上的位置 0.25λ 。由 A 点表示的负载在 WTG 标度上为 0.135λ 。因此, 负载和第一电压最大值之间的距离是

$$d_{\max} = (0.25 - 0.135)\lambda = 0.115\lambda.$$

第一个电压最小值位于 C 点。在 A 点和 C 点之间移动 W TG 刻度可得出

$$d_{\min} = (0.5 - 0.135)\lambda = 0.365\lambda,$$

$d_{\max}$ 过去了 0.25λ 。

(d) 线长 3.3λ ; 减去 0.5λ 的倍数得到 0.3λ 。根据 WTG 标度上 0.135λ 处的负载, 线的输入位于 $(0.135 + 0.3)\lambda = 0.435\lambda$ 。这被标记为 SWR 圆上的 D 点, 归一化阻抗为

$$z_{\text{in}} = 0.28 - j0.40,$$

这产生

$$Z_{\text{in}} = z_{\text{in}}Z_0 = (0.28 - j0.40)50 = (14 - j20) \, \Omega.$$

(e) 归一化输入导纳 y_{in} 是通过将史密斯圆图上的 0.25λ 移动到穿过圆的 z_{in} 的图像点 (SWR 圆上标记为点 E) 来找到的。点 E 的坐标给出

$$y_{\text{in}} = 1.15 + j1.7,$$

相应的输入导纳为

$$Y_{\text{in}} = y_{\text{in}}Y_0 = \frac{y_{\text{in}}}{Z_0} = \frac{1.15 + j1.7}{50} = (0.023 + j0.034) \, \text{S}.$$

示例 2-12: 使用史密斯圆图确定 Z_L

这个问题与示例 2-6 类似, 只不过现在我们使用史密斯圆图演示其解决方案。

鉴于 $50 \, \Omega$ 线上的电压驻波比 $S = 3$, 第一个电压最小值出现在距离

负载, 并且连续最小值之间的距离为 20 cm, 求负载阻抗。

解: 连续最小值之间的距离等于 $\lambda/2$ 。因此, $\lambda = 40$ 厘米。以波长为单位, 第一个电压最小值为

$$d_{\min} = \frac{5}{40} = 0.125\lambda.$$

图2-31中史密斯圆图上的A点对应于S= 3。使用圆规, 通过A点画出常数S圆。B点对应于电压最小值的位置。在 WTL 标尺上从 B 点向负载移动 0.125λ 后 (逆时针), 我们到达 C 点, 它代表负载的位置。 C 点的归一化负载阻抗为

$$z_L = 0.6 - j0.8.$$

乘以 $Z_0 = 50 \, \Omega$, 我们得到

$$Z_L = 50(0.6 - j0.8) = (30 - j40) \, \Omega.$$

概念问题 2-19: 史密斯圆图的外周代表 $|\Gamma|$ 的值? 史密斯圆图上的哪一点代表匹配负载?

概念问题 2-20: 什么是 SWR 圆? 对于 SWR 圆上的所有点来说, 哪些量是恒定的?

概念问题 2-21: 围绕史密斯圆图旋转一圈对应的线长是多少? 为什么?

概念问题 2-22: SWR 圆上的哪些点对应于线上电压最大值和最小值的位置, 为什么?

概念问题 2-23: 给定一个归一化阻抗 z_L , 如何使用史密斯圆图找到相应的归一化导纳 $y_L = 1/z_L$?

2-11 阻抗匹配

传输线通常将一端的发电机电路连接到另一端的负载。负载可以是天线、计算机终端或具有等效输入阻抗 Z_L 的任何电路。

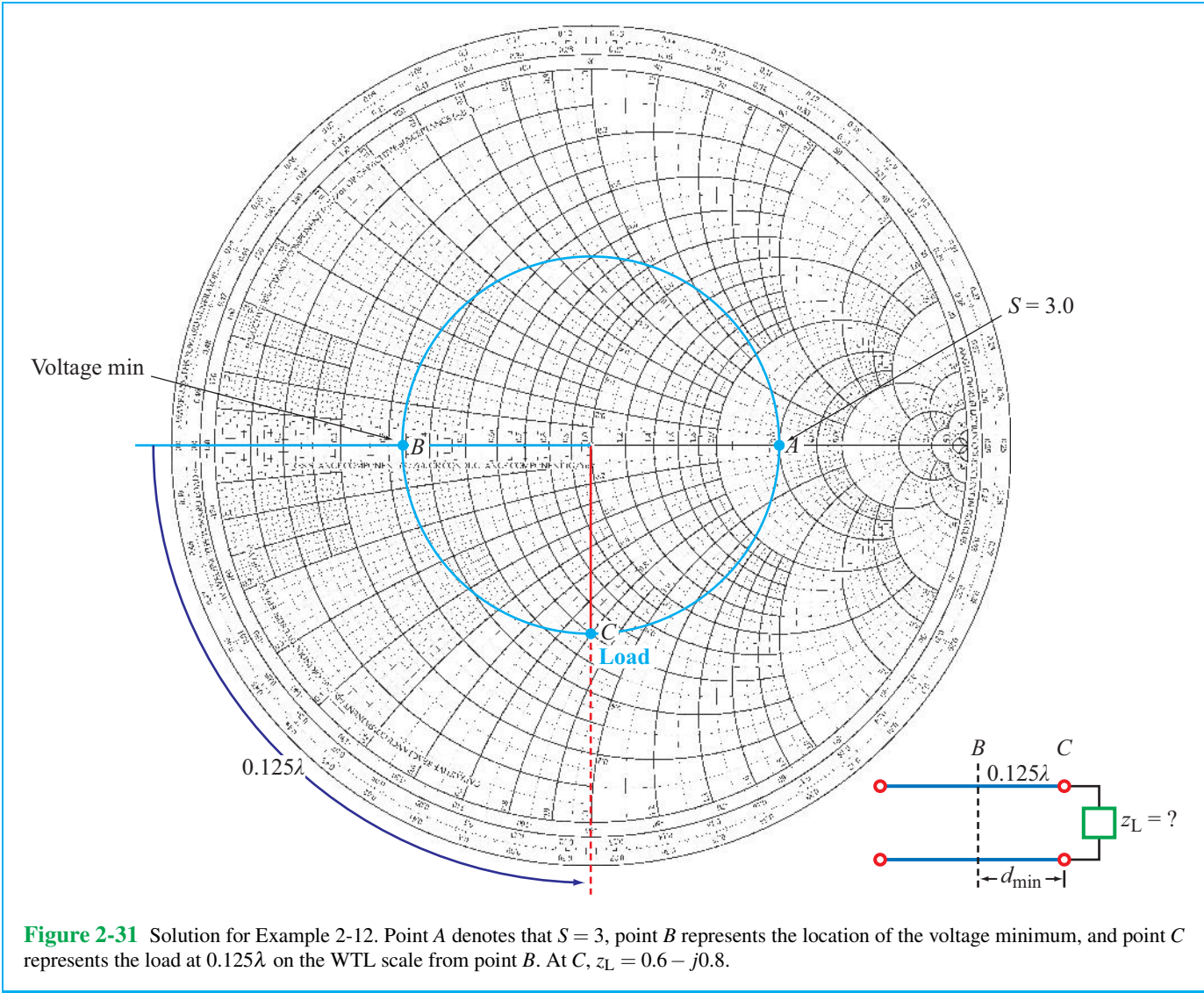


Figure 2-31 Solution for Example 2-12. Point A denotes that $S = 3$, point B represents the location of the voltage minimum, and point C represents the load at 0.125λ on the WTL scale from point B. At C, $z_L = 0.6 - j0.8$.

► The transmission line is said to be matched to the load when its characteristic impedance $Z_0 = Z_L$, in which case waves traveling on the line towards the load are not reflected back to the source. ◀

Since the primary use of a transmission line is to transfer power or transmit coded signals (such as digital data), a matched load ensures that all of the power delivered to the transmission line by the source is transferred to the load (and no *echoes* are relayed back to the source).
The simplest solution to matching a load to a transmission line is to design the load circuit such that its impedance

$Z_L = Z_0$. Unfortunately, this may not be possible in practice because the load circuit may have to satisfy other requirements. An alternative solution is to place an **impedance-matching network** between the load and the transmission line as illustrated in Fig. 2-32.

► The purpose of the matching network is to eliminate reflections at terminals MM' for waves incident from the source. Even though multiple reflections may occur between AA' and MM' , only a forward traveling wave exists on the feedline. ◀

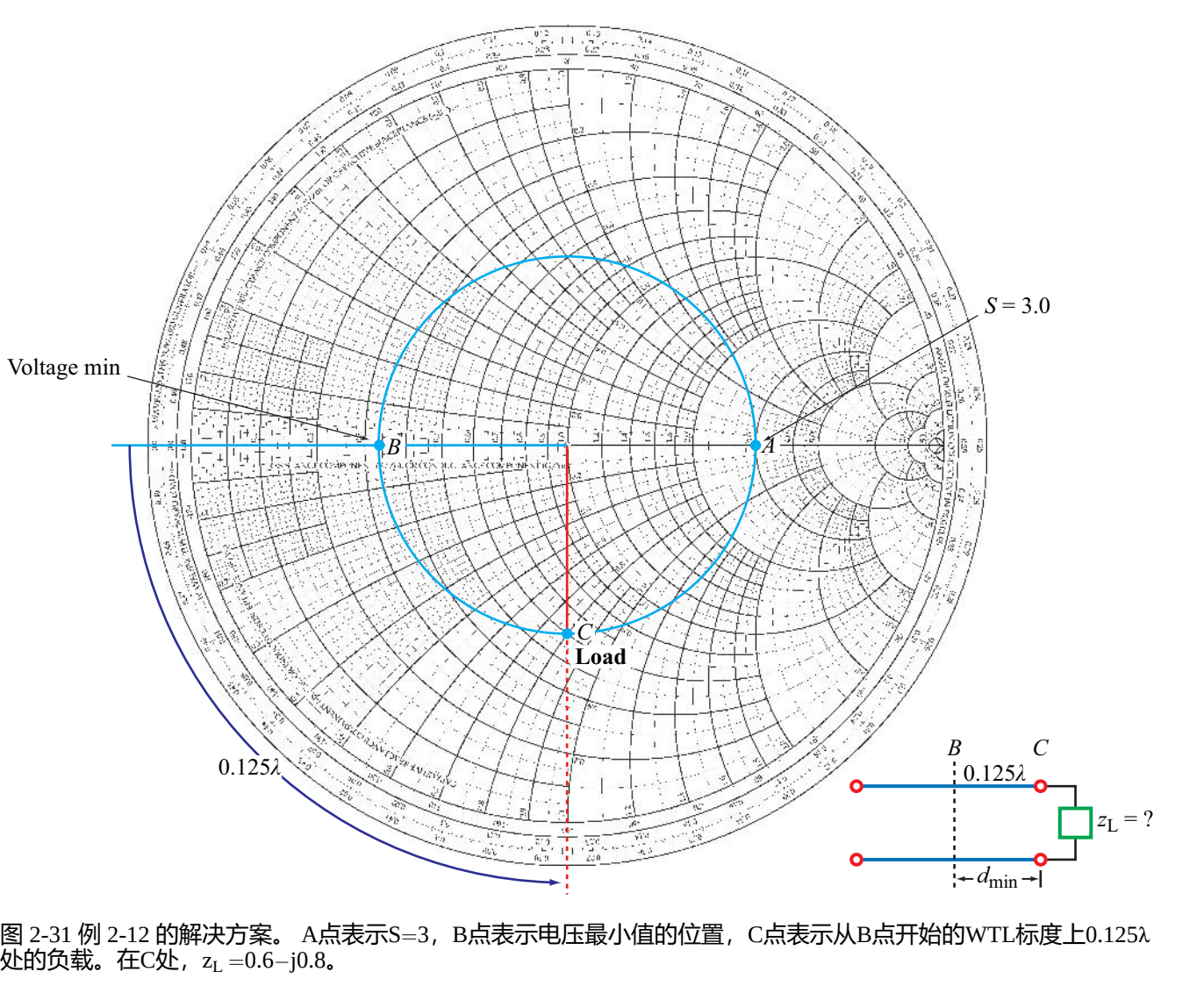


图 2-31 例 2-12 的解决方案。A 点表示 $S=3$ ，B 点表示电压最小值的位置，C 点表示从 B 点开始的 WTL 标度上 0.125λ 处的负载。在 C 处， $z_L = 0.6 - j0.8$ 。

► 当传输线的特性阻抗 $Z_0 = Z_L$ 时，传输线被称为与负载匹配，在这种情况下，沿传输线向负载传播的波不会反射回源。◀

由于传输线的主要用途是传输功率或传输编码信号（例如数字数据），因此匹配的负载可确保源传递到传输线的所有功率都传输到负载（并且不会将回波中继回源）。
将负载与传输线匹配的最简单解决方案是设计负载电路，使其阻抗

$Z_L = Z_0$ 。不幸的是，这在实践中可能是不可能的，因为负载电路可能必须满足其他要求。另一种解决方案是在负载和传输线之间放置一个阻抗匹配网络，如图 2-32 所示。

► 匹配网络的目的是消除终端 MM' 对从源入射的波的反射。尽管 AA' 和 MM' 之间可能发生多次反射，但馈线上仅存在前向行波。◀

Module 2.6 Interactive Smith Chart Locate the load on the Smith chart; display the corresponding reflection coefficient and SWR circle; “move” to a new location at a distance d from the load; read the wave impedance $Z(d)$ and phase-shifted reflection coefficient Γ_d ; perform impedance to admittance transformations and vice versa; and use all of these tools to solve transmission-line problems via the Smith chart.

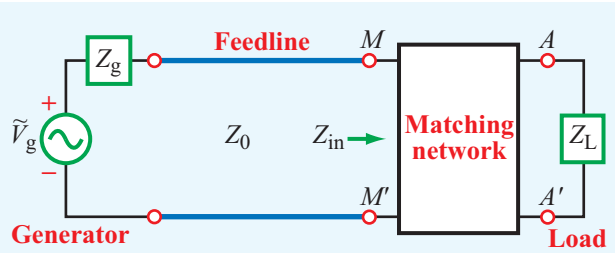
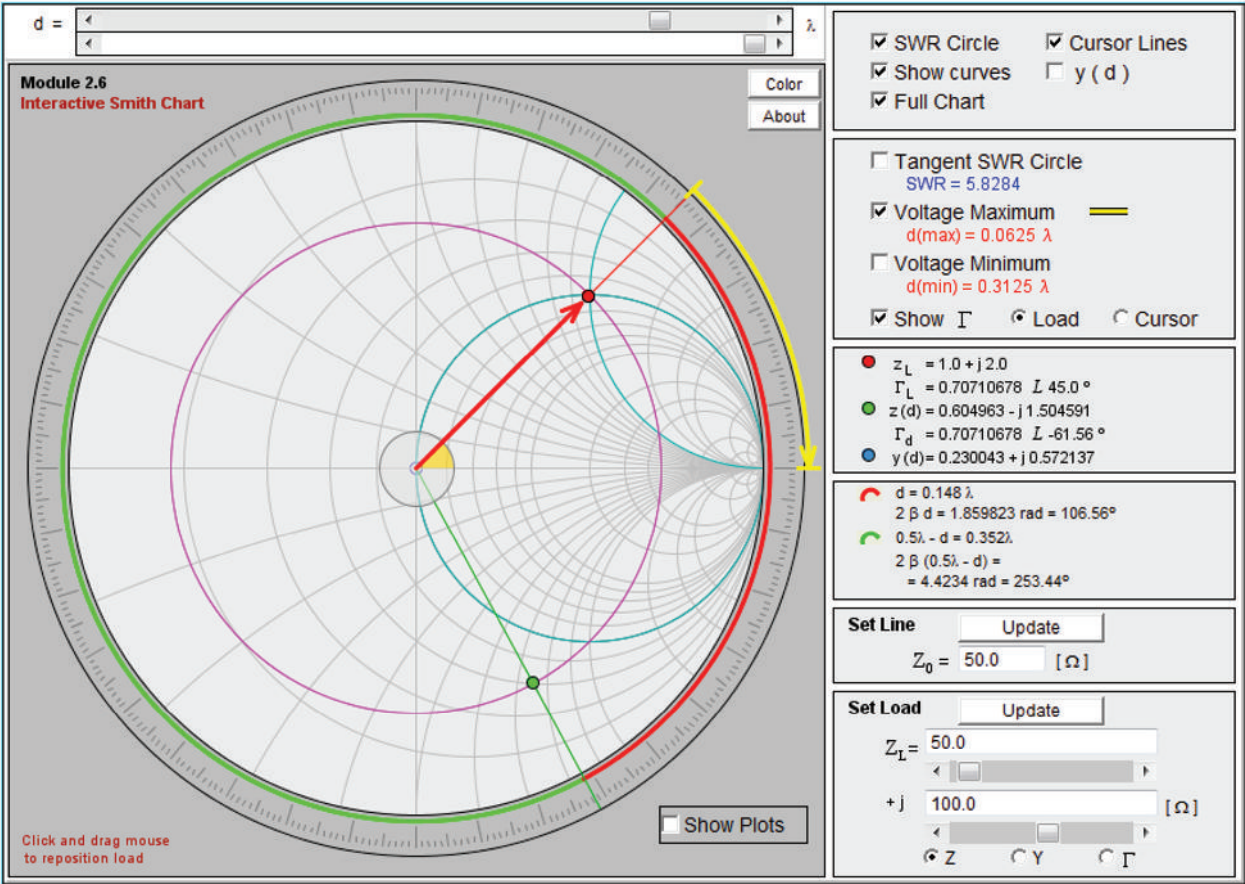


Figure 2-32 The function of a matching network is to transform the load impedance Z_L such that the input impedance Z_{in} looking into the network is equal to Z_0 of the feedline.

Within the matching network, reflections can occur at both terminals (AA' and MM'), creating a standing-wave pattern, but the net result (of all of the multiple reflections within the matching network) is that the wave incident from the source experiences no reflection when it reaches terminals MM' . This is achieved by designing the matching network to exhibit an impedance equal to Z_0 at MM' when looking into the network from the transmission line side. If the network is lossless, then all of the power going into it will end up in the load.

模块 2.6 交互式史密斯圆图 在史密斯圆图上定位负载；显示相应的反射系数和驻波比圆；“移动”到距负载距离 d 的新位置；读取波阻抗 $Z(d)$ 和相移反射系数 Γ_d ；执行阻抗到导纳的变换，反之亦然；并使用所有这些工具通过史密斯圆图解决传输线问题。

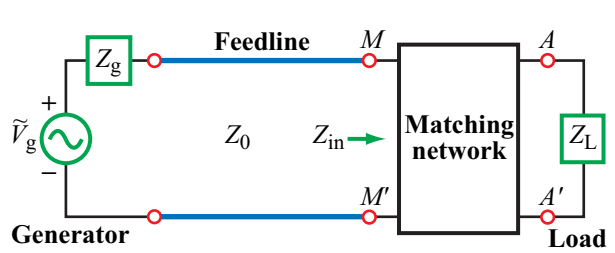
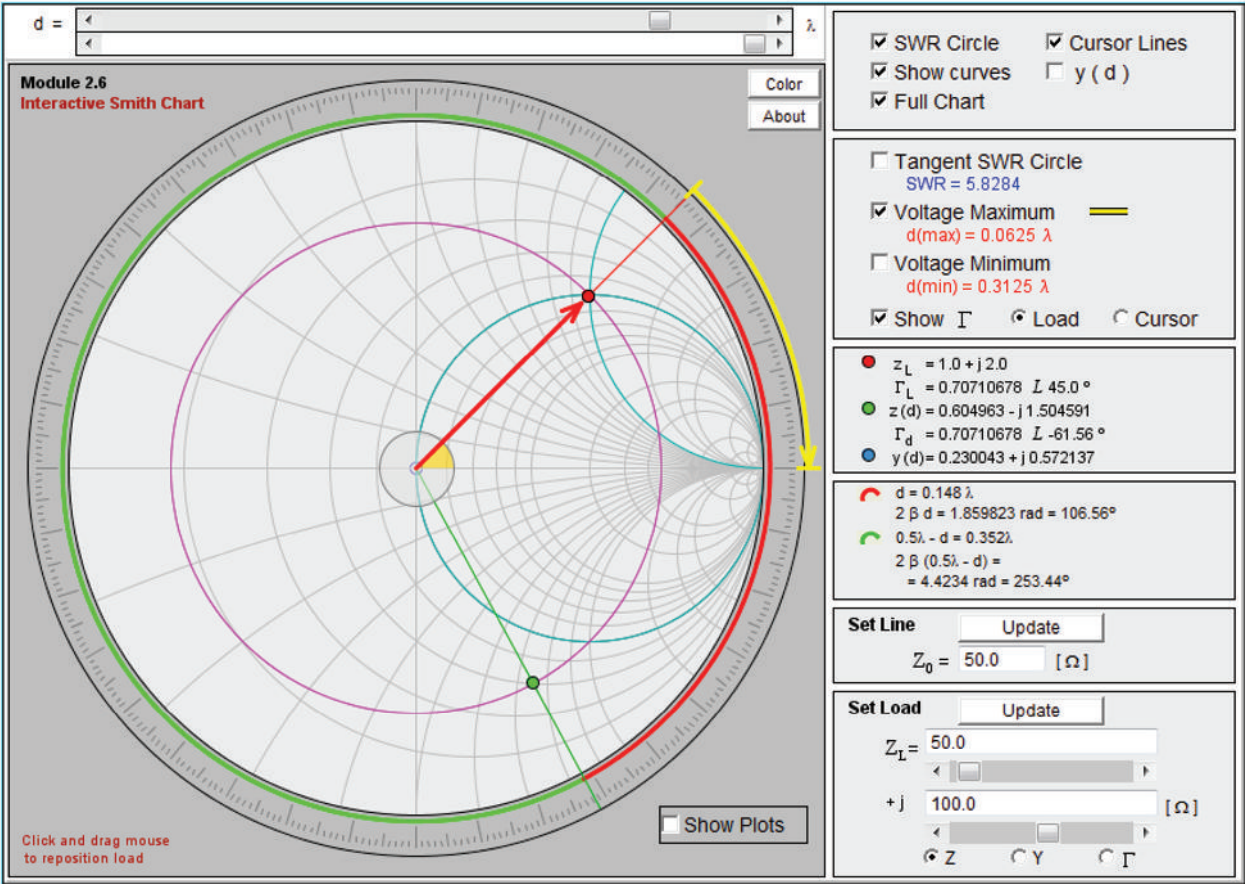


图2-32 匹配网络的功能是变换负载阻抗 Z_L ，使得网络中的输入阻抗 Z_{in} 等于馈线的 Z_0 。

在匹配网络内，反射可能发生在两个终端（ AA' 和 MM' ），从而产生驻波模式，但最终结果（匹配网络内的所有多次反射）是从源入射的波在到达终端 MM' 时不会经历反射。这是通过设计匹配网络来实现的，当从传输线侧观察网络时，匹配网络在 MM' 处表现出等于 Z_0 的阻抗。如果网络是无损的，那么进入网络的所有功率最终都会进入负载。

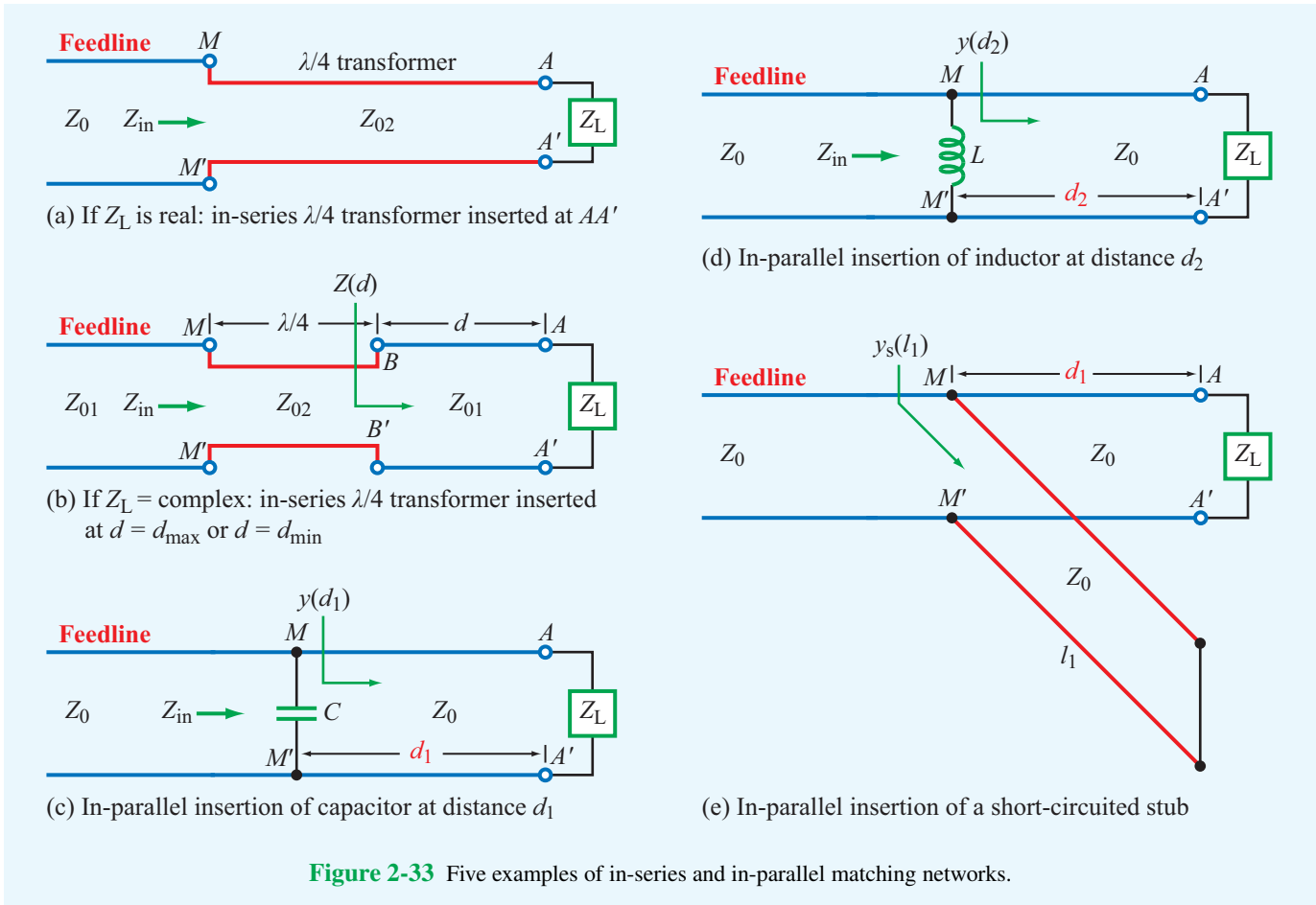


Figure 2-33 Five examples of in-series and in-parallel matching networks.

► Matching networks may consist of lumped elements, such as capacitors and inductors (but not resistors because resistors incur ohmic losses), or of sections of transmission lines with appropriate lengths and terminations. ◀

The matching network, which is intended to match a load impedance $Z_L = R_L + jX_L$ to a lossless transmission line with characteristic impedance Z_0 , may be inserted either in series (between the load and the feedline) as in Fig. 2-33(a) and (b) or in parallel (Fig. 2-33(c) to (e)). In either case, the network has to transform the real part of the load impedance from R_L (at the load) to Z_0 at MM' in Fig. 2-32 and transform the reactive part from X_L (at the load) to zero at MM' . To achieve these two transformations, the matching network must have at least two degrees of freedom (that is, two adjustable parameters).

If $X_L = 0$, the problem reduces to a single transformation, in which case matching can be realized by inserting

a quarter-wavelength transformer (Section 2-8.5) next to the load (Fig. 2-33(a)).

► For the general case where $X_L \neq 0$, a $\lambda/4$ transformer can be designed to provide the desired matching, but it has to be inserted at a distance $d_{\max}$ or $d_{\min}$ from the load (Fig. 2-33(b)), where $d_{\max}$ and $d_{\min}$ are the distances to voltage maxima and minima, respectively. ◀

The design procedure is outlined in Module 2.7. The in-parallel insertion networks shown in Fig. 2-33(c)–(e) are the subject of Examples 2-13 and 2-14.

2-11.1 Lumped-Element Matching

In the arrangement shown in Fig. 2-34, the matching network consists of a single lumped element, either a capacitor or an

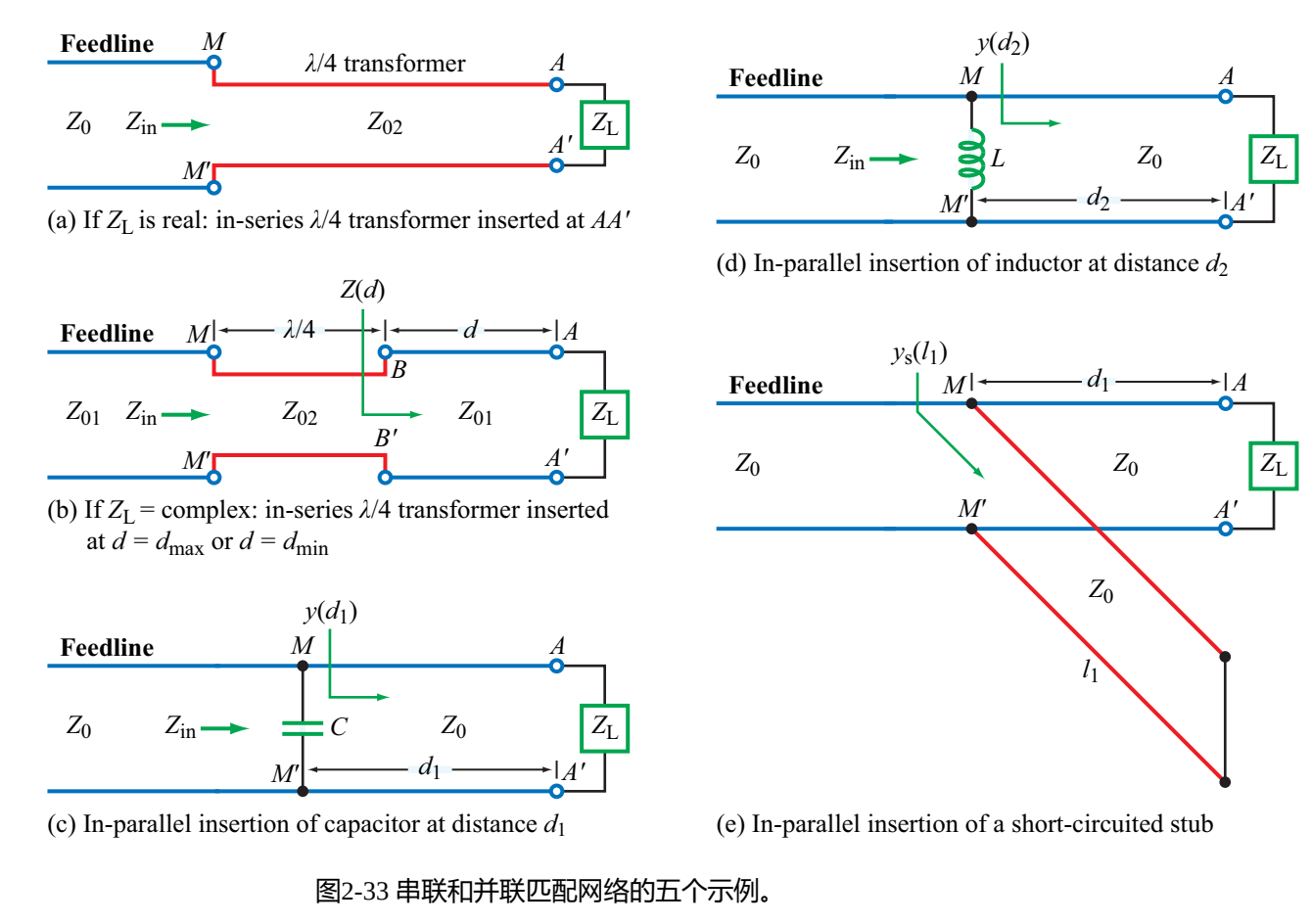


图2-33 串联和并联匹配网络的五个示例。

► 匹配网络可能由集总元件组成，例如电容器和电感器（但不是电阻器，因为电阻器会产生欧姆损耗），或者由具有适当长度和终端的传输线部分组成。◀

匹配网络的目的是将负载阻抗 $Z_L = R_L + jX_L$ 与具有特性阻抗 Z_0 的无损传输线相匹配，可以如图 2-33(a) 和 (b) 所示串联插入（在负载和馈线之间），也可以如图 2-33(c) 至 (e) 所示并联插入。无论哪种情况，网络都必须将负载阻抗的实部从负载处的 R_L （转换为图 2-32 中 MM' 处的 Z_0 ），并将无功部分从负载处的 X_L （转换为 MM' 处的零）。为了实现这两种变换，匹配网络必须至少具有两个自由度（即两个可调整的参数）。

如果 $X_L = 0$ ，则问题简化为单个变换，在这种情况下，可以通过插入来实现匹配

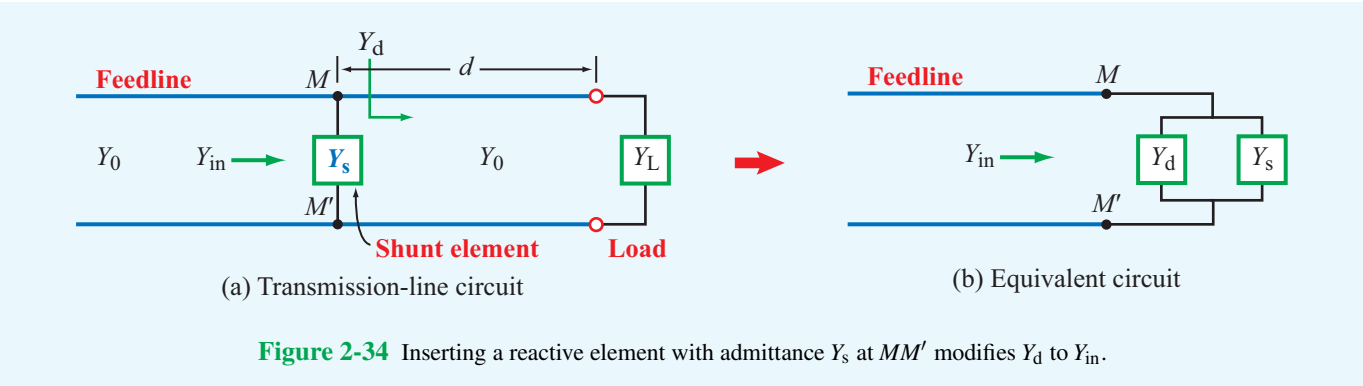
负载旁边的四分之一波长变压器（第 2-8.5 节）（图 2-33(a)）。

► 对于 $X_L \neq 0$ 的一般情况，可以设计一个 $\lambda/4$ 变压器来提供所需的匹配，但必须将其插入到距负载的距离 $d_{\max}$ 或 $d_{\min}$ 处（图 2-33(b)），其中 $d_{\max}$ 和 $d_{\min}$ 分别是到电压最大值和最小值的距离。◀

设计过程在模块 2.7 中概述。图 2-33(c)–(e) 中所示的并行插入网络是示例 2-13 和 2-14 的主题。

2-11.1 集总元件匹配

在图 2-34 所示的布置中，匹配网络由单个集总元件（电容器或电感器）组成。



inductor, connected in parallel with the line at a distance d from the load. Parallel connections call for working in the admittance domain. Hence, the load is denoted by an admittance Y_L , and the line has a characteristic admittance Y_0 . The shunt element has an admittance Y_s . At MM' , Y_d is the admittance due to the transmission-line segment to the right of MM' . The input admittance Y_{in} (referenced at a point just to the left of MM') is equal to the sum of Y_d and Y_s :

$$Y_{in} = Y_d + Y_s. \quad (2.138)$$

In general, Y_d is complex, and Y_s is purely imaginary because it represents a reactive element (capacitor or inductor). Hence, Eq. (2.138) can be written as

$$Y_{in} = (G_d + jB_d) + jB_s = G_d + j(B_d + B_s). \quad (2.139)$$

When all quantities are normalized to Y_0 , Eq. (2.139) becomes

$$y_{in} = g_d + j(b_d + b_s). \quad (2.140)$$

To achieve a matched condition at MM' , it is necessary that $y_{in} = 1 + j0$, which translates into two specific conditions, namely

$$g_d = 1 \quad (\text{real-part condition}), \quad (2.141a)$$

$$b_s = -b_d \quad (\text{imaginary-part condition}). \quad (2.141b)$$

The real-part condition is realized through the choice of d , which is the distance from the load to the shunt element, and the imaginary-part condition is realized through the choice of lumped element (capacitor or inductor) and its value. These two choices are the two degrees of freedom needed in order to match the load to the feedline.

Example 2-13: Lumped Element

A load impedance $Z_L = 25 - j50 \, \Omega$ is connected to a $50 \, \Omega$ transmission line. Insert a shunt element to eliminate reflections towards the sending end of the line. Specify the insert location d (in wavelengths), the type of element, and its value given that $f = 100 \text{ MHz}$.

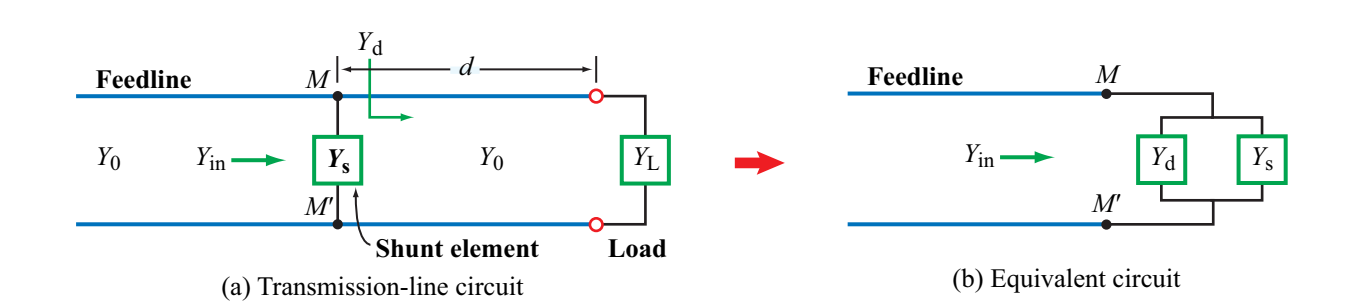
Solution: The normalized load impedance is

$$z_L = \frac{Z_L}{Z_0} = \frac{25 - j50}{50} = 0.5 - j1,$$

which is represented by point A on the Smith chart of **Fig. 2-35**. Next, we draw the constant S circle through point A . As alluded to earlier, to perform the matching task, it is easier to work with admittances than with impedances. The normalized load admittance y_L is represented by point B , which is obtained by rotating point A over 0.25λ or equivalently by drawing a line from point A through the chart center to the image of point A on the S circle. The value of y_L at B is

$$y_L = 0.4 + j0.8,$$

and it is located at position 0.115λ on the WTG scale. In the admittance domain, the r_L circles become g_L circles, and the x_L circles become b_L circles. To achieve matching, we need to move from the load toward the generator a distance d such that the normalized input admittance y_d of the line terminated in the load (**Fig. 2-34**) has a real part of 1. This condition is satisfied by either of the two **matching points** C and D on the Smith charts of **Figs. 2-35** and **2-36**, respectively, corresponding to intersections of the S circle with the $g_L = 1$ circle. Points C and D represent two possible solutions for the distance d in **Fig. 2-34(a)**.



电感器，与负载距离为 d 的线路并联。并行连接要求在准入域中工作。因此，负载用导纳 Y_L 表示，线路具有特征导纳 Y_0 。分流元件的导纳为 Y_s 。在 MM' 处， Y_d 是 M M' 右侧传输线段引起的导纳。以 MM' 左侧点为参考的输入导纳 Y_{in} (等于 Y_d 和 Y_s 之和)：

$$Y_{in} = Y_d + Y_s. \quad (2.138)$$

一般来说， Y_d 是复数， Y_s 是纯虚数，因为它代表电抗元件（电容器或电感器）。因此，等式 (2.138) 可以写成

$$Y_{in} = (G_d + jB_d) + jB_s = G_d + j(B_d + B_s). \quad (2.139)$$

当所有量都归一化为 Y_0 时，方程 (2.139) 变为

$$y_{in} = g_d + j(b_d + b_s). \quad (2.140)$$

为了在 MM' 处实现匹配条件，需要 $y_{in} = 1 + j0$ ，它转化为两个特定条件，即

$$g_d = 1 \quad (\text{real-part condition}), \quad (2.141a)$$

$$b_s = -b_d \quad (\text{imaginary-part condition}). \quad (2.141b)$$

实部条件通过选择 d (负载到分流元件的距离) 来实现，虚部条件通过选择集总元件（电容器或电感器）及其值来实现。这两个选择是使负载与馈线相匹配所需的两个自由度。

例 2-13: 集总元件

负载阻抗 $Z_L = 25 - j50 \, \Omega$ 连接到 $50 \, \Omega$ 传输线。插入分流元件以消除线路发送端的反射。指定插入位置 d (以波长为单位)、元素类型及其值 (假定 $f = 100 \text{ MHz}$)。

解：归一化负载阻抗为

$$z_L = \frac{Z_L}{Z_0} = \frac{25 - j50}{50} = 0.5 - j1,$$

其由图2-35的史密斯圆图上的A点表示。接下来，我们通过 A 点绘制常数 S 圆。如前所述，为了执行匹配任务，使用导纳比使用阻抗更容易。归一化负载导纳 y_L 用 B 点表示， B 点是将 A 点旋转 0.25λ 或等效地从 A 点穿过图表中心到 S 圆上 A 点的图像画一条线得到的。 B 处 y_L 的值为

$$y_L = 0.4 + j0.8,$$

它位于WTG标尺上的位置 0.115λ 。在导纳域中， r_L 圆变为 g_L 圆， x_L 圆变为 b_L 圆。为了实现匹配，我们需要从负载向发电机移动距离 d ，使得负载端接线路 (图 2-34) 的归一化输入导纳 y_d 的实部为 1。图 2-34 的史密斯圆图上的两个匹配点 C 和 D 中的任何一个都可以满足这个条件。2-35和2-36分别对应于S圆与 $g_L = 1$ 圆的交点。 C 点和 D 点代表图2-34(a)中距离 d 的两种可能的解。

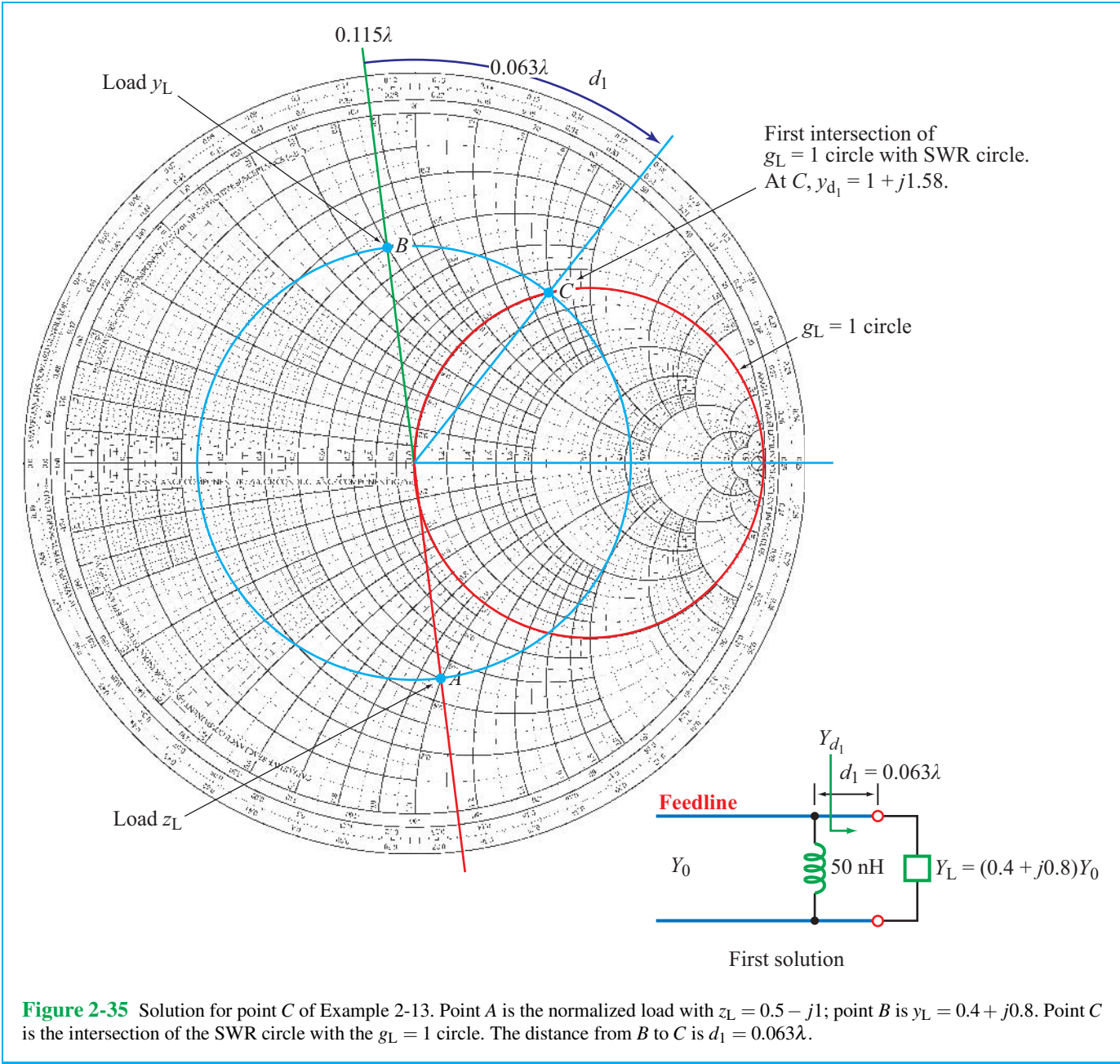


Figure 2-35 Solution for point C of Example 2-13. Point A is the normalized load with $z_L = 0.5 - j1$; point B is $y_L = 0.4 + j0.8$. Point C is the intersection of the SWR circle with the $g_L = 1$ circle. The distance from B to C is $d_1 = 0.063\lambda$.

Solution for Point C (Fig. 2-35): At C,

$$y_{d_1} = 1 + j1.58,$$

which is located at 0.178λ on the WTG scale. The distance between points B and C is

$$d_1 = (0.178 - 0.115)\lambda = 0.063\lambda.$$

Looking from the generator toward the parallel combination of the line connected to the load and the shunt element, the normalized input admittance at terminals MM' is

$$y_{in1} = y_{s1} + y_{d_1},$$

(continued on p. 101)

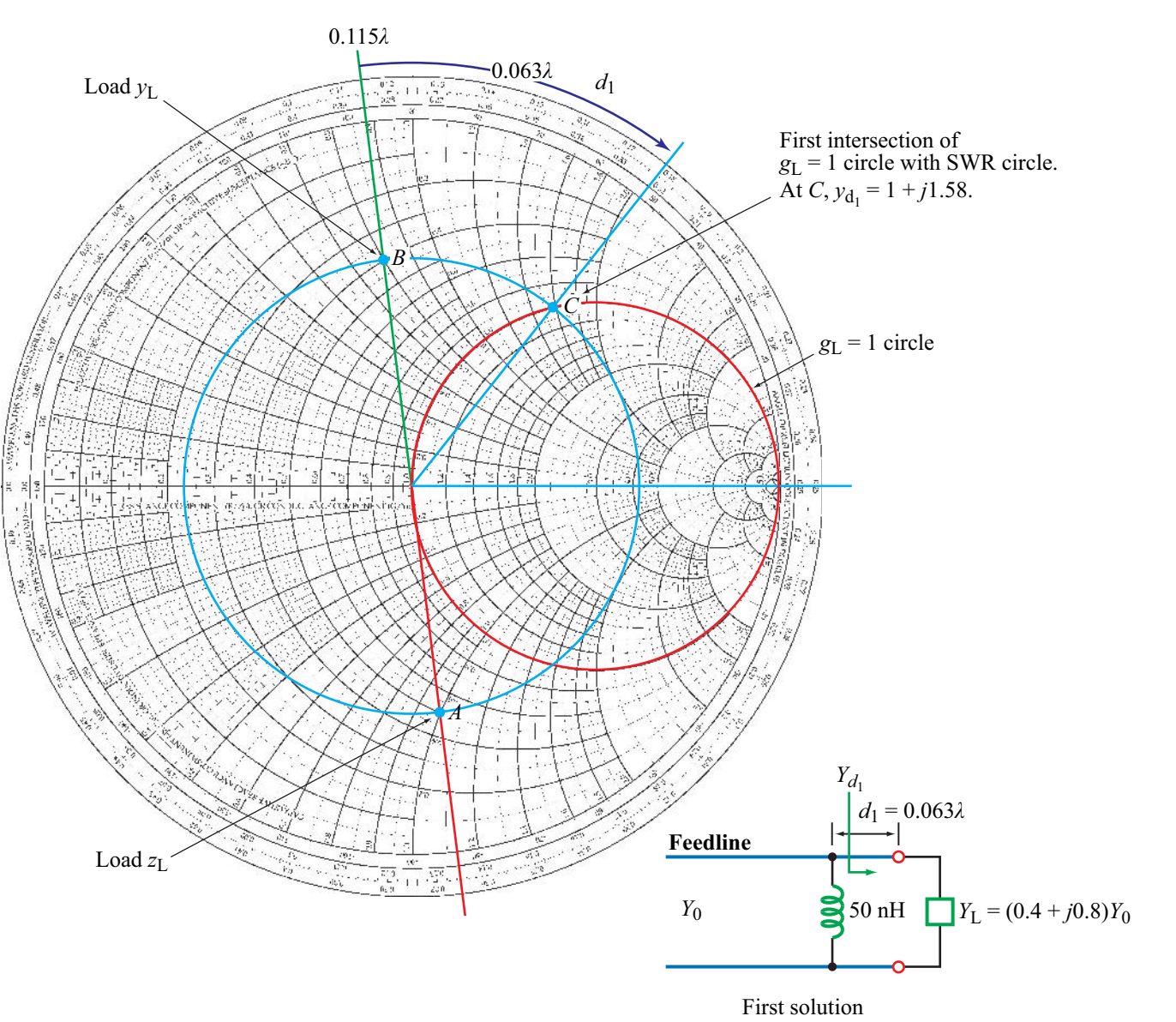


图 2-35 例 2-13 中 C 点的解法。A 点为 $z_L = 0.5 - j1$ 的归一化载荷；B 点为 $y_L = 0.4 + j0.8$ 。C 点是 SWR 圆与 $g_L = 1$ 圆的交点。B 到 C 的距离为 $d_1 = 0.063\lambda$ 。

C 点的解 (图 2-35)：在 C 处，

$$y_{d_1} = 1 + j1.58,$$

它位于 WTG 尺度上的 0.178λ 。B 点和 C 点之间的距离为

$$d_1 = (0.178 - 0.115)\lambda = 0.063\lambda.$$

从发电机看向连接到负载的线路和并联元件的并联组合，端子 MM' 处的归一化输入导纳为

$$y_{in1} = y_{s1} + y_{d_1},$$

(继续上页。101)

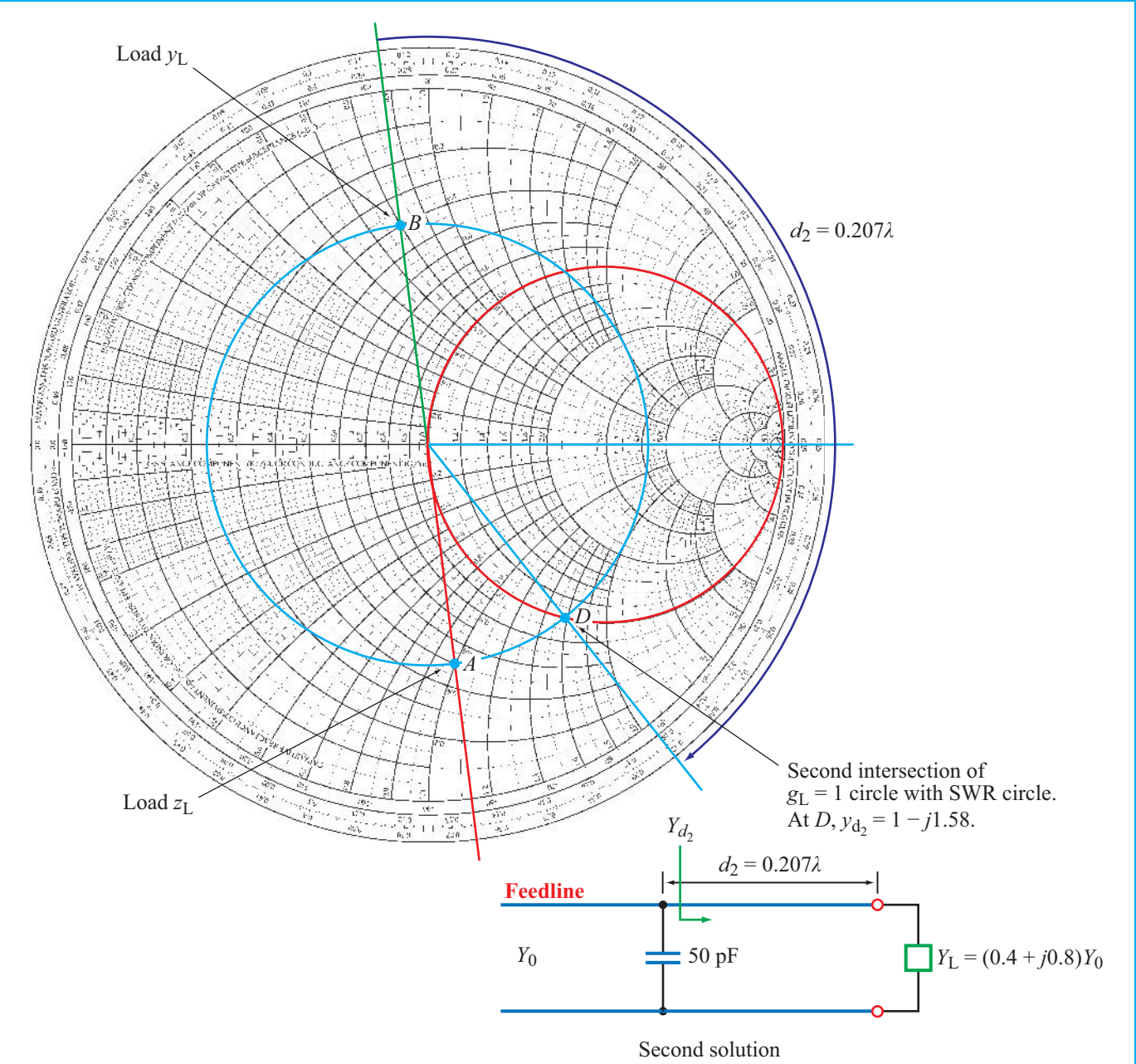


Figure 2-36 Solution for point D of Example 2-13. Point D is the second point of intersection of the SWR circle and the $g_L = 1$ circle. The distance B to D gives $d_2 = 0.207\lambda$.

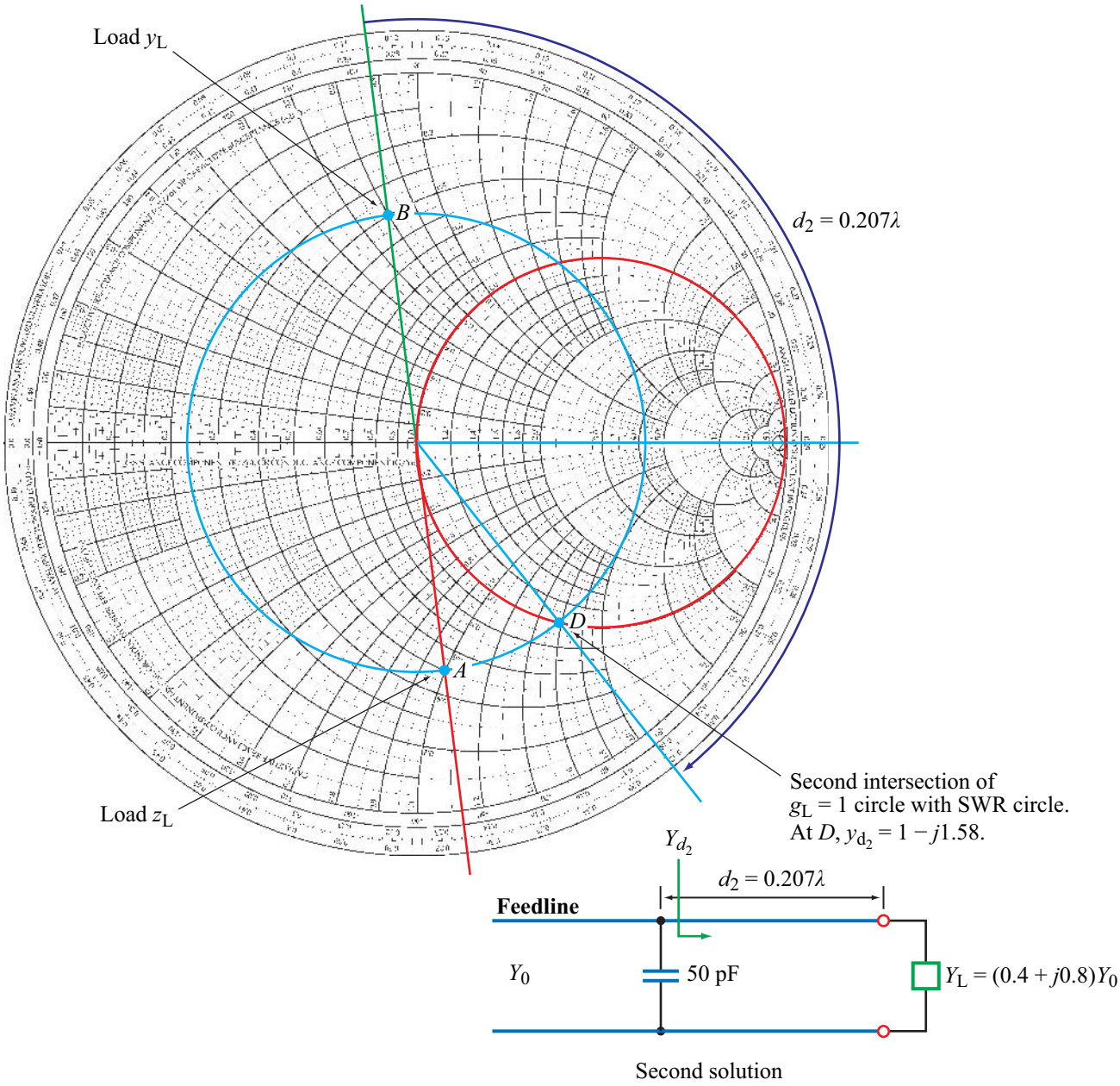


图 2-36 例 2-13 中 D 点的解法。点 D 是 SWR 圆和 $g_L = 1$ 圆的第二个交点。点 B 到 D 的距离为 $d_2 = 0.207\lambda$ 。

2-11 IMPEDANCE MATCHING

101

where y_{s1} is the normalized input admittance of the shunt element. To match the feed line to the parallel combination, we need $y_{in1} = 1 + j0$. Hence, we need y_{s1} to cancel the imaginary part of y_{d1} ; that is,

$$y_{s1} = -j1.58.$$

The corresponding impedance of the lumped element is

$$Z_{s1} = \frac{1}{Y_{s1}} = \frac{1}{y_{s1}Y_0} = \frac{Z_0}{jb_{s1}} = \frac{Z_0}{-j1.58} = \frac{jZ_0}{1.58} = j31.62 \Omega.$$

Since the value of Z_{s1} is positive, the element to be inserted should be an inductor and its value should be

$$L = \frac{31.62}{\omega} = \frac{31.62}{2\pi \times 10^8} = 50 \text{ nH}.$$

The results of this solution have been incorporated into the circuit of Fig. 2-35.

Solution for Point D (Fig. 2-36): At point D,

$$y_{d2} = 1 - j1.58,$$

and the distance between points B and D is

$$d_2 = (0.322 - 0.115)\lambda = 0.207\lambda.$$

The needed normalized admittance of the reactive element is

$$y_{s2} = +j1.58.$$

Hence,

$$Z_{s2} = -j31.62 \Omega,$$

which is the impedance of a capacitor with

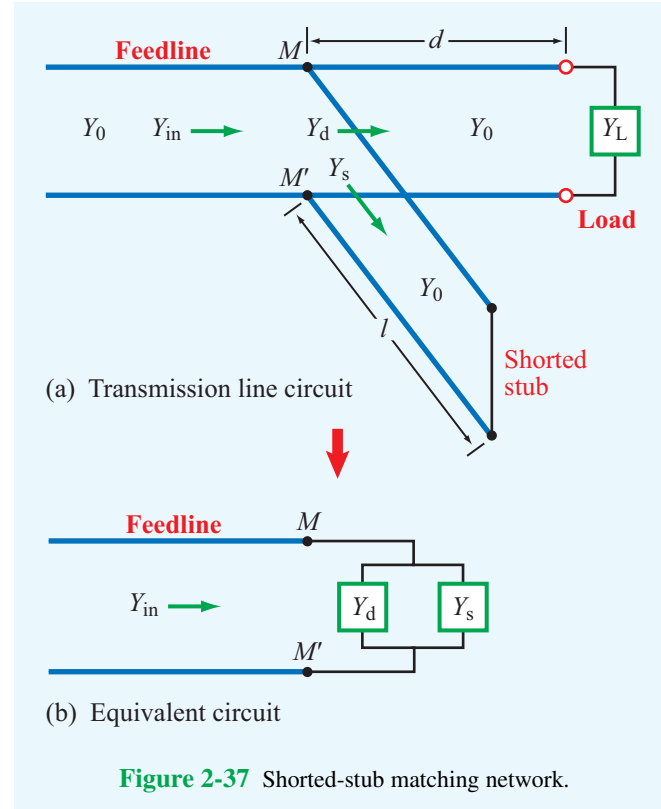
$$C = \frac{1}{31.62\omega} = 50 \text{ pF}.$$

Figure 2-36 displays the circuit solution for d_2 and C .

2-11.2 Single-Stub Matching

The single-stub matching network shown in Fig. 2-37(a) consists of two transmission line sections: one of length d connecting the load to the feedline at MM' and another of length l connected in parallel with the other two lines at MM' . This second line is called a **stub**, and it is usually terminated in either a short or open circuit; hence, its input impedance and admittance are purely reactive. The stub shown in Fig. 2-37(a) has a short-circuit termination.

► The required two degrees of freedom are provided by the length l of the stub and the distance d from the load to the stub position. ◀



Because at MM' the stub is added in parallel to the line (which is why it is called a **shunt** stub), it is easier to work with admittances than with impedances. The matching procedure consists of two steps. In the first step, the distance d is selected to transform the load admittance, $Y_L = 1/Z_L$, into an admittance of the form $Y_d = Y_0 + jB$ when looking toward the load at MM' . Then in the second step, the length l of the stub line is selected so that its input admittance Y_s at MM' is equal to $-jB$. The parallel sum of the two admittances at MM' yields Y_0 , which is the characteristic admittance of the line. The procedure is illustrated by Example 2-14.

Example 2-14: Single-Stub Matching

Repeat Example 2-13, but use a shorted stub (instead of a lumped element) to match the load impedance $Z_L = (25 - j50) \Omega$ to the 50Ω transmission line.

Solution: In Example 2-13, we demonstrated that the load can be matched to the line via either of two solutions:

1. $d_1 = 0.063\lambda$, and $y_{s1} = jb_{s1} = -j1.58$,

2-11 IMPEDANCE MATCHING

101

其中 y_{s1} 是分流元件的归一化输入导纳。为了将馈线与并联组合相匹配，我们需要 $y_{in1} = 1 + j0$ 。因此，我们需要 y_{s1} 来抵消 y_{d1} 的虚部；那是，

$$y_{s1} = -j1.58.$$

集总元件的相应阻抗为

$$Z_{s1} = \frac{1}{Y_{s1}} = \frac{1}{y_{s1}Y_0} = \frac{Z_0}{jb_{s1}} = \frac{Z_0}{-j1.58} = \frac{jZ_0}{1.58} = j31.62 \Omega.$$

由于 Z_{s1} 的值为正，因此要插入的元件应该是电感器，其值应该是

$$L = \frac{31.62}{\omega} = \frac{31.62}{2\pi \times 10^8} = 50 \text{ nH}.$$

该解决方案的结果已合并到图 2-35 的电路中。

D 点的解 (图 2-36)：在 D 点，

$$y_{d2} = 1 - j1.58,$$

B 点和 D 点之间的距离为

$$d_2 = (0.322 - 0.115)\lambda = 0.207\lambda.$$

电抗元件所需的归一化导纳为

$$y_{s2} = +j1.58.$$

因此，

$$Z_{s2} = -j31.62 \Omega,$$

这是电容器的阻抗

$$C = \frac{1}{31.62\omega} = 50 \text{ pF}.$$

图 2-36 显示了 d_2 和 C 的电路解决方案。

2-11.2 单存根匹配

图2-37 (a) 所示的单短截线匹配网络由两个传输线部分组成：一个长度为 d ，在 MM' 处将负载连接到馈线，另一个长度为 l ，在 MM' 处与其他两条线路并联连接。第二条线称为短截线，通常以短路或开路终止；因此，其输入阻抗和导纳是纯电抗性的。图2-37(a)所示的短截线具有短路终端。

► 所需的两个自由度由短截线的长度 l 和从负载到短截线位置的距离 d 提供。 ◀

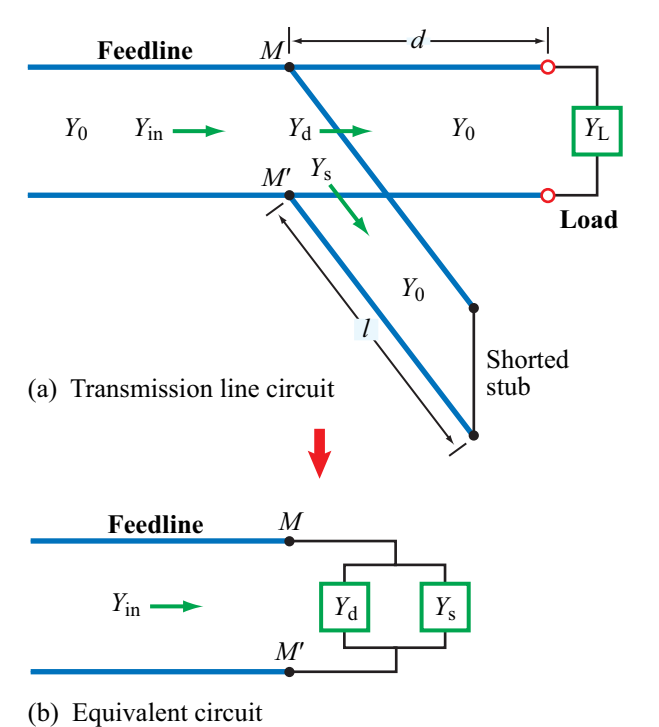


图2-37 短截线匹配网络。

因为在 MM' 处，短截线与线路并联添加（这就是它被称为并联短截线的原因），因此使用导纳比使用阻抗更容易。匹配过程由两个步骤组成。第一步，选择距离 d ，将负载导纳 $Y_L = 1/Z_L$ 转换为 $Y_d = Y_0 + jB$ 形式的导纳（当观察 MM' 处的负载时）。然后在第二步中，选择短截线的长度 l ，使其在 MM' 处的输入导纳 Y_s 等于 $-jB$ 。 MM' 处两个导纳的并联和得出 Y_0 ，它是线路的特征导纳。示例 2-14 说明了该过程。

示例 2-14: 单存根匹配

重复示例 2-13，但使用短路短截线（而不是集总元件）将负载阻抗 $Z_L = (25 - j50) \Omega$ 与 50Ω 传输线相匹配。

解决方案：在示例 2-13 中，我们演示了可以通过以下两种解决方案之一将负载与线路匹配：

1. $d_1 = 0.063\lambda$, and $y_{s1} = jb_{s1} = -j1.58$,

2. $d_2 = 0.207\lambda$, and $y_{s_2} = jb_{s_2} = j1.58$.

The locations of the insertion points at distances d_1 and d_2 remain the same, but now our task is to select corresponding lengths l_1 and l_2 of shorted stubs that present the required admittances at their inputs.

To determine l_1 , we use the Smith chart in **Fig. 2-38**. The normalized admittance of a short circuit is $-j\infty$, which is represented by point E on the Smith chart with position 0.25λ on the WTG scale. A normalized input admittance of $-j1.58$ is located at point F with position 0.34λ on the WTG scale. Hence,

$$l_1 = (0.34 - 0.25)\lambda = 0.09\lambda.$$

Similarly, $y_{s_2} = j1.58$ is represented by point G with position 0.16λ on the WTG scale of the Smith chart in **Fig. 2-39**. Rotating from point E to point G involves a rotation of 0.25λ plus an additional rotation of 0.16λ or

$$l_2 = (0.25 + 0.16)\lambda = 0.41\lambda.$$

Example 2-15: $\lambda/4$ Transformer for Complex Load

Design a quarter-wavelength transformer to match a load with $Z_L = (100 + j100)\,\Omega$ to a $50\,\Omega$ line.

Solution: The required transmission-line circuit is shown in **Fig. 2-40(a)** on p. 105. We need to insert a $\lambda/4$ transformer to eliminate reflections at MM' . To that end, we need to specify the insertion distance d and the characteristic impedance Z_{0_2} of the $\lambda/4$ section so that $Z_{\text{in}} = Z_{0_1}$.

A $\lambda/4$ section with characteristic impedance Z_{0_2} transfers an impedance Z_a at one end of the section to impedance Z_b at the other end such that

$$Z_a Z_b = Z_{0_2}^2. \tag{2.142}$$

In the present case (**Fig. 2-40(a)**), $Z_a = Z(d)$, the input impedance at BB' looking towards the load, and $Z_b = Z_{\text{in}} = Z_{0_1} = 50\,\Omega$. Since the $\lambda/4$ transformer is a lossless line, its impedance Z_{0_2} is purely real. Consequently, to satisfy Eq. (2.142), it is necessary that $Z(d)$ be purely real as well, which is the key to how we select the distance d . Given this rationale, we proceed as follows.

1. The normalized load impedance is

$$z_L = \frac{Z_L}{Z_{0_1}} = \frac{100 + j100}{50} = 2 + j2,$$

which is shown as point A in the Smith chart of **Fig. 2-40(b)**.

2. Move on the SWR circle until $z(d)$ becomes purely real, which occurs at points B and C .

(a) Point B:

$$\begin{aligned} d_1 &= (0.25 - 0.209)\lambda \\ &= 0.041\lambda, \end{aligned}$$

$$\begin{aligned} z(d_1) &= 4.27, \\ Z(d_1) &= 4.27 \times 50 = 213.5\,\Omega, \end{aligned}$$

$$\begin{aligned} Z_{0_2} &= \sqrt{Z(d_1) Z_{0_1}} \\ &= \sqrt{213.5 \times 50} \\ &= 103.3\,\Omega. \end{aligned}$$

(b) Point C:

$$\begin{aligned} d_2 &= d_1 + \frac{\lambda}{4} \\ &= (0.041 + 0.25)\lambda = 0.291\lambda, \end{aligned}$$

$$\begin{aligned} z(d_2) &= 0.23, \\ Z(d_2) &= 0.23 \times 50 = 11.5\,\Omega, \end{aligned}$$

$$\begin{aligned} Z_{0_2} &= \sqrt{Z(d_2) Z_{0_1}} \\ &= \sqrt{11.5 \times 50} \\ &= 24.2\,\Omega. \end{aligned}$$

Concept Question 2-24: To match an arbitrary load impedance to a lossless transmission line through a matching network, what is the required minimum number of degrees of freedom that the network should provide?

Concept Question 2-25: In the case of the single-stub matching network, what are the two degrees of freedom?

Concept Question 2-26: When a transmission line is matched to a load through a single-stub matching network, no waves are reflected toward the generator. What happens to the waves reflected by the load and by the shorted stub when they arrive at terminals MM' in **Fig. 2-37**?

2. $d_2 = 0.207\lambda$, and $y_{s_2} = jb_{s_2} = j1.58$.

距离 d_1 和 d_2 处的插入点位置保持不变，但现在我们的任务是选择短路短截线的相应长度 l_1 和 l_2 ，以在其输入端提供所需的导纳。

为了确定 l_1 ，我们使用图 2-38 中的史密斯圆图。短路的归一化导纳为 $-j\infty$ ，它由史密斯圆图上 WTG 标度上位置 0.25λ 上的点 E 表示。归一化输入导纳 $-j1.58$ 位于 WTG 标度上位置 0.34λ 的 F 点。因此，

$$l_1 = (0.34 - 0.25)\lambda = 0.09\lambda.$$

类似地， $y_{s_2} = j1.58$ 由图 2-39 中史密斯圆图 WTG 标度上位置 0.16λ 的 G 点表示。从 E 点旋转到 G 点需要旋转 0.25λ 加上额外的 0.16λ 旋转，或者

$$l_2 = (0.25 + 0.16)\lambda = 0.41\lambda.$$

示例 2-15: 复杂负载的 $\lambda/4$ 变压器

设计一个四分之一波长变压器，将 $Z_L = (100 + j100)\,\Omega$ 的负载与 $50\,\Omega$ 线路相匹配。

解：所需的传输线电路如图2-40(a)所示。105. 我们需要插入一个 $\lambda/4$ 变压器来消除 MM' 处的反射。为此，我们需要指定插入距离 d 和 $\lambda/4$ 部分的特性阻抗 Z_{0_2} ，以便 $Z_{\text{in}} = Z_0$

具有特性阻抗 Z_{0_2} 的 $\lambda/4$ 部分将该部分一端的阻抗 Z_a 传输到另一端的阻抗 Z_b ，使得

$$Z_a Z_b = Z_{0_2}^2. \tag{2.142}$$

在本例中（图 2-40(a)）， $Z_a = Z(d)$ ， BB' 处朝向负载的输入阻抗，以及 $Z_b = Z_{\text{in}} = Z_{0_1} = 50\,\Omega$ 。由于 $\lambda/4$ 变压器是无损线路，因此其阻抗 Z_{0_2} 是纯实数。因此，满足方程 (2.142)， $Z(d)$ 也必须是纯实数，这是我们如何选择距离 d 的关键。考虑到这个理由，我们进行如下操作。

1. 归一化负载阻抗为

$$z_L = \frac{Z_L}{Z_{0_1}} = \frac{100 + j100}{50} = 2 + j2,$$

如图2-40(b)史密斯圆图中的A点所示。

2. 在 SWR 圆上移动，直到 $z(d)$ 变为纯实数，这发生在 B 点和 C 点。

(a) B 点：

$$\begin{aligned} d_1 &= (0.25 - 0.209)\lambda \\ &= 0.041\lambda, \end{aligned}$$

$$\begin{aligned} z(d_1) &= 4.27, \\ Z(d_1) &= 4.27 \times 50 = 213.5\,\Omega, \end{aligned}$$

$$\begin{aligned} Z_{0_2} &= \sqrt{Z(d_1) Z_{0_1}} \\ &= \sqrt{213.5 \times 50} \\ &= 103.3\,\Omega. \end{aligned}$$

(b) C 点：

$$\begin{aligned} d_2 &= d_1 + \frac{\lambda}{4} \\ &= (0.041 + 0.25)\lambda = 0.291\lambda, \end{aligned}$$

$$\begin{aligned} z(d_2) &= 0.23, \\ Z(d_2) &= 0.23 \times 50 = 11.5\,\Omega, \end{aligned}$$

$$\begin{aligned} Z_{0_2} &= \sqrt{Z(d_2) Z_{0_1}} \\ &= \sqrt{11.5 \times 50} \\ &= 24.2\,\Omega. \end{aligned}$$

概念问题 2-24：为了通过匹配网络将任意负载阻抗与无损传输线相匹配，网络应提供的最小自由度数是多少？

概念问题 2-25：在单根匹配网络的情况下，两个自由度是多少？

概念问题 2-26：当传输线通过单短截线匹配网络与负载匹配时，不会有波反射到发电机。当负载和短路短截线反射的波到达图 2-37 中的端子 MM' 时，会发生什么情况？

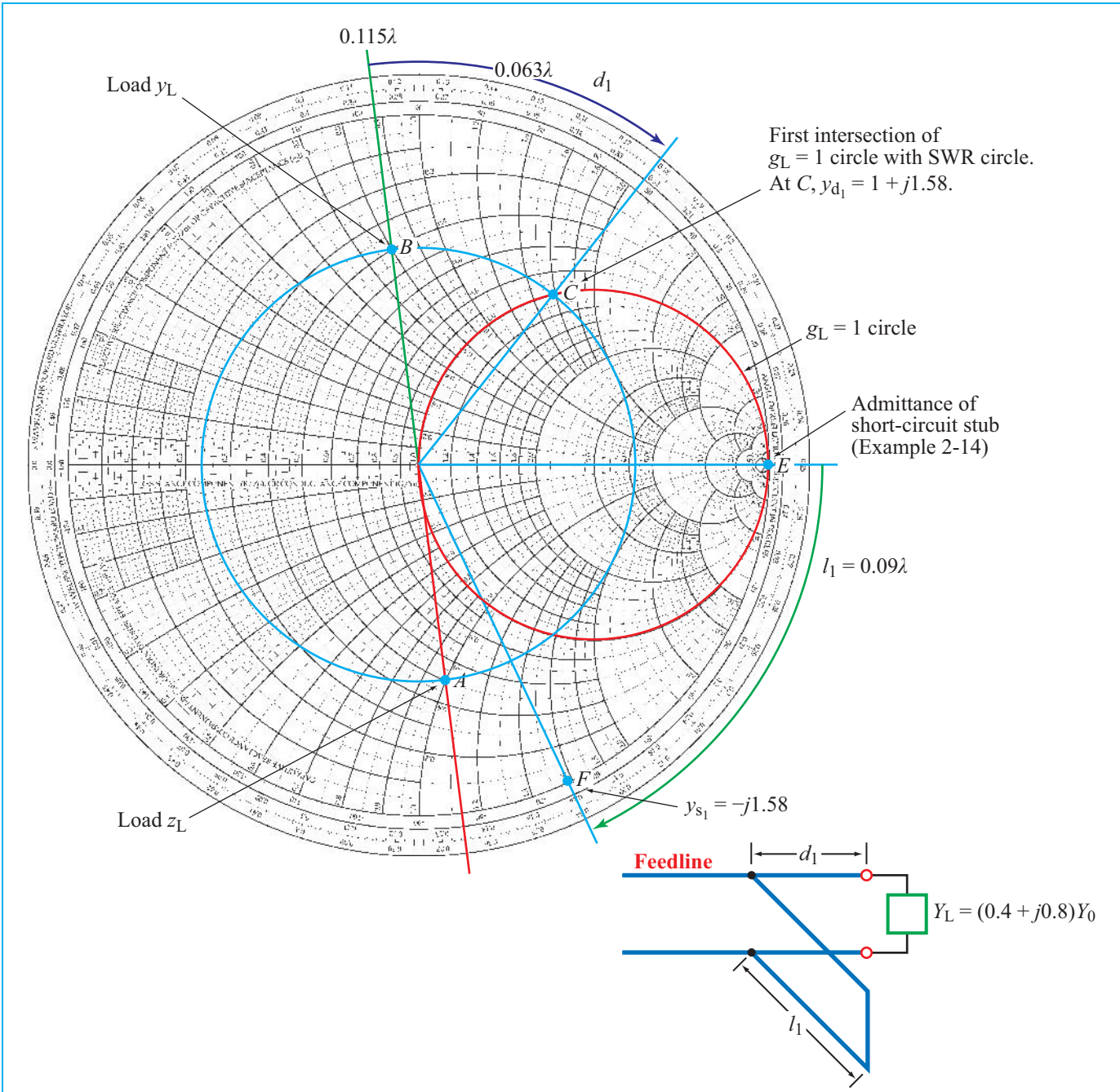


Figure 2-38 Solution for point C of Example 2-14. Point A is the normalized load with $z_L = 0.5 - j1$; point B is $y_L = 0.4 + j0.8$. Point C is the intersection of the SWR circle with the $g_L = 1$ circle. The distance from B to C is $d_1 = 0.063\lambda$. The length of the shorted stub (E to F) is $l_1 = 0.09\lambda$.

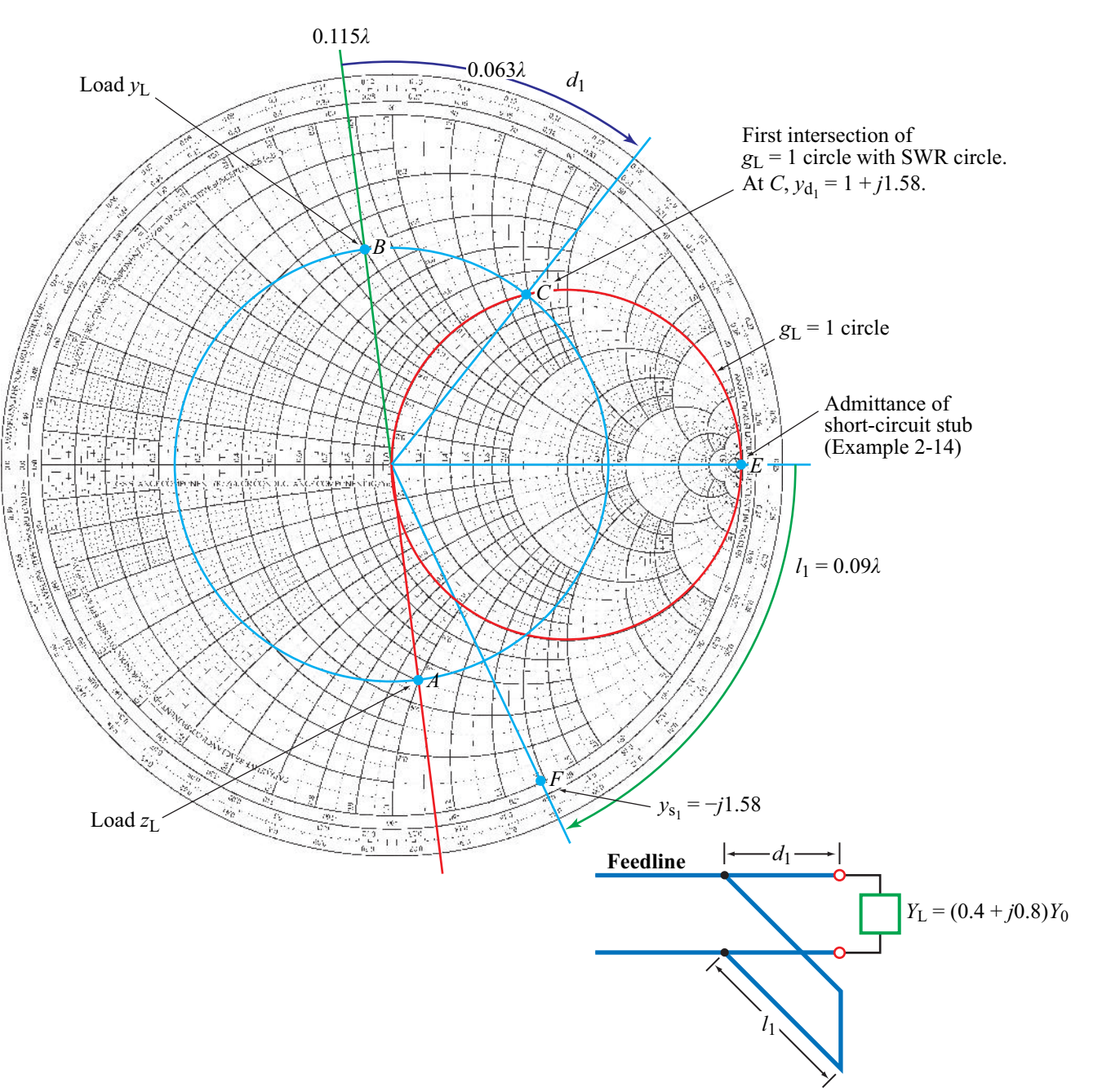
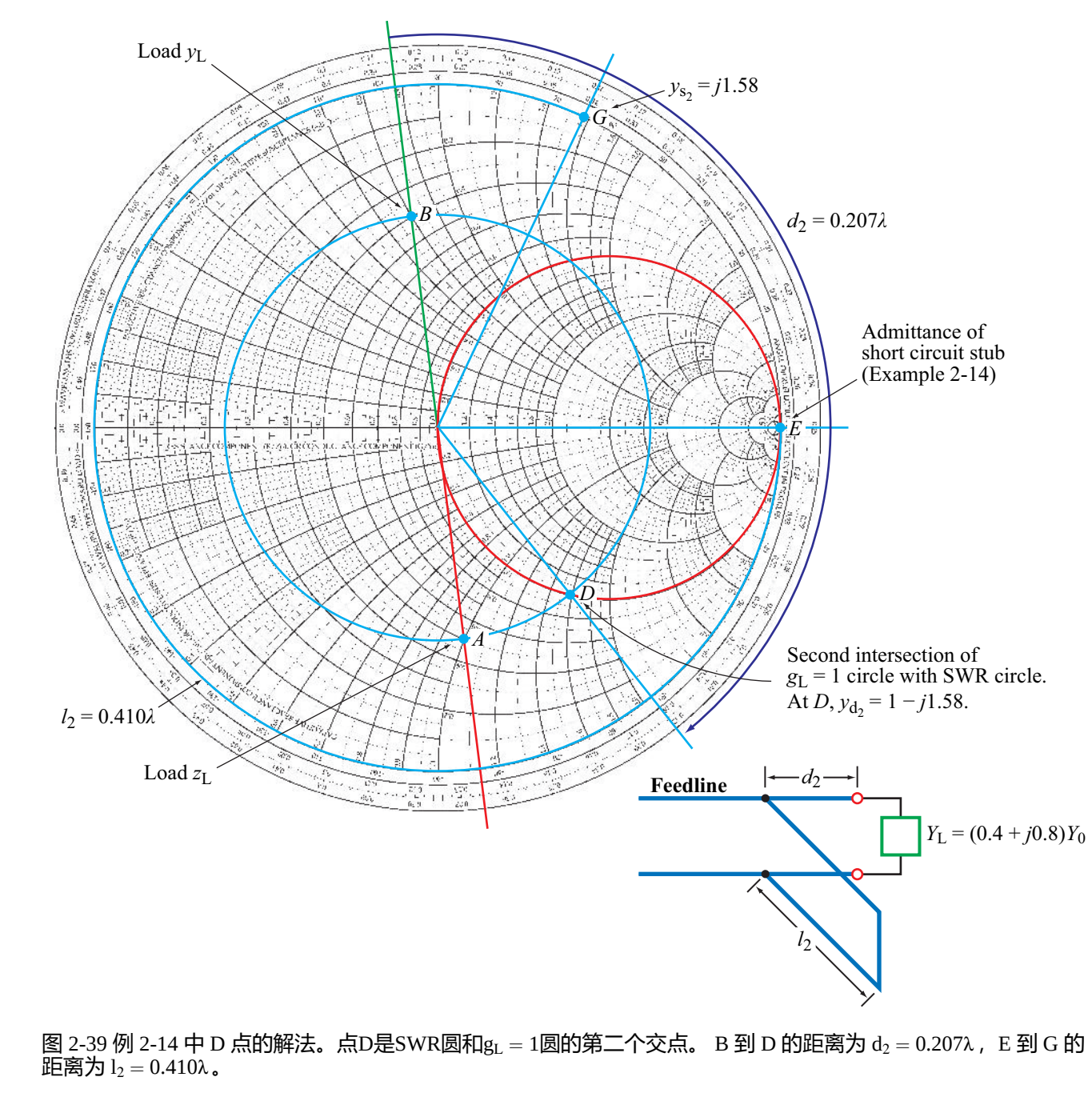
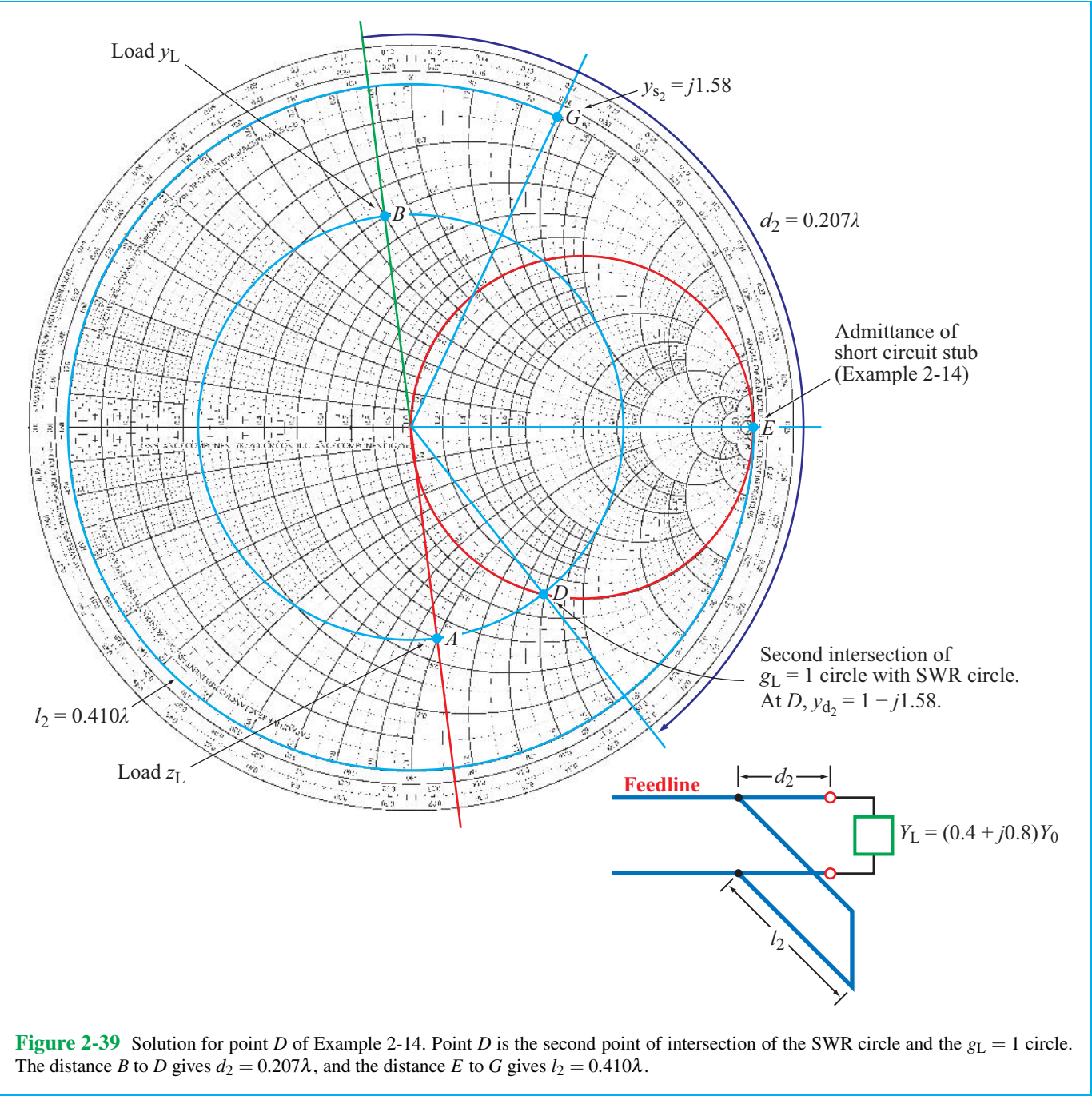


图 2-38 例 2-14 中 C 点的解法。A 点为 $z_L = 0.5 - j1$ 的归一化载荷；B 点为 $y_L = 0.4 + j0.8$ 。C 点是 SWR 圆与 $g_L = 1$ 圆的交点。B 到 C 的距离为 $d_1 = 0.063\lambda$ 。短路短截线 (E 至 F) 的长度为 $l_1 = 0.09\lambda$ 。



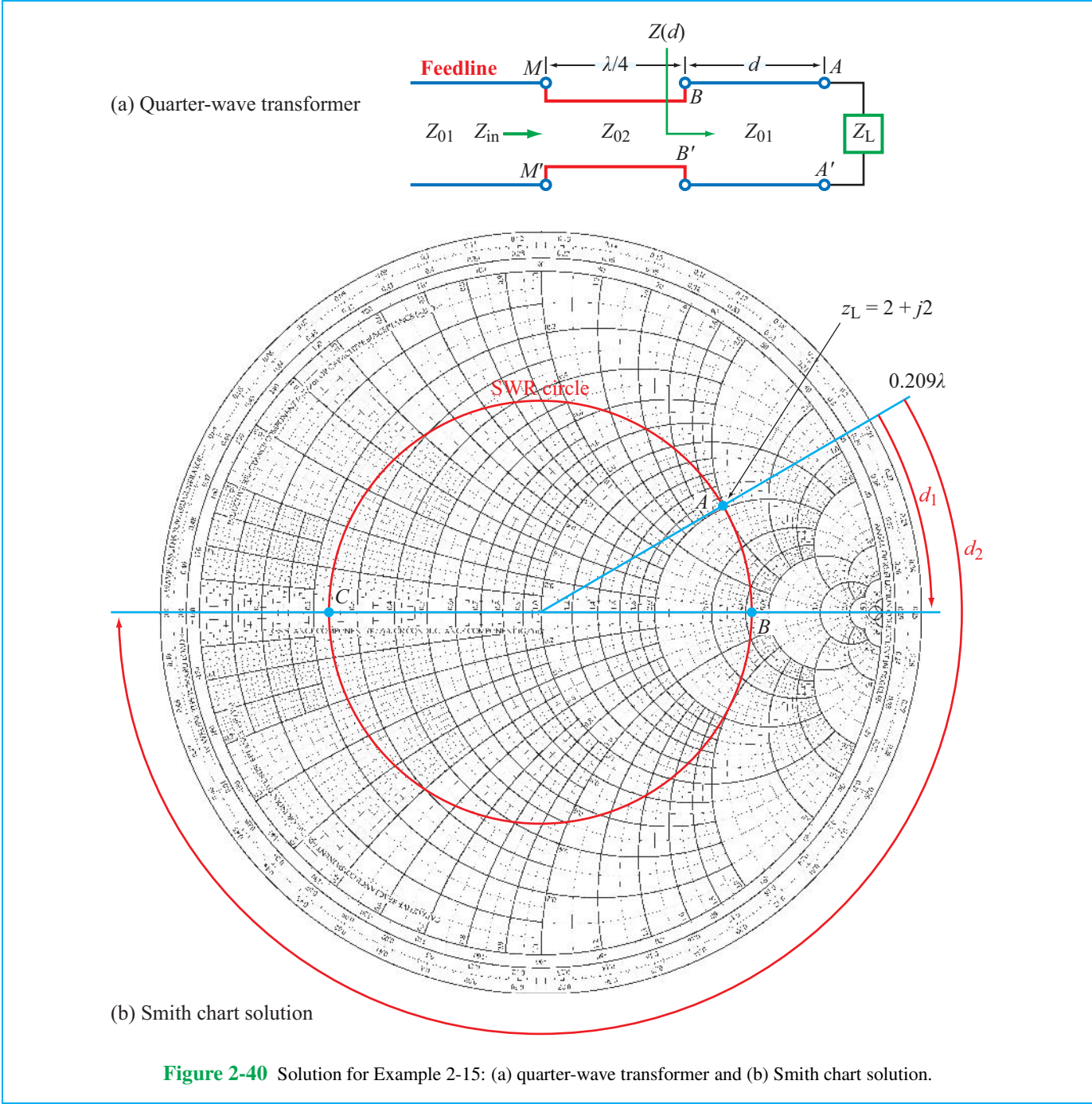


Figure 2-40 Solution for Example 2-15: (a) quarter-wave transformer and (b) Smith chart solution.

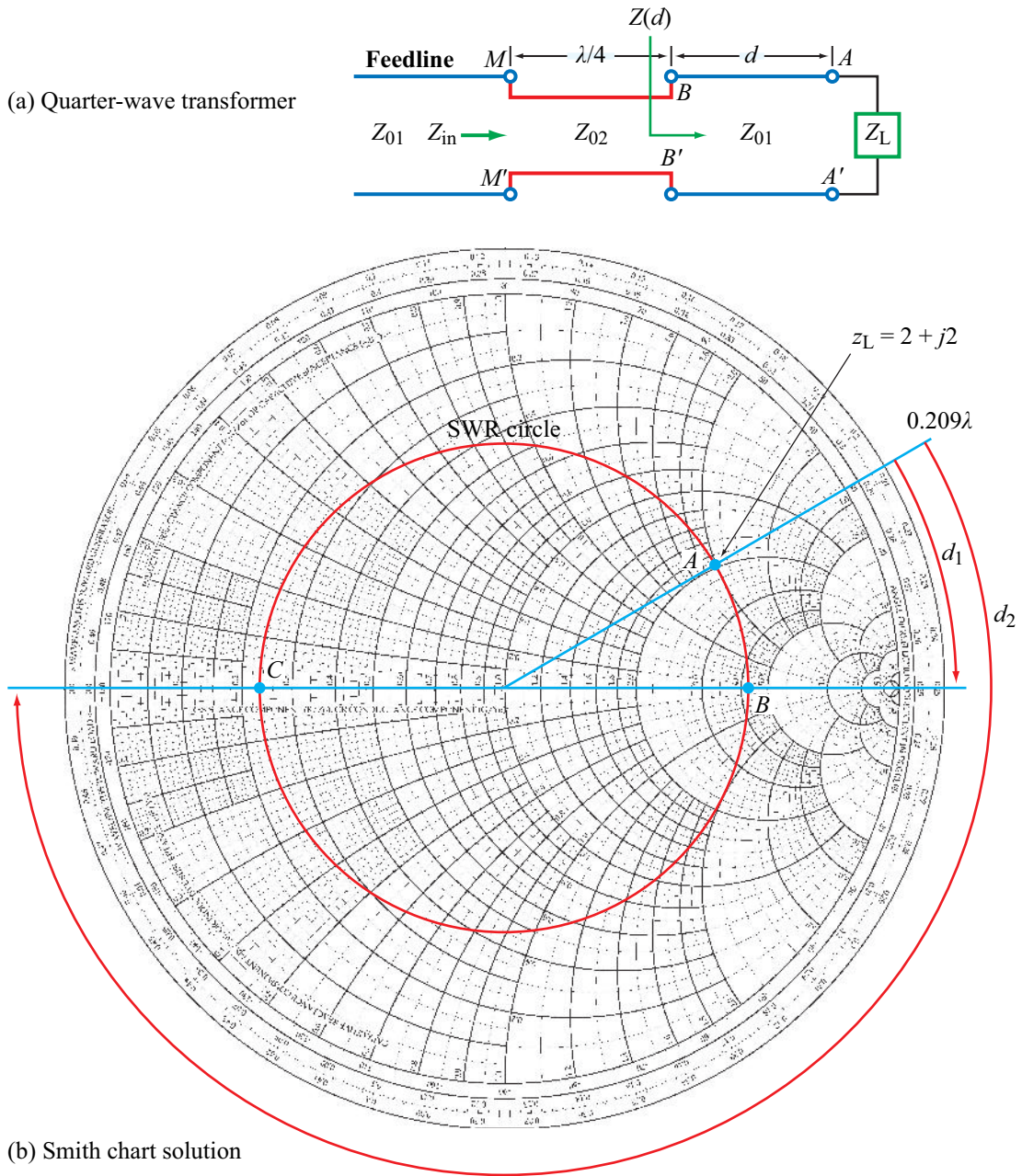


图 2-40 示例 2-15 的解决方案：(a) 四分之一波长变压器和 (b) 史密斯圆图解决方案。